

BS Software Engineering Course Outlines

*(For Sessions - Spring 2023 & before)
Semester IV to VIII*



**DEPARTMENT OF SOFTWARE ENGINEERING
FACULTY OF ENGINEERING & COMPUTING
NATIONAL UNIVERSITY OF MODERN LANGUAGES
ISLAMABAD
2024**

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~~**Scheme of Study for BS Software Engineering**~~

Semester – I

Code	Courses	Credit Hrs	
SEPF-121	Programming Fundamentals	3	1
SEIT-111	Introduction to Information & Communication Technology	2	1
IS-112	Islamic Studies	2	0
SEAP-101	Applied Physics	3	0
SEEC-113	English Composition & Comprehension	3	0
SEAF-161	Accounting Fundamentals	3	0
Math-1 for Pre-Medical students (3,0)		16	2

Semester – II

Code	Courses	Credit Hrs	
SECP-114	Communication & Presentation Skills	3	0
SEOO-122	Object Oriented Programming	3	1
SEDS-123	Discrete Structures	3	0
SESE-124	Software Engineering	3	0
SECA-102	Calculus & Analytical Geometry	3	0
PS-115	Pakistan Studies	2	0
Math-2 for Pre-Medical students (3,0)		17	1

Semester – III

Code	Courses	Credit Hrs	
SEDA-225	Data Structures & Algorithms	3	1
SESR-241	Software Requirement Engineering	3	0
SELA-203	Linear Algebra	3	0
SEIP-273	Introduction to Psychology	3	0
SEFM-262	Formal Methods in Software Engineering	3	0
		15	1

Semester – IV

Code	Courses	Credit Hrs	
SEDB-227	Database Systems	3	1
SESD-242	Software Design & Architecture	2	1
SEOS-226	Operating Systems	3	1
SEPS-204	Probability & Statistics	3	0
SEBP-263	Business Process Engineering	3	0
		14	3

Semester – V

Code	Courses	Credit Hrs	
SESC-344	Software Construction & Development	2	1
SEWE-346	Web Engineering	2	1
SEHC-343	Human Computer Interaction	3	0
SEAI-365	Artificial Intelligence* (SE Elec. 1)	2	1
SECN-328	Computer Networks	3	1
		12	4

Semester – VI

Code	Courses	Credit Hrs	
SEIS-329	Information Security	3	0
SETB-317	Technical & Business Writing	3	0
SESQ-345	Software Quality Engineering	3	0
SEVP-364	Visual Programming* (SE Elec. 2)	2	1
SEMA-366/SEUS-366	Track Elective 1* (SE Elec. 3)	2	1
SESS-367	Social Service	1	0
		14	2

Semester – VII

Code	Courses	Credit Hrs	
SESR-448	Software Re-Engineering	3	0
SESP-447	Software Project Management	3	0
SEOR-468	Operations Research **	3	0
SEFP-499	Final Year Project	0	3
SETE-469	Technical Entrepreneurship***	3	0
SEWE-470/SEST-470	Track Elective 2* (SE Elec. 4)	2	1
		14	4

Semester – VIII

Code	Courses	Credit Hrs	
SEFP-499	Final Year Project	0	3
SEFL-472	Foreign Language***	2	0
SEPP-416	Professional Practices	3	0
SEGD-471/SEPI-471	Track Elective 3* (SE Elec. 5)	2	1
		7	4

~~Course Outlines~~

~~Semester I~~

~~PROGRAMMING FUNDAMENTALS (SEPF-121)~~

~~I. Course Details~~

Credit Hours	4 (3+1)
Pre-requisites	-
Course Leader	Ms. Saadia Shabbir
Recommended Textbook(s)	1. C++ How to Program by Deitel and Deitel, 10th Edition, Pearson, (2016)
Recommended Reference (Books/Websites/Articles)	1. Starting out with Python, Tony Gaddis, Pearson; 4th edition (2017) 2. Starting out with Programming Logic & Designs, Tony Gaddis, Pearson; 4th edition (2015) 3. Introduction to Computation and Programming Using Python: With Application to Understanding Data, 2nd Edition by Guttag, John, MIT Press; 2nd edition (2016) 4. Practice of Computing Using Python by William Punch & Richard Enbody, Pearson; 2nd edition (2012) 5. A first book of C++, Garry Bronson, Cengage Learning, 4th edition, 2011

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Understand basic concepts of problem analysis and solving techniques	Cognitive	2	2	A1, Q1, Mid Term
CLO-2	Apply fundamental programming concepts of conditional and repetitive structures	Cognitive	3	2	A2, Q2, Mid Term, Final Term
CLO-3	Implement concepts of modular design, data handling, and pointers	Cognitive	3	2	A3, Q3, Final Term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Introduction to problem solving, a brief review of Von-Neumann architecture, Introduction to programming, role of compiler and linker, introduction to algorithms, basic data types and variables, input/output constructs, arithmetic, comparison and logical operators, conditional statements and execution flow for conditional statements, repetitive statements and execution flow for repetitive statements, lists and their memory organization, multi-dimensional lists, introduction to modular programming, function definition and calling, stack rolling and unrolling, string and string operations, pointers/references, static and dynamic memory allocation, File I/O operations.

VI. Weekly Breakdown

Week No.	CLOs	Topics	Reference
1	CLO-1	Introduction to Problem Solving, Introduction to Algorithms and Flowcharts, Introduction to Programming Languages, Benefits gained from Programming Languages. Generations of Programming Languages, Use of different languages in different generations, Working of Compilers, and Linkers.	Chapter 1
2		Introduction to C++, Advantages of using C++, First program in C++, Naming rules in C++, Data types, Types of C++ Instructions.	Chapter 1, 2
3		Preprocessor directive, The Main Method, Compiling and Executing the program.	Chapter 2
4		Variable definition and declaration, Escape sequences, Arithmetic Operators, Input gathering	Chapter 2
5	CLO-2	Decision making statements, if, if else and nested if else statements.	Chapter 4
6		Loops, for loop, nested for loop, Using for loop with single and multiple statements, while and nested while loop.	Chapter 4, 5
7		The do while loop, Difference between while and do while loop, Using switch and break statements, Logical and Conditional Operators, Operators Precedence, continue and goto statements.	Chapter 4, 5

8	CLO-3	Local and global variables, Introduction to Arrays and defining Arrays. Arrays initialization, Operations, Passing to functions, Multidimensional Arrays.	Chapter 7
9	Midterm Exams		
10	CLO-3	Functions, Functions Declaration, Function Calling, Function Definition. Eliminating function declaration, Passing arguments to functions (by value).	Chapter 6
11		Passing variables to functions, returning values from functions, Return statement, Overloaded functions, Inline functions.	Chapter 6
12		Structures, Defining structures and structure variables, Accessing structure members.	Chapter 4
13		Returning structure variables, Passing arguments to functions (with respect to structure)	Chapter 4
14		Pointers, Pointers of Arrays, Pointer constants and pointer variables, Passing pointers as arguments,	Chapter 8
15		Pointer and functions, Basic concept of Pass by reference. Function Pointers	Chapter 8
16		File Handling, Introduction to ifstream and ofstream, Writing to /from file, Simple Programs: word count, sentence count.	Chapter 14

~~PROGRAMMING FUNDAMENTALS (SEPF-121) -LAB~~

~~I.Course Details~~

Credit Hours	4 (3+1)
Pre-requisites	-
Course Leader	Ms. Saadia Shabbir
Recommended Textbook(s)	1. C++ How to Program by Deitel and Deitel, 10th Edition, Pearson, (2016)
Recommended Reference (Books/Websites/Articles)	1. Starting out with Python, Tony Gaddis, Pearson; 4th edition (2017) 2. Starting out with Programming Logic & Designs, Tony Gaddis, Pearson; 4th edition (2015) 3. 4. Introduction to Computation and Programming Using Python: With Application to Understanding Data, 2nd Edition by Guttag, John, MIT Press; 2nd edition (2016) 4. Practice of Computing Using Python by William Punch & Richard Enbody, Pearson; 2nd edition (2012) 5. A first book of C++, Garry Bronson, Cengage Learning, 4th edition, 2011

~~II.Lab Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Develop simple programs to provide solution for stated specifications	Cognitive	6	4	Lab tasks (reports), Mid-term exam, Project
CLO-2	Manipulate state-of the art tools to develop, debug and test simple applications	Psychomotor	3	5	Lab tasks (reports), Mid-term exam, Project

~~III.Lab Assessment~~

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25
Total	100

~~IV.Weekly Breakdown (All CLOs shall be assessed in every lab)~~

Week No.	Topics
1	Introduction to C++, Basic features of Microsoft Visual Studio, first program in C++, Compiling and Executing the program.
2	Variable declaration and initialization, Assignment Operator
3	Arithmetic Operators, Input gathering, Operators Precedence
4	Decision making statements, if, if else and nested if else statements,
5	Using switch and break statements, Logical and Conditional Operators
6	Loops, While Loop
7	Do-while loop, Difference between while and do while loop
8	For Loop (Basic Examples)
9	For Loop (Advanced Examples)
8	Functions, Functions Declaration and Definition, Function Calling
9	Midterm Exams
10	Parameters of Functions, Argument Passing
11	Return Type, Return Value, Inline Functions, Overloaded Functions, Local and Global Variables
12	Basic concepts of arrays, Traversing arrays
13	Arrays (Advanced Examples)
14	Pointers, Pointers of Arrays, Pointer constants and pointer variables, Passing pointers as arguments, Pointer and functions, Basic concept of Pass by reference
15	Structures, defining structures and structure variables, Accessing structure members, Returning structure variables, Passing arguments to functions (with respect to structure)
16	File Handling in C++

~~INTRODUCTION TO INFO. AND COMMUNICATION TECH. (SEIT-111)~~

~~I. Course Details~~

Credit Hours	3 (2+1)
Pre-requisites	-
Course Leader	Mr. Zohaib Ahmad
Recommended Textbook(s)	Discovering Computers, Shelly Cashman series, 2016.
Recommended Reference (Books/Websites/Articles)	1. Introduction to Computers, Peter Norton, 6th edition. 2. Understanding computers: today and tomorrow, comprehensive, Deborah Morley, Charles S. Parker, 15th Edition, Cengage Learning, 2014. 3. Using information technology, Brian K. Williams, Stacey C. Sawyer.

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Define various types and components of computer including input/output devices, memory, storage media	Cognitive	1	1	Q1, A1, Midterm
CLO-2	Describe different types of software from operating systems to system utilities and productivity apps	Cognitive	2	2	Q2, A2, Midterm, Final Term
CLO-3	Demonstrate the basic concepts regarding computer networks, database management and computer security from a user point of view, components of computer including input/output devices, memory, storage media	Cognitive	3	4	Q3, A3, Final Term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

~~IV. Grading Policy~~

~~For students admitted in Fall 2021 and onwards~~

Grade	A+	A	B+	B	C+	C	D+	D	F
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%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

~~Introducing Today's Technologies Computers, Devices, and the Web: Computing Components, Processors, Memory, the Cloud, and More. Input and Output Extending Capabilities of Computers and Mobile Devices. Digital Storage Preserving Content Locally and, on the Cloud. Operating Systems Managing, Coordinating, and Monitoring Resources, GUI and CLI with CLI commands and their usages. Storing and retrieving Information & Input/output media and devices. Communicating Digital Content Wired and Wireless Networks and Devices. Connecting and Communicating Online the Internet, Websites, and Media. Computers and Mobile Devices Evaluating Options for Home and Work. Programs and Apps Productivity, Graphics, Security, and Other Tools. Digital Security, Ethics, and Privacy Threats, Issues, and Defenses. Building Solutions Database, System, and Application Development Tools. Numbering Systems: Binary, Octal, Decimal and Hexadecimal, Base inter conversions, Binary arithmetic operations.~~

VI. Weekly Breakdown

Week No.	Topics	CLOs
1	Technology, Types of Computers, Mobile & Game Devices, Data and Information, The Web ,Web Searching, Online Social Networks, Internet Communications	CLO-1
2	Digital Security and Privacy, Viruses and Other Malware, Privacy, Health Concerns, Environmental Issues, Programs and Apps, Operating Systems, Applications, Installing and Running Programs, Developing Programs and Apps, Communications and Networks, Wired and Wireless Communications, Networks, Technology Uses, Government, Finance, Retail, Entertainment, Health Care, Science, Travel, Publishing, Manufacturing, Technology Users	CLO-1
3	Online: The Internet, Websites and Media, The Internet , Evolution of the Internet , Connecting to the Internet, Internet Service Providers , How Data Travels the Internet , IP & MAC Addresses and Domain Names , The World Wide Web ,Navigating the Web, Web Addresses, Web Apps and Mobile Apps	CLO-1
4	Types of Websites, Digital Media on the Web, Graphics ,Audio , Video, Plug-Ins , Other Internet Services, Email , Email Lists , Internet Messaging, Chat Rooms , Online Discussions , VoIP, FTP ,Netiquette	CLO-1

5	Evaluating Computers and Mobile Devices, Mobile Computers and Desktops, Laptops, Tab and Other Mobile Computers, Handheld Computers, Desktops and All-in-Ones, Servers, Terminals, Point-of-Sale Terminals, Wearable Devices, Game Devices, Embedded Computers	CLO-1
6	Inside the Case, The Motherboard, Processors, The Control Unit, The Arithmetic Logic Unit, Machine Cycle, Registers, The System Clock, Personal Computer and Mobile Device Processors, Processor Cooling, The Internet of Things, Cloud Computing, Cloud Computing Services	CLO-2
7	Memory, Bytes and Addressable Memory, Types of Memory, RAM, Cache, ROM, Flash Memory, CMOS, Memory Access Times, Adapters, Adapter Cards, USB Adapters, Buses, Bus Width, Types of Buses, Power Supply and Batteries, Data Representation & Number System	CLO-2
8	Programs and Apps, Role of the Operating System, Obtaining Software, Installing Software, Categories of Programs and Apps, Multimedia and Interest Applications, Communications Applications, File, Disk, and System Management Tools	CLO-2
9	Midterm Exams	
10	Operating Systems, Operating System Functions, Starting Computers and Mobile Devices, Shutting Down Computers and Mobile Devices, Providing a User Interface, Managing Programs, Managing Memory	CLO-2
11	Types of Operating Systems, Desktop Operating Systems, Windows/Mini Feature, Mac OS/Mini Feature, Unix, Linux, Chrome Os, Running Multiple, Systems, Android, iOS, Windows Phone Mobile versus Desktop Operating Systems	CLO-2
12	Communications, Networks, LANs, MANs, WANs, and PANs, Network Architectures, Communications Software, Communications Network, Communications Standards and Protocols	CLO-3
13	Communications Lines, Transmission Media, Physical Transmission Media, Twisted-Pair Cable, Coaxial Cable, Fiber-Optic Cable, Wireless, Infrared, Broadcast Radio, Cellular Radio, Microwaves	CLO-3
14	CH 5 :Digital Security Ethics, and Privacy: Threats, Issues, and Defenses Digital Security Risks, Cybercrime, Internet and Network Attacks Malware, Botnets, Encryption, Digital Signatures and Certificates, Hardware Theft, Vandalism, and Failure	CLO-3
15	Backing Up – The Ultimate Safeguard, Wireless Security, Ethics and Society, Information Accuracy, Intellectual Property Rights, Codes of Conduct, Cookies, Phishing, Spyware and Adware, Social Engineering, Privacy Laws, Employee Monitoring, Content Filtering	CLO-3
16	What is data base and data base management system, Evolution of database management, Data Concepts and Characteristics, Data Organization	CLO-3

~~INTRODUCTION TO INFO. & COMM. TECH. (SEIT-111) - LAB~~

~~I.Course Details~~

Credit Hours	3 (2+1)
Pre-requisites	-
Course Leader	Mr. Zohaib Ahmad
Recommended Textbook(s)	Discovering Computers, Shelly Cashman series, 2016.
Recommended Reference (Books/Websites/Articles)	1. Introduction to Computers, Peter Norton, 6th edition. 2. Understanding computers: today and tomorrow, comprehensive, Deborah Morley, Charles S. Parker, 15th Edition, Cengage Learning, 2014. 3. Using information technology, Brian K. Williams, Stacey C. Sawyer.

~~II.Lab Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Discuss problems, participate and provide understanding of MS Office (Word, Excel, Power point)	Cognitive	2	2	Participation, Discussion, Viva
CLO-2	Revise the Tasks and obtain feedback of previous Labs in the form of lab reports	Affective	3	5	Report writing, OEL, Final term assessment
CLO-3	Apply the understanding, operating, and designing to real problems, develop problem solving skills, promote creativity	Psychomotor	2	7	Lab tasks, OEL, Final term assessment

~~III.Lab Assessment~~

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25
Total	100

~~IV. Weekly Breakdown (All CLOs shall be assessed in every lab)~~

Week No.	Topics
1	Introduction to the lab equipment and SOPs of the labs
2	Introduction to computer hardware and system information
3	Learning of different functions of Microsoft Word
4	Working with Insert tab (table, smart art , pictures, shapes)
5	Working with Insert tab (page no., header/footer, page break, symbols and equations comments, links, track changes.)
6	Working with Design and page layout tabs and table figures and contents.
7	Referencing, PDF conversion, views, reviews
8	Report writing and CV making in Microsoft Word
9	Mid Term Exam
10	Introduction to Excel (different functions of Microsoft Excel) Cell merging, sum, average, cell sizing, sorting swapping, designing, filters
11	Graph making, Chart making and formula implementation in Microsoft Excel
12	Pivot table in Microsoft Excel
13	Introduction to Power Point Designing presentation in Power point
14	Presentation making
15	Open Ended Lab
16	Lab End Term Exam

ISLAMIC STUDIES (IS-112)

I.Course Details

Credit Hours	2 (2+0)
Pre-requisites	-
Course Leader	Mr. Zohaib Ahmed
Recommended Textbook(s)	1. Islamic education for BA B-SC B-COM. By M.D.Zafar. Aziz book depot Urdu Bazar Lahore 2. Islamic studies for BA B-SC B-COM. by Zia urRahman Ahmad Aziz book depot Urdu Bazar Lahore 3. A study of Islamic History by prof. K. Ali. Aziz book depot Urdu bazaar, Lahore
Recommended Reference (Books/Websites/Articles)	1. Islam in the light of the final testament and traditions by Shaukatomari Taurus Publications Karachi. 2. Islamic Ideology by Khalifa Abdul Hakim Institute of Islamic culture 2-club road, Lahore. 3. Whatever everyone should know about Islam and Muslims by Suzanne Haneef Kazi publications 121 zulgarnain chambers Ganpat road Lahore

II. Course Learning Outcomes (CLO)

CLO	Description	Domain	Taxonomy Level	PLO	Assessment Artifact
CLO-1	Discuss basic concept of Islam (faith, pillars and systems etc.) and express their impact on society and describe the religion of Islam importance in the human life in the light of guidance provided by Quran and Sunnah	Cognitive	2	1	Q1, A1, Midterm
CLO-2	Demonstrate the challenges of modern science and contemporary world	Cognitive	3	10	A2, Q2 Midterm, Final Term
CLO-3	Analyze Islamic Ethics and code of personal practice in social life	Cognitive	4	9	A2, A3, Q3, Q4 Midterm, Final Term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5

Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Basic Themes of Quran, Introduction to Sciences of Hadith, Introduction to Islamic Jurisprudence, Primary & Secondary Sources of Islamic Law, Makken & Madnian life of the Prophet, Islamic Economic System, Political theories, Social System of Islam.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Importance of Quran to be a noble human Introduction of Quran, Fazail-e-Quran, Duties of a Muslim during recitation of the Quran, Importance of Understanding of the Quran	Textbook 1
2		Characteristics of true human being and a beloved servant of Al-Rehman (Surh al-Furqan 63-78) Humbleness in dealings/ walk, Focusing on the Vision, Spending nights in prayer, Pray for protection from the hell, Pray for the Family (spouse and child)	
3		Social norms: Towards Rasool (ﷺ)/ elders & authorities (Surah Al-Hujrat 1-5) Lowering voice in front of Rasoolﷺ/ Authority, Presentation of matters in respectful words, Manners of Calling Rasoolﷺ / Authority Conduct of a human towards Media (al-Hujrat 5-8)	

4		Why the Sunnah is important for a Human? Literal Meaning of the Sunnah, Importance of the Sunnah according to the Quran, Relativity among the Quran and the Sunnah	
5		Twenty selected Ahadith with translation	
6		Twenty selected Ahadith with translation	
7	CLO-2	Islam in the Light of the Quran and Hadith(Theological Section) Tauheed, Risalat, Aakhirat	
8		Islam in the Light of the Quran and Hadith (Teaching section) Prayer, Fasting, Charity (Zakat/Sadaqat), Hajj, Jihad	
9	Midterm Exams		
10	CLO-2	Study of Seerat-un-Nabi(ﷺ) Makkah Life Birth / early childhood, Before & after revelation of the Quran, The preaching and its different stages	Textbook 1
11		Study of Seerat un Nabi (ﷺ) Madina Life; Hijrah; Qualities of the Leadership and the Nation ,Fatah-e-Makkah and conduct of the Prophet towards his opponents, HajatOul Wida	
12	CLO-3	Importance of Quran to be a noble human Introduction of Quran, Fazail-e-Quran, Duties of a Muslim during recitation of the Quran, Importance of Understanding of the Quran	Textbook 1 and Textbook 2
13		Characteristics of true human being and a beloved servant of Al-Rehman (Surh al-Furqan 63-78) Humbleness in dealings/ walk, Focusing on the Vision, Pray for the Family (spouse and child) , Spending within limits (wealth/time/education/ energies.	Textbook 1 and Textbook 2
14		Pray Allah in hard times, Don't harm any thing in the universe , Don't make unlawful relations , Don't attend falsehood meetings , Pass by from irrelevant things	Textbook 1 and Textbook 2
15		Social norms: Towards Rasool (ﷺ)/ elders & authorities (Surah Al-Hujrat 1-5) Lowering voice in front of Rasoolﷺ/ Authority Presentation of matters in respectful words , Manners of Calling Rasoolﷺ / Authority	Textbook 1 and Textbook 2
16		Why the Sunnah is important for a Human? Literal Meaning of the Sunnah, Importance of the Sunnah according to the Quran, Relativity among the Quran and the Sunnah	Textbook 2 and Textbook 3

~~APPLIED PHYSICS (SEAP-101)~~

~~I. Course Details~~

Credit Hours	3 (3+0)
Pre-requisites	-
Course Leader	Dr. Naveed Ahmed
Recommended Textbook(s)	Fundamentals of Physics (Extended), 10th edition, Resnick and Walker
Recommended Reference (Books/Websites/Articles)	Electronic Devices (Conventional Current Version), 10th Edition Thomas L. Floyd

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Apply acquired knowledge to solve series and parallel resistor networks by using KVL and KCL	Cognitive	3	1	A1, Q1, Midterm
CLO-2	Understand and explain the basic operation and working of semiconductor devices	Cognitive	3	1	A2, Q2 Midterm
CLO-3	Apply acquired knowledge to solve circuits which consists of semiconductor devices	Cognitive	3	1	A3, Q3, Final Term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

~~IV. Grading Policy~~

~~For students admitted in Fall 2021 and onwards~~

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

~~For students admitted before Fall 2021~~

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Charge, Negative and Positive charge, and Electric field, Basic concept of Voltage, Current and Resistor, Series and Parallel combinations of Resistors, Ohm's Law, KCL and KVL, Introduction of materials, conductor insulator semiconductor, electronic configuration, P type and N type materials, doping process, Introduction to diode theory, Approximations of diodes in circuits, some mathematical problems, Voltage and current relationship in diode, forward and reverse biasing of diodes, Zener diode and its applications, Rectifiers, Introduction to rectifiers, Half wave and full wave rectifier, Power supplies, Ripple factor and its calculations, Introduction of BJT and its operation, Collector characteristics of BJTs and region of operations, Continuation with BJT operation, Biasing of BJT, Common emitter configuration of BJT and concept of load line and Q point, Continuation with load line and Q point with some problems, Emitter bias configuration and stability of Q point, Voltage divider bias configuration and its analysis, Introduction of JFETs and MOSFETs, Basic theory of JFETs and MOSFETs, Basic calculations of JFETs and MOSFETs.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Basis of Electricity, Charge, Negative and Positive charge, and Electric field Basic concept of Voltage, Current and Resistor	Chapter 1
2		Series and Parallel combinations of Resistors, Ohm's Law	Chapter 1
3		KCL and KVL	Chapter 2, 3
4	CLO-2	Basis of Diode Theory, Introduction of materials, conductor insulator semiconductor, electronic configuration	Chapter 4
5		P type and N type materials, doping process, Introduction to diode theory	Chapter 4
6		Approximations of diodes in circuits, some mathematical problems Voltage and current relationship in diode, forward and reverse biasing of diodes	Chapter 5
7		Zener diode and its applications, Rectifiers	Chapter 5
8		Introduction to rectifiers, Half wave and full wave rectifier	Chapter 5
9	Midterm Exams		
10	CLO-2	Power supplies, Ripple factor and its calculations	Chapter 5
11		Bipolar Junction Transistors, Introduction of BJT and its operation	Chapter 6
12		Collector characteristics of BJTs and region of operations	Chapter 6

13		Continuation with BJT operation, Biasing of BJT	Chapter 6
14	CLO-3	Common emitter configuration of BJT and concept of load line and q point, Continuation with load line and Q point with some problems	Chapter 6
15		Emitter bias configuration and stability of Q point, Voltage divider bias configuration and its analysis	Chapter 6
16		Introduction of JFETs and MOSFETs, Basic theory of JFETs and MOSFETs, Basic calculations of JFETs and MOSFETs	Chapter 7

ENGLISH COMPOSITION AND COMPREHENSION (SEEC-113)

I.Course Details

Credit Hours	3 (3+0)
Pre-requisites	-
Course Leader	Mr. Safee Ullah Safee
Recommended Textbook(s)	High school English grammar and composition by Wren and Martin
Recommended Reference (Books/Websites/Articles)	1.College Writing Skills with reading by John Langan, Mc Graw-Hill, 5th Edition https://www.britishcouncilfoundation.id/en/english/articles/british-and-american-english 2. https://www.butte.edu/departments/cas/tipsheets/readingstrategies/skimming_scanning 3. https://www.dyslexia-reading-well.com/44-phonemes-in-english.html https://grammar.yourdictionary.com/punctuation/punctuation-rules-help.html

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Identify the parts of speech, phrase clause and sentence	Cognitive	1	7	A1, Q1, Midterm
CLO-2	Change the voice and narration of the sentence	Cognitive	2	7	A2, Q2 Midterm
CLO-3	Use English vocabulary and skills to in writing paragraphs, essays, letters and applications	Cognitive	2	7	A3, Q3, Final Term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

~~IV. Grading Policy~~

~~For students admitted in Fall 2021 and onwards~~

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

~~For students admitted before Fall 2021~~

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

~~V. Course Contents~~

~~Parts of speech, Modals and Articles, Tenses and Conditional sentence, Types of Sentences, Change of voice, Change of Narration, Writing Strategies, Writing strategies Paragraph writing, Use of articles and Punctuation marks, Reading Comprehension, Essay Writing, Phonemes-vowels, consonants and diphthongs. Rules of Pronunciation, American and British sound differences, Rules of Spellings, Reading: extensive/intensive, skimming/scanning~~

~~VI. Weekly Breakdown~~

Week No.	CLOs	Topics	Reference
1	CLO-1	Parts of speech	Chapter-4, 43 [Textbook]
2		Modals and Articles	Chapter 14, 32 [Textbook 2]
3		Tenses and Conditional sentence	Chapter 3, 24 [Textbook]
4	CLO-2	Change of voice	Chapter 22 [Textbook]
5		Change of narration	Chapter 16 [Textbook Part 1]
6		Strategies for effective writing	Chapter 33 [Textbook Part IV]
7	CLO-2	Paragraph writing	//
7	CLO-2	Phonemes-vowels, consonants	//

8	CLO-2	Parts of speech	Chapter5, [Textbook 2]
9	Midterm Exams		
10	CLO-2	Rules of Pronunciation, American and British sound differences, Rules of Spellings)	//
11	CLO-3	Rules of spellings and use of articles	[Textbook 1]
12	CLO-3	Reading : extensive/intensive, skimming/scanning	//
13	CLO-3	Use of punctuation marks	//
14	CLO-3	Reading comprehension	//
15	CLO-3	Essay writing	[Textbook 2]
16	CLO-3	Note taking/note making and preparing for exams	[Textbook 1]

~~ACCOUNTING FUNDAMENTALS (SEAF-161)~~

~~I. Course Details~~

Credit Hours	3 (3+0)
Pre-requisites	-
Course Leader	Mr. Zohaib Ahmad
Recommended Textbook(s)	1. Financial & managerial accounting (The basis for business decisions), Meigs & Meigs, 15th edition, McGraw Hill/Irwin, 2009.
Recommended Reference (Books/Websites/Articles)	1. Software Engineering, A practitioner's approach, Pressman R. S. & Maxim B. R., 8th Edition, McGraw-Hill, 2015. 2. Fundamentals of accounting, Libby & Libby, 8th edition, McGraw Hill, 2013.

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Identify why accounting is a necessary skill. Describe the key concepts and elements of accounting.	Cognitive	1	1	A1, Q1, Midterm
CLO-2	Describe accounting cycle: Preparation of Journal, ledger, trial balance, income statement, statement of retained earnings, statement of owner's equity and balance sheet.	Cognitive	2	2	A2, Q2 Midterm, Final Term
CLO-3	Demonstrate accounting entries in different concerns like adjustments of entries, merchandising concerns etc.	Cognitive	3	3	A3, Q3, Midterm, Final Term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Weekly Breakdown

Week No.	CLOs	Topics	Reference
1	CLO-1	Introduction to Accounting. Accounting Information. Types of Accounting Information. Role of Accounting Information in making decisions. Importance of Accounting Information for internal and External users. Accounting Systems.	Chapter 1
2		Basic Terms. Business and its types. Types of Accounts (Assets, Liabilities, Capital, Expense and Revenue. Types of Accounts. Accounting Cycle. Accounting Period, Credit, Debit, Fiscal Year,	Chapter 1
3		How to balance the fundamental accounting equation; Debits and Credits; "T" Accounting; Double Entry Accounting; Transactions. Cash and credit transaction. Effect of Transactions on Accounting Equation. Source Documents; General Journal. Journalizing the events of increase in capital, increase and decrease in Assets and Liabilities.	Chapter 2
4	CLO-3	General Journal. Accrual Basis Accounting Applying Matching Principle and Realization Principle in recording expenses and Revenue. General Journal and its relationship to ledger. Posting (Process of transferring information from the journal to the individual accounts in the Ledger. Types of Ledgers.	Chapter 2
5	CLO-2	Trial Balance, Its Uses and Limitations. The Accounting Cycle. Journal, Ledgers and Trial balance.	Chapter 3
6		Financial Statements. Nature and general purpose of Financial Statements. Relevant Accounting Principles. Purpose of the Income Statement; Multi-Step Income Statement; What are Retained Earnings, Revenue, Expenses, Net Income, Income Tax,	Chapter 3

7		Statement of Retained Earnings and Classified Balance Sheet. Relation of Income statement, Statement of Owners Equity and Balance Sheet.	Chapter 4
8		Practice from journal to Balance Sheet. Accounting Cycle.	Chapter 4
9	Midterm Exams		
10	CLO-3	Need for adjusting entries, Types of Adjusting Entries. The concept of Depreciation. Accumulated Depreciation. Book Value. Contra Asset Account. Depreciate able Assets. Useful Life. Converting Assets to Expenses; Prepaid Expenses, Accrued Expenses	Chapter 5
11		Converting liabilities to Revenue. Accruing uncollected revenue. Adjusted trial balance. Purpose of Adjusted Trial Balance. Effects of Adjusting Entries on Financial Statements	Chapter 5
12	CLO-2	Preparing a Work Sheet and its uses. Closing The Temporary Accounts. Preparation of After closing Trial Balance.	Chapter 5
13	CLO-1	GAAP (Generally Accepted Accounting principles). Introduction to Merchandising Concerns. Operating cycle of Merchandising Companies	Chapter 5
14	CLO-3	Journal Entries in Periodic & Perpetual Systems Accounting for Merchandising Concerns	Chapter 6
15		Cash and credit purchase transactions, Cost Transaction, Transportation cost), Purchases return and purchases Discount transaction journal entries.	Chapter 6
16		Special cases in Merchandising Companies Net method. Special cases in Merchandising Companies Gross method. Credit terms, Cash discounts, Returns of Unsatisfactory, Merchandising Exercise. Practice of the entries for all merchandising concerns.	Chapter 6

~~MATH 1~~

~~I. Course Details~~

Credit Hours	3 (3+0)
Pre-requisites	-
Course Leader	Dr. Asad Husain
Recommended Textbook(s)	Thomas and Finney- Calculus and Analytical Geometry, illustrated Publisher, Pearson Addison Wesley; edition 11, 2007
Recommended Reference (Books/Websites/Articles)	-

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment
CLO-1	Understand the concepts of sets, binary operations, and number theory a line, curves, functions, and be able to draw the graphs.	Cognitive	2	1	A1, Q1, Midterm
CLO-2	Evaluate the limits, continuity, differentiation	Cognitive	3	1	A2, Q2, Midterm, Final term
CLO-3	Evaluate the limits, continuity, differentiation, and Integration.	Cognitive	3	2	A3, Q3, Final term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

~~IV. Grading Policy~~

~~For students admitted in Fall 2021 and onwards~~

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	≥ 90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	< 50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Sets definition, types of sets, Venn diagrams, Number theory; Properties of real numbers and real line, coordinate plane; Complex numbers properties and operations; Introduction to functions, Types of functions, graphs, the inverse of functions, slope tangent and normal. Introduction to limits, Techniques of finding limits. Continuous and discontinuous functions Partial Fraction; Concept and idea of differentiation; Geometrical and Physical meaning of derivatives; Rules of differentiation, Chain rule, Techniques of differentiation, Maxima and Minima of a function for single-variable, Concavity; Antiderivatives, Concept, and idea of Integration Rules and techniques of integration; The area under the curve a graphical perspective, Definite Integrals; Introduction to differential equations

VI. Weekly Breakdown

Week No.	CLOs	Topics	Reference
1	CLO-1	Sets definition , types of sets,	Chapter 1
2		Venn diagrams, Number theory	Chapter 1
3		Properties of real numbers and real line, coordinate plane	Chapter 1
4		Complex numbers, properties, and operations	Chapter 1
5		Introduction to functions, Types of functions, graphs, the inverse of functions, slope tangent, and normal	Chapter 1
6	CLO-2	Introduction to limits, Techniques of finding limits	Chapter 2
7		Continuous and discontinuous functions	Chapter 2
8		Partial Fraction	
9	Midterm Exams		
10	CLO-2	Concept and idea of differentiation	Chapter 3
11		Geometrical and Physical meaning of derivatives	Chapter 3
12	CLO-3	Rules of differentiation, Chain rule, Techniques of differentiation	Chapter 3
13		Maxima and Minima of a function for single-variable, Concavity	Chapter 4
14		Antiderivatives, Concept, and idea of Integration, Rules and techniques of integration	Chapter 5

15		The area under the curve a graphical perspective, definite Integrals	Chapter6
16		Introduction to differential equations	Chapter 9

~~**Semester II**~~

~~COMMUNICATION & PRESENTATION SKILLS (SECP-114)~~

~~I. Course Details~~

Credit Hours	3 (3+0)
Pre-requisites	-
Course Leader	Mr. Safee Ullah Safee
Recommended Textbook(s)	1. Effective business communication, hamurphy 8th edition. 2. Business communications-strategy and Skills, Mary Munter. 3. Business English and communication, Lyn R. Clark and Kenneth Zimmer-8th edition.
Recommended Reference (Books/Websites/Articles)	1. Technical report writing today by Daniel G Riordan, 8th edition. 2. Patterns of college writing, Laurie G. Kirsznner and Stephen R. Mandell, 4th edition. 3. English Phonetics and phonology, Peter Roach.

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Define basic concepts and terms about communication	Cognitive	1	7	A1, Q1, Midterm
CLO-2	Discuss different theories and principles about communication	Cognitive	2	7	A2, Q2 Midterm
CLO-3	Apply different techniques to prepare a presentation	Cognitive	3	7	A3, Q3, Final Term
CLO-4	Participate in class presentations and volunteer to share knowledge in the class individually or as a group member	Affective	2	6	A4, Presentation

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

~~IV. Grading Policy~~

~~For students admitted in Fall 2021 and onwards~~

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

~~For students admitted before Fall 2021~~

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

~~V. Course Contents~~

~~Definition, importance, components, seven Cs, barriers and theories of communication, Ethical and Global Communication, Team Communication, Job search communication Writing resume and cover letters, Types and importance of interviews, preparation for interviews, Types and importance of meetings, planning meetings, Manners and principles to participate in a meeting, presentation Skills-steps to prepare a presentation, ways of oral delivery, verbal and nonverbal strategies. Remedies to overcome stage fear, Rhetoric-art of persuasion.~~

~~VI. Weekly Breakdown~~

Week No.	CLOs	Topics	Reference
1	CLO-1	Definition, importance, components, seven Cs of Communication	Chapter 1 [Textbook 2]
2		Ethical and Global Communication,	Chapter 1,2 [Textbook 1]
3		Team Communication	[Ref. Book 3]
3		Job search communication	//
4		Writing resume and cover letters	//
5	CLO-2	Types and importance of interviews, preparation for interviews	//
6		Giving interviews and steps after interviews	[Textbook 3]
7		Types and importance of meetings, planning meetings Manners and princinples to participate in a meeting,	Chapter 14 [Textbook 1]
8		Definition, importance, components, seven Cs of Communication	//

9	Midterm Exams		
10	CLO-2	Manners and principles to participate in a meeting	//
11	CLO-3	Steps to prepare a presentation, ways to deliver an oral message	Chapter 10 [Textbook 3]
12		Verbal and nonverbal strategies	Chapter 10 [Textbook 3]
13		Remedies to overcome the stage fear Preparation and delivery of the presentation	//
14	CLO-4	Rhetoric- art of persuasion	//
15		Final presentations	//
16		Presentation	

OBJECT ORIENTED PROGRAMMING (SEOO-122)

I.Course Details

Credit Hours	4 (3+1)
Pre-requisites	Programming Fundamentals
Course Leader	Ms. Saadia Shabbir
Recommended Textbook(s)	1. JAVA: The Complete Reference Object Oriented Programming by Herbert Schildt, 11 th Edition, McGraw-Hill Osborne Media; (2018)
Recommended Reference (Books/Websites/Articles)	1. Starting Out with C++ from Control Structures to Objects, 9 th Edition, Tony Gaddis, Pearson; 8 th edition (2014) 2. C++ How to Program, Deitel & Deitel, Pearson; 10 th edition (2016) 3. Object Oriented Programming in C++ by Robert Lafore, 4 th edition, ISBN-13: 978-0672323089, Sams; (2001) 4. Java: How to Program by Paul Deitel, Pearson College Div; 9 th edition (2011)

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Understand basic concepts of object-oriented paradigm	Cognitive	2	2	A1, Q1, Mid Term
CLO-2	Identify the objects & their relationships to build object-oriented solution	Cognitive	4	3	A2, Q2, Mid Term, Final Term
CLO-3	Develop object-oriented solutions for small systems	Cognitive	6	4	A3, Q3, Final Term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
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%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

~~V. Course Contents~~

~~Introduction to object oriented design, history and advantages of object oriented design, introduction to object oriented programming concepts, classes, objects, data encapsulation, constructors, destructors, access modifiers, const vs non-const functions, static data members & functions, function overloading, operator overloading, identification of classes and their relationships, composition, aggregation, inheritance, multiple inheritance, polymorphism, abstract classes and interfaces, generic programming concepts, function & class templates, standard template library, object streams, data and object serialization using object streams, exception handling.~~

~~VI. Weekly Breakdown~~

Week No.	CLOs	Topics	Reference
1	CLO-1	Procedural programming Vs object-oriented programming, Revision of Programming Fundamental Concepts. History of Java, Java programming environment, Fundamental programming structure in java, Lexical issues (white spaces, identifiers, literals, comments, separators and java keywords, Data types	Chapter 1,2,3
2		Structure of Java program, Java Compilation Process Compiling and running a Java program. Methods, Introducing methods, Method signatures, Arguments, and parameters	Chapter 5,6
3		Introduction to classes and its importance, Types of Classes Objects creation and handling, Anonymous object, Utilizing methods of classes	Chapter 6
4	CLO-2	Controlling access to members, Constructors, Constructor overloading, Static class members, Static methods	Chapter 6
5		Inheritance, Types of Inheritance (Multiple , Multilevel, Hierarchical and Hybrid), Superclass and Subclass, Relationship between super class and subclass, Use of super keyword for using variables, function and constructor of super class	Chapter 8
6		Polymorphism, Abstraction and Encapsulation, Generalization, specialization, Aggregation, Composition, Nested Classes, Run-time Polymorphism, Compile-time Polymorphism, Late binding and Early Binding.	Chapter 8
7		Method overriding, use of super in override functions, using abstract classes, using final and protected keyword with	Chapter 8

		inheritance, Abstract and non-abstract methods. Dynamic and dispatch methods.	
8	CLO-3	Packages Defining package, Package access protection, Importing packages. Use of Object class.	Chapter 9
9	Midterm Exams		
10	CLO-3	Interfaces, defining an interface, implementing and applying interfaces, Variables in interfaces, Interface with Multiple Inheritance	Chapter 9
11		Exception Handling Fundamentals, Types of Exceptions, using exception clauses (try, catch, throw, throws and finally)	Chapter 10
12		Graphical User Interface Components, Introduction to Swing. Controls Button, Labels, Text Fields, Text Area, Checkbox and Radio buttons, Event Handling, The delegation event model (events, event sources and event listeners), Introduction to AWT, AWT classes	Chapter 23, 24
13		Creating a window program, working with graphics, Layout managers and menus, Control fundamentals	Chapter 25
14		Enums, Generics, Boxing, auto Boxing, Collections, File Handling	Chapter 26
15		Week For Open Ended Lab For Lab Final Exams, Projects, Presentations	
16		Event Handling Practice	Chapter 12, 23

~~OBJECT ORIENTED PROGRAMMING (SEOO-122) – LAB~~

~~I.Course Details~~

Credit Hours	4 (3+1)
Pre-requisites	Programming Fundamentals
Course Leader	Ms. Saadia Shabbir
Recommended Textbook(s)	1. JAVA: The Complete Reference Object Oriented Programming by Herbert Schildt, 11th Edition, McGraw-Hill Osborne Media; (2018)
Recommended Reference (Books/Websites/Articles)	1. Starting Out with C++ from Control Structures to Objects, 9th Edition, Tony Gaddis, Pearson; 8th edition (2014) 2. C++ How to Program, Deitel & Deitel, Pearson; 10th edition (2016) 3. Object Oriented Programming in C++ by Robert Lafore, 4th edition, ISBN-13: 978-0672323089, Sams; (2001) 4. Java: How to Program by Paul Deitel, Pearson College Div; 9th edition (2011)

~~II.Lab Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Develop programs using object-oriented programming concepts	Cognitive	6	4	Lab tasks (reports), Mid-term exam, Project
CLO-2	Manipulate the use of Java Compiler and Eclipse IDE to create java applications	Psychomotor	3	5	Lab tasks (reports), Mid-term exam, Project

~~III.Lab Assessment~~

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25
Total	100

~~IV. Weekly Breakdown (All CLOs shall be assessed in every lab)~~

Week No.	Topics
1	Introduction to Eclipse, Writing Java programs, Basic programming structure of Java
2	Methods, Method signatures, Parameters and Arguments
3	Classes and objects, Properties, and behavior
4	Constructors of classes and their utility, Constructor Overloading
5	Static Variables and Static Methods, Basic concept of Final Keyword
6	Encapsulation as concept and its implementation, Access modifiers, Data Hiding
7	Basic concept of inheritance, Parent and Child Classes
8	Polymorphism through method overriding, Use of Super Keyword, Use of Protected
9	Midterm Exams
10	Use of Final Keyword with respect to inheritance, Final classes, Final methods
11	Object Casting and Dynamic Method Dispatch
12	Abstract Classes and Abstract Methods
13	Interfaces, defining an interface, implementing interfaces, Variables in interfaces
14	Exception Handling Fundamentals, Types of Exceptions, using exception clauses (try, catch, throw, throws and finally)
15	Graphical User Interface Components, Introduction to AWT, AWT classes
16	Final Assessment

~~DISCRETE STRUCTURES (SEDS-123)~~

I. Course Details

Credit Hours	3 (3+0)
Pre-requisites	-
Course Leader	Dr. Naveed Ahmad
Recommended Textbook(s)	Discrete Mathematics and Its Applications, 7th edition by Kenneth H. Rosen year 2012
Recommended Reference (Books/Websites/Articles)	1. Discrete mathematics with applications, Susanna S. Epp, 5th Edition, 2019. 2. Discrete mathematics, Richard Johnson Baugh, Global edition, 2018

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Apply the elements of propositional logic statements and logical operations.	Cognitive	3	2	A1, Q1, Midterm
CLO-2	Solve basic problems demonstrating the understanding of fundamental for sets, functions, relations and counting principles	Cognitive	2	3	A2, Q2, Final Term
CLO-3	Apply some properties to graphs, trees and related discrete structures, and be able to discover their relationship with practical examples	Cognitive	3	2	A3, Q3, Final Term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	5
Assignments	10
Presentation/Project	10
Midterm	25
Final Term	50
Total	100

~~IV. Grading Policy~~

~~For students admitted in Fall 2021 and onwards~~

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

~~For students admitted before Fall 2021~~

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

~~V. Course Contents~~

~~Mathematical reasoning, propositional and predicate logic, rules of inference, proof by induction, proof by contraposition, proof by contradiction, proof by implication, set theory, relations, equivalence relations and partitions, partial orderings, recurrence relations, functions, mappings, function composition, inverse functions, recursive functions, Number Theory, sequences, series, counting, inclusion and exclusion principle, pigeonhole principle, permutations and combinations, elements of graph theory, planar graphs, graph coloring, euler graph, Hamiltonian path, rooted trees, traversals.~~

~~VI. Weekly Breakdown~~

Week No.	CLOs	Topics	Reference
1	CLO-1	Introduction to Logic, Conditional Statement, converse, Contrapositive, Inverse,	Chapter 1
2		Logic and bit operations, Proofs, valid and invalid arguments, Propositional Equivalences, Constructing new logical equivalences	Chapter 1
3		Propositional logic, Translating from English into logical expressions involving propositions.	Chapter 1
4		Predicates and Quantifiers, The universal quantifier, the existential quantifier, Nested Quantifier, Order of Quantifier.	Chapter 1
5		Translating from English into logical expressions involving predicates and quantifiers.	Chapter 1
6	CLO-2	Rules of Inference, Modus Ponens, Modus Tollens, Hypothetical Syllogism, Disjunctive Syllogism, Addition, Simplification, Conjunction, Resolution	Chapter 1
7		Problem Solving Using Rules of Inference.	Chapter 1

8		Proof by induction, proof by contraposition, proof by contradiction, proof by implication.	Chapter 2
9	Midterm Exams		
10	CLO-2	Set theory, Set operations, Set identities and Proofs, Set builder notation, Empty set, Null set, Singleton set, Proper and Improper subsets.	Chapter 2
11		Applications and definition of functions, domain, co-domain, image, range, preimage.	Chapter 2
12		Sequence and summations, arithmetic progression, Special integer sequences, geometric progressions.	Chapter 2
13	CLO-3	The basics of counting, applications, product rule, sum rule, complex counting problems.	Chapter 6
14		Relations, Properties of relations, reflexive relations.	Chapter 9
15		Types of graphs, Graph Terminology, adjacent vertices, degree of a vertex, Graph Models, Undirected graphs. Walk, path, trail, circuit, Euler Paths and Circuits, Hamilton Path and Circuits.	Chapter 10 Online sources
16		Introduction to Trees, Applications of Trees, Tree Traversal, Post Order, PreOrder and Inorder Traversals.	Chapter 11

~~SOFTWARE ENGINEERING (SESE-124)~~

~~I. Course Details~~

Credit Hours	3 (3+0)
Pre-requisites	None
Course Leader	Dr. Huma Hayat Khan
Recommended Textbook(s)	1. Software Engineering, A practitioner's approach, Pressman R. S. & Maxim B. R., 9th Edition, McGraw-Hill, 2019. 2. Software Engineering, Sommerville I., 10th Edition, Pearson Inc., 2014.
Recommended Reference (Books/Websites/Articles)	The new software engineering, Sue Conger, 2008.

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Describe the software engineering concepts, principles, and process flow of the SDLC	Cognitive	2	2	A1, Q1, Midterm
CLO-2	Explain various software development processes/methodologies	Cognitive	2	2	A2, Q2 Midterm, Final Term
CLO-3	Analyze software engineering key areas	Cognitive	4	3	A3, Q3, Presentation / Project, Final Term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	5
Presentation/Project	10
Midterm	25
Final Term	50
Total	100

~~IV. Grading Policy~~

~~For students admitted in Fall 2021 and onwards~~

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50

GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00
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For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Nature of Software, Overview of Software Engineering, Professional software development, Software engineering practice, Software process structure, Software process models, Agile software Development, Agile process models, Agile development techniques, Kanban software development, lean software development, Requirements engineering process, Functional and non-functional requirements, UML diagrams, Context models, behavioral models, model driven engineering, Architectural design, Design and implementation, Risk Management, Software testing and quality assurance, Project management, configuration management, Software Process improvement.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Software Crisis. What is a software? The Nature of Software, Defining Software, Software Application Domains, Legacy Software, The Changing Nature of Software (web apps, mobile apps, cloud computing, product line software), Software Engineering, The Software Process, The Process Framework, Umbrella Activities, Software Engineering Principles.	Chapter 1 [Textbook 1]
2		Software Development Life Cycle, A Generic Process Model, Defining a Framework Activity, Software Process Flow, Identifying Task sets, Process Patterns, Process Assessment and Improvement	Chapter 2 [Textbook 1]
3	CLO-2	What is process Model?, process flow, prescriptive vs descriptive process models, Prescriptive Process Model; The Waterfall Model, Incremental Process Models, Evolutionary Process Models, Concurrent Models.	Chapter 2,4 [Textbook 1]
4		Specialized Process Models; Component-Based Development, The Formal Methods Model, Aspect-Oriented Software Development, The Unified Process, A Brief History, Phases of the Unified Process, Personal and Team Process Models, Personal Software Process, Team Software Process.	Chapter 4 [Textbook 2]

5		What Is Agility? Agile Software Development, Predictive vs descriptive, Agile manifesto, Agility and the Cost of Change, What Is an Agile Process?, Characteristics of Agile Software Development, Agility Principles, User Stories, template, good user stories, spike, user story writing workshop, story mapping	Chapter 3 [Textbook 1]
6		Agile Planning, concepts of effort, duration, accuracy, precision, relative, and absolute, estimation styles (planning poker, card sorting), velocity in agile, release planning its types, Scrum process model, Extreme Programming	Chapter 3 [Textbook 1]
7		Dynamics system development method, Feature driven development, Kanban, pair programming	Chapter 3 [Textbook 1]
8	CLO-3	Human aspects of software engineering (Software teams, stakeholders): Characteristics of a Software Engineer, The Psychology of Software Engineering, The Software Team, Team Structures, Agile Teams	Chapter 5 [Textbook 1]
9	Midterm Exams		
10	CLO-3	Software requirement engineering, Functional vs non-functional requirement, RE process, Developing Use case	Chapter 7,8 [Textbook 1] Chapter 8 [Textbook 2]
11		Software Design and Modelling-Introduction to UML, use case modelling, Context Models (Data Flow Diagrams) and behavioral models (activity Diagram)	Chapter 9, Appendix 1 [Textbook 1]
12		Architectural Design- Software Architecture, Architectural Styles Architectural Descriptions, Architectural Decisions, What is quality? Software quality, software quality dilemma, achieving software quality, Review techniques, informal reviews, formal reviews, post mortem evaluation, software quality assurance, formal approaches to SQA, software reliability, Software process improvement- CMM and CMMI.	Chapter 10, 15,17,28 [Textbook 1], Chapter 13 [Textbook 2]
13		Software Testing fundamentals, testing types, testing levels	Chapter 19, 20 [Textbook 1], Chapter 22, 23 [Textbook 2]

14		Risk Management: Reactive versus Proactive Risk Strategies, Software Risks ,Risk Identification, Assessing Overall Project Risk, Risk Components and Drivers, Risk Projection, Developing a Risk Table, Assessing Risk Impact, Risk Refinement, Risk Mitigation, Monitoring, and Management, The RMMM Plan	Chapter 26 [Textbook1], Chapter 35 [Textbook 2]
15		Maintenance and Reengineering: Software Maintenance, Software Supportability, Reengineering, Business Process Reengineering, Business Processes, A BPR Model, Software Reengineering, A Software Reengineering Process Model, Software Reengineering Activities, Reverse Engineering ,Reverse Engineering to Understand Data, Reverse Engineering to Understand Processing, Reverse Engineering User Interfaces	Chapter 36 [Textbook 2]
16		Presentation Week	

~~CALCULUS & ANALYTICAL GEOMETRY (SECA-102)~~

~~I.Course Details~~

Credit Hours:	3 (3+0)
Pre-requisites:	None
Course Leader	Dr. Asad Husain
Recommended Textbook(s)	Calculus and analytical Geometry, Thomas and Finney, 11th edition, Pearson Addison Wesley, 2007.
Recommended Reference (Books/Websites/Articles)	Advanced engineering mathematics, Erwin Kreyszig, John Wiley & Sons, 2019.

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment
CLO-1	Apply the concepts of Graphs of Functions, limits and continuity to solve problems	Cognitive	3	1	A1,Q1, Midterm
CLO-2	Solve problems related to derivation, Maxima, and Minima.	Cognitive	3	1	A2,Q2, Midterm, Final Term
CLO-3	Solve the problems related to integration, and conic section	Cognitive	3	2	A3,Q3, Final Term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

~~IV. Grading Policy~~

~~For students admitted in Fall 2021 and onwards~~

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50

GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00
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For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	≥ 90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	< 50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Limits and Continuity; Introduction to functions, Introduction to limits, Techniques of finding limits, Indeterminate forms of limits, Continuous and discontinuous functions and their applications, Differential calculus; Concept and idea of differentiation, Geometrical and Physical meaning of derivatives, Rules of differentiation, Techniques of differentiation, Rates of change, Tangents and Normal lines, Chain rule, implicit differentiation, linear approximation, Applications of differentiation; Extreme value functions, Mean value theorems, Maxima and Minima of a function for single variable, Concavity, Integral calculus; Concept and idea of Integration, Indefinite Integrals, Techniques of integration, Riemann sums and Definite Integrals, Applications of definite integrals, Improper integral, Applications of Integration; Area under the curve, Analytical Geometry; Straight lines in R^3 , Equations for planes.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Introduction to functions, Even and Odd Functions, Domain and Range of Functions, Piecewise Continuous Functions, Periodic Functions	Chapter 1
2		Increasing and Decreasing Functions, Graph of the functions	Chapter 1
3		Limit of a function, Graphical approach, Properties of limits, Theorems of limits	Chapter 2
4		Limits of various types of functions	Chapter 2
5		Continuity of a function at a point, Continuous and discontinuous functions, Continuity by Graphical approach	Chapter 2
6	CLO-2	Derivatives, geometrical meaning of the derivative, general Theorems of derivatives	Chapter 3
7		Trigonometric functions, explicit and implicit functions and its derivatives, second order and higher order derivatives	Chapter 3
8		Tangents and Normal Lines, Application of Derivatives, Max-Min Theorem, Absolute Extrema, Local Extrema, Concavity Examples	Chapter 4
9	Midterm Exams		
10	CLO-2	Integration, Techniques of Integration, Basic Integration Formulas, Substitution method	Chapter 5
11		Partial Fraction, Integration by Parts	Chapter 7

12	CLO-3	Trigonometric Substitution	Chapter 7
13		Riemann sums and Definite Integrals	Chapter 8
14		Improper integrals	Chapter 7,8
15		Properties of definite integral, Fundamental theorems of calculus	Chapter 7,8
16		Application of Integration: Area under the curves, Solids of revolution, Straight lines in $\mathbb{R}^3$, Equations for planes	Chapter 6

PAKISTAN STUDIES (PS-115)

I. Course Details

Credit Hours	2 (2+0)
Pre-requisites	-
Course Leader	Mr. Zohaib Ahmad
Recommended Textbook(s)	1. Sh. Muhammad Rafiq, Pakistan Studies
Recommended Reference (Books/Websites/Articles)	2. Pakistan Affairs by Ikram Rabbani 3. Empire at Retreat by Rabia Umer Ali

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Discuss Pakistan's historical perspective; ideological basis; environmental outlook; political and constitutional developments; contemporary and future Pakistan.	Cognitive	2	1	Q1, A1, Midterm
CLO-2	Apply the guiding principle driven out from study of Pakistan's history; ideology; environment; politics; constitutional developments; social and economic challenges in fulfilment of Pakistan's Vision 2025 with special reference to the role of Pakistani youth.	Cognitive	3	10	A2, Q2 Midterm, Final Term
CLO-3	Analyze the impact based social activism to enable students play their respective role in development of prosperous Pakistan alongside their personal growth.	Cognitive	4	10	A3, Q3, Midterm, Final Term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Historical background of Pakistan: Muslim society in Indo-Pakistan, the movement led by the societies, the downfall of Islamic society, the establishment of British Raj- Causes and consequences. Political evolution of Muslims in the twentieth century: Sir Syed Ahmed Khan; Muslim League; Nehru; Allama Iqbal; Independence Movement; Lahore Resolution; Pakistan culture and society, Constitutional and Administrative issues, Pakistan and its geo-political dimension, Pakistan and International Affairs, Pakistan and the challenges ahead.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Evolution and growth of Muslim society in Subcontinent.	1 st
2		Factors leading to Muslim nationalism in Subcontinent.	
3		Definition, sources and significance of Ideology.	
4		Basis of Pakistan's Ideology with particular reference to the Values of Islam, economic deprivation of Muslims and role of All India Muslim League (AIML) in Subcontinent.	
5		Two Nation Theory: Origin and Explication.	
6		Ideological rationale with special reference to Sir Syed Ahmad Khan, Allama Muhammad Iqbal and Quaid-i-Azam Muhammad Ali Jinnah.	
7	CLO-2	Landforms, climate and water resources.	
8		Environmental hazards and remedies.	
9	Midterm Exams		

10	CLO-2	Constitutional and political developments in Pakistan.	1st
11		Geo-political significance of Pakistan.	
12	CLO-3	Economic potential and its utilization.	1st and 2nd
13		Social issues, their gravity and resolution.	1st and 2nd
14		World affairs: challenges and contributions.	1st and 2nd
15		Human rights in Pakistan.	1st and 2nd
16		Pakistan's Vision 2025, Youth's Role in Development of Pakistan, Projects	2nd and 3rd

MATH 2

I. Course Details

Credit Hours	3 (3+0)
Pre-requisites	-
Course Leader	Dr. Asad Husain
Recommended Textbook(s)	Advanced Engineering Mathematics, Author Erwin Kreyszig, John Wiley & Sons, Limited, ISBN 111957109X, 9781119571094, 2019
Recommended Reference (Books/Websites/Articles)	-

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment
CLO-1	Understand the concepts of System Linear Equations and operation of matrices	Cognitive	2	1	A1,Q1, Midterm
CLO-2	Solve the Homogeneous and nonhomogeneous Systems of Equations	Cognitive	3	1	A2,Q2, Midterm, Final Term
CLO-3	Understand the concepts of vectors and fields.	Cognitive	2	2	A3,Q3, Final Term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	≥ 90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	≥ 90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	< 50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Introduction to linear algebra, Systems of Linear Equations; Solution of a system of linear equations; Concept of Matrices, and their types; Basic operation of Matrices; Techniques and concept of Matrix multiplication; Concept of the determinant of matrix and Nonsingular Matrices; Inverse of a matrix and Cramer's Rule; Elementary row operation of a matrix; Echelon form and Gauss's elimination method; Reduce Echelon form and Gauss's Jordan elimination method; Elimination Method for solving system of linear equations; Homogeneous Systems of Equations $Ax=0$; Non-Homogeneous Systems of Equations $Ax=b$; Introduction to Vectors spaces; Dot product; Cross product; Gradient of a scalar field; Divergence of a vector field; Curl of a vector field; Concepts of Stokes's theorem and the Divergence theorem.

VI. Weekly Breakdown

Week No.	CLOs	Topics	Reference
1	CLO-1	Introduction to linear algebra, Systems of Linear Equations	Chapter 7-1
2		Solution of a system of linear equations	Chapter 7-1
3		Concept of Matrices, and their types	Chapter 7-1
4		The basic operation of Matrices	Chapter 7-2
5		Techniques and concept of Matrix multiplication	Chapter 7-2
6	CLO-2	Concept of the determinant of a matrix and Nonsingular Matrices	Chapter 7-7
7		The inverse of a matrix and Cramer's Rule	Chapter 7-8
8		Elementary row operation of a matrix	Chapter 7-7
9	Midterm Exams		
10	CLO-2	Echelon form and Gauss's elimination method	Chapter 7-3
11		Reduce Echelon form and Gauss's Jordan elimination method	Chapter 7-8
12		Elimination Method for solving system of linear equations	Chapter 7-5
13		Homogeneous Systems of Equations $Ax=0$, Non-Homogeneous Systems of Equations $Ax=b$	Chapter 7-5
14	CLO-3	Introduction to Vectors spaces, Dot product, Cross product	Chapter 7-9
15		The gradient of a scalar field, Divergence of a vector field, The curl of a vector field	Chapter 9
16		Concepts of Stokes's theorem and the Divergence theorem	Chapter 9

~~Semester III~~

~~DATA STRUCTURE & ALGORITHMS (SEDA-225)~~

~~I.Course Details~~

Credit Hours:	4 (3+1)
Pre-requisites:	Programming Fundamentals
Course Leader:	Dr. Jaweria Kanwal
Textbook:	Data Structures and Algorithm Analysis in C++ by Mark Allen Weiss Fourth Edition, 2013
Reference Books:	1. C++ Plus Data Structures by Nell Dale, 5th Edition. 2. C++ Programming: Program Design including Data Structures by D. S. Malik, 7th Edition. 3. Data Structures and Program Design in C++ by Robert L. Kruse and Alexander J. Ryba 4. Introduction to Algorithms by Thomas, 2nd ed., MIT Press, 2007

~~II. Course Learning Outcomes (CLOs)~~

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment Artifacts
CLO-1	Understand well known linear data structures	Cognitive	2	2	A1, Q1, Mid Term
CLO-2	Demonstrate well known algorithms: searching and sorting algorithms	Cognitive	3	2	A2, Q2, Final Term
CLO-3	Explain strategies for efficient storage of data with non-linear data structures and analysis of the respective algorithms	Cognitive	2	2	A3, Q3, Final Term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

~~Abstract data types, Array and List, linked list & its various types, sorted linked list, searching an unsorted array, binary search for sorted arrays,, Stacks (linked lists and array implementations), Recursion and recursive algorithms, Sorting algorithms (selection, insertion, merge, quick, bubble, radix, bucket), queue, dequeuer, priority queues (linked and array implementations of queues), trees and tree traversals, binary search trees, heaps, M-way tress, balanced trees, graphs, breadth-first and depth-first traversal, topological order, shortest path, Divide & Conquer algorithms. Adjacency matrix and adjacency list implementations and their tradeoffs, Analysis of Algorithm.~~

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Introduction to Data structures, need for data structures, Array data structure and its operations	Chapter 2 Chapter 8
2	CLO-2	Searching: Linear Search, Binary Search, Sorting introduction: Bubble Sort, Selection Sort, Insertion Sort.	Chapter 2, 7
3	CLO-3	Analysis of Algorithm, Time Analysis, Asymptotic Notations for algorithmic complexity analysis	Chapter 2
4	CLO-1	Recursion, Generic Class: Template Class, Dynamic Memory Allocation, Pointers concept, Stack: concepts, operations, Array based Stack implementation	Chapter 3
5		Application of Stacks, Stack and queue differentiation, Queue concepts and operations, Array Based Queue Implementation	Chapter 3
6		Circular Queue concepts and operations	Chapter 3
7		Linked List Data structure, array and link list differentiation and its operations (Insertion beginning, middle and end)	Chapter 3
8		Doubly Linked List: concepts, operations	Chapter 3

9	Midterm Exams		
10	CLO-1	Linked List Based Stack and Queue Implementation	Chapter 3
11	CLO-2	Divide and Conquer : Merge Sort, Bucket Sort, Radix Sort, Quick Sort	Chapter 7
12		Trees and Binary Trees: Basic Concepts and terminologies, Tree Traversals (Inorder, postorder, preorder)	Chapter 4
13	CLO-3	Binary Search tree concepts and different function (Search insert, Traversal, Predecessor and successor etc.)	Chapter 4
14		AVL Trees- Different balancing rotations	Chapter 4
15		Hashing introduction, hash tables, Collision Resolution techniques. (Open Addressing, Closed Addressing)	Chapter 5
16		Graphs: Basic concepts and properties, types of graphs, Graph representations matrix and linked implementations and their tradeoffs. Graph Search algorithms: DFS, BFS	Chapter 9

~~DATA STRUCTURE AND ALGORITHMS (SEDA-225) – LAB~~

~~I.Course Details~~

Credit Hours:	4 (3+1)
Pre-requisites:	Programming Fundamentals
Course Leader:	Dr. Jaweria Kanwal
Textbook:	Data Structures and Algorithm Analysis in C++ by Mark Allen Weiss Fourth Edition, 2013
Reference Books:	1. C++ Plus Data Structures by Nell Dale, 5th Edition. 2. C++ Programming: Program Design including Data Structures by D. S. Malik, 7th Edition. 3. Data Structures and Program Design in C++ by Robert L. Kruse and Alexander J. Ryba 4. Introduction to Algorithms by Thomas, 2nd ed., MIT Press, 2007

~~II. Lab Learning Outcomes (CLOs)~~

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment Artifacts
CLO-1	Analyze best and efficient data structures and algorithms to solve problems	Cognitive	4	3	Lab tasks, Exam/Viva
CLO-2	Demonstrate linear data structures and algorithms	Psychomotor	4	4	Lab tasks, Project, Exam/Viva
CLO-3	Express the experimental data in the appropriate format in the form of a LAB report	Affective	3	7	Lab Report

~~III. Lab Assessment~~

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25
Total	100

~~IV. Weekly Breakdown (All CLOs shall be assessed in every lab)~~

Week No.	Topics
1	Array Implementation, Array Searching (Linear Search), Matrix addition and Multiplication
2	Binary Search, Bubble Sort, Selection Sort, Insertion Sort
3	Recursion (Factorial, Binary Search)
4	Array Based Stack Implementation
5	Array Based Linear Queue Implementation
6	Array Based Circular Queue Implementation
7	Linked List Implementation (Insertion from Beg, Middle, End, Display)
8	Linked List Implementation (Deletion from Beg, Middle, End)
9	Midterm Exams
10	Linked List based Stack & Queue Implementation
11	Doubly Linked List Insertion & Deletion
12	Reverse Display using Linked List
13	Divide and Conquer algorithms implementation (Merge Sort, Quick Sort)
14	Binary Search Tree operations and traversal
15	Graph and basic algorithms implementation
16	Projects and Viva

~~SOFTWARE REQUIREMENT ENGINEERING (SESR-241)~~

~~I. Course Details~~

Credit Hours	3 (3+0)
Pre-requisites	Software Engineering
Course Leader	Dr. Huma Hayat Khan
Recommended Textbook(s)	Requirement engineering fundamentals by Klaus Pohl & Chris Rupp, , 2nd edition, Rocky Nook publisher, 2015.
Recommended Reference (Books/Websites/Articles)	Handouts & Software Requirements, 3rd Edition, By Karl Wiegers, Joy Beatty, Microsoft press, 2013.

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Explain the fundamental concepts and activities of requirement engineering.	Cognitive	2	2	Q1, A1, Midterm
CLO-2	Illustrate requirements using templates and modelling techniques to document requirements	Cognitive	3	3	Q2, A2, Midterm, Final term
CLO-3	Apply the techniques and tools required for requirements validation, negotiations, prioritization, management	Cognitive	3	5	Q3, A3, Presentation / Project, Final term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

~~IV. Grading Policy~~

~~For students admitted in Fall 2021 and onwards~~

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

~~V. Course Contents~~

~~Introduction to Requirements Engineering, Software Requirements, classification of requirements, Requirements process, Levels/layers of requirements, Requirement characteristics, Analyzing quality requirements, Software requirements in the context of systems engineering, Requirement evolution, requirement traceability, requirement prioritization, trade-off analysis, risk analysis and impact analysis, Requirement management, interaction between requirement and architecture, Requirement elicitation, elicitation sources and techniques, Requirement specification and documentation, specification sources and techniques, Requirements validation and techniques, Management of Requirements, Introduction to Management, Requirements Management Problems, and Tools for requirements.~~

~~VI. Weekly Breakdown~~

Week No.	CLOs	Topics	Reference
1	CLO1	Figures and Facts that support Requirements Engineering, what is RE?, Embedding Requirements Engineering into Process Models, Fundamentals of Communication Theory, Characteristics of RE	Chapter 1
2		System and Context Boundaries: System Context, Defining System and Context Boundaries, Defining the System Boundary, Defining the Context Boundary, Documenting the System Context	Chapter 2
3		Eliciting Requirements, Requirements Sources, Stakeholders and Their Significance, Requirements Categorization According to the Kano Model, RE process overview	Chapter 3
4		Elicitation techniques: Types of Elicitation Techniques; Survey Techniques, Creativity Techniques	Chapter 3
5		Document-centric Techniques, Observation Techniques, Support Techniques	Chapter 3
6	CLO 2	Documenting requirement; Types of Documentation;Requirements Documentation using Natural Language,Requirements Documentation using Conceptual Models,Hybrid Requirements Documents, Document Structures, quality Criteria for Requirements Documents, sample required document.	Chapter 4
7		Documenting Requirements in Natural Language,Effects of Natural Language, Nominalization, Nouns without Reference Index,Universal	Chapter 5

		Quantifiers ,Incompletely Specified Conditions,Incompletely Specified Process Verbs,Requirement Construction using Templates	
8		Model-Based Requirements Documentation,The Term Model,Goal Model,Goal Documentation Using AND/OR Trees,Example of AND/OR Trees, Use Cases , UML Use Case Diagrams,Use Case Specifications,Three Perspectives on the Requirements, Requirements modelling in data perspective through ERD	Chapter 6
9	Midterm Exams		
10	CLO3	Requirements Validation and Negotiation,Fundamentals of Requirements Validation,Fundamentals of Requirements Negotiation ,Quality Aspects of Requirement, Quality Aspect “Content”,Quality Aspect “Documentation”,Quality Aspect “Agreement”, Requirement Validation Techniques; Commenting, Inspection,Walk-Through,Perspective-Based Reading,Validation through Prototypes,Using Checklists for Validation ,Requirements Negotiation, Conflict Identification, Conflict Analysis, Conflict Resolution,Documentation of the Conflict Resolution	Chapter 7
11		Requirements Management,Assigning Attributes to Requirements , Attributes for Natural Language Requirements and Models,Attribute Scheme,Attribute Types of Requirements,Views on Requirements, Selective Views on the Requirements,Condensed Views on the Requirements, prioritizing Requirements,Method for Requirements Prioritization, Techniques for Requirements Prioritization	Chapter 8
12		Traceability of Requirements, Advantages of Traceable Requirements, Purpose-Driven Definition of Traceability, Classification of Traceability Relations, Representation of Requirements Traceability, Versioning of Requirements, Requirements Versions, Requirements Configurations, Requirements Baselines	Chapter 8
13		Management of Requirements Changes, Requirements Changes, The Change Control Board, The Change Request, Classification of Incoming Change Requests, Basic Method for Corrective and Adaptive Changes, Measurement of Requirements, Product vs. Process Metric, Examples of Product and Process Metrics	Chapter 8
14		Tool support, general tool support, modeling tools, requirements management tools	Chapter 9

15		Evaluating Tools: Evaluating Tools, Project View, User View, Product View, Process View, Provider View, Technical View, Economic View	Chapter 9
16		Presentations week	

~~LINEAR ALGEBRA (SELA-203)~~

~~I. Course Details~~

Credit Hours:	3 (3+0)
Pre-requisites:	None
Course Leader:	Dr. Asad Husain
Recommended Book(s):	1. Elementary Linear Algebra by Howard Anton 2. Applied Linear Algebra by Peter J. Olver, Chehrzad Shakiban Second Edition, 2018
Recommended Reference (Books/Websites/Articles)	1. Linear Algebra and its Applications by Gilbert Strang 2. Advanced Engineering Mathematics, Author Erwin Kreyszig, Tenth Edition, Publisher John Wiley & Sons, Limited, 2019

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment
CLO-1	Solve a system of linear equations using Matrices	Cognitive	3	1	A1, Q1, Midterm
CLO-2	Apply the basic knowledge of vector spaces, eigenvalue and eigenvectors	Cognitive	3	1	A2, Q2, Midterm, Final Term
CLO-3	Solve a system of linear equations using different methods	Cognitive	3	2	A3, Q3, Final Term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

~~IV. Grading Policy~~

~~For students admitted in Fall 2021 and onwards~~

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	≥ 90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	< 50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

System of Linear Equations and Matrices: Introduction to system of linear equations, Matrix form of system of Linear Equations, Gaussian Elimination method, Gauss-Jordan Method, Consistent and inconsistent systems, Homogeneous system of equations. Vector Equations: Introduction to vector in plane, Vector in R_n , Vector form of straight line, Linear Combinations, Geometrical interpretation of solution of Homogeneous and Non-homogeneous equations, Applications of Linear Systems: Traffic Flow Problem, Electric circuit Problem, Economic Model, Linear transformations: Introduction to linear transformations, Matrix transformations, Domain and range of linear transformations, Geometric interpretation of linear transformations, Matrix of linear transformations, Inverse of a matrix: Definition of inverse of a matrix, Algorithm to find the inverse of matrices, LU factorization, Determinants: Introduction to determinants, Geometric meaning of determinants, Properties of determinants, Cramer Rule, Cofactor method for finding the inverse of a matrix, Vector Spaces: Definition of vector spaces, Subspaces, Spanning set, Null Spaces and column spaces of linear transformation, Linearly Independent sets and basis, Bases for Null space, Dimension of a vector space EigenValues and Eigenvectors: Introduction to Eigenvalue and Eigenvectors, Computing the Eigenvalues, Properties of Eigenvalues, Diagonalization, applications of Eigenvalues. Numerical Linear Algebra, Gauss Elimination, Inner product, Cross product, Gram-Schmidt Process, QR – Decomposition, $AV = b$ by LU – decomposition, Elimination Matrix, Orthogonal Matrices, Least square Vector, Least Square Error, Hessenberg's Theorem, Schur's theorem, Singular value Decomposition (SVD).

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Introduction to the Course System of Linear Equations	Chapter 1
2		System of Linear Equations and Matrices: Introduction to system of linear equations Matrix form of system of Linear Equations.	Chapter 1
3		Gaussian Elimination method Gauss-Jordan Method, Consistent and inconsistent systems.	Chapter 1
4		Homogeneous system of equations. Linear Combinations, Applications of Linear Systems: Traffic Flow Problem, Electric circuit Problem	Chapter 1
5		Determinants: Introduction to determinants, Geometric meaning of determinants, Properties of determinants, Cramer Rule	Chapter 2

6	CLO-2	Cofactor method for finding the inverse of a matrix, Vector Equations: Introduction to vector in plane, Vector in R^n, Vector form of straight line	Chapter 2, 3
7		Matrix transformations, Domain and range of linear transformations, Inverse of a matrix: Definition of inverse of a matrix	Chapter 3
8		Vector Spaces: Definition of vector spaces, Subspaces, Spanning set, Null Spaces and column spaces of linear transformation, Bases for Null space, Dimension of a vector space, Linearly Independent sets and basis	Chapter 4
9	Midterm Exams		
10	CLO-2	Introduction to Eigenvalue and Eigenvectors, Eigen values and Eigenvectors: Examples, Computing the Eigenvalues, Properties of Eigenvalues	Chapter 5
11	CLO-3	Diagonalization, Diagonalization, Inner product, Cross product	Chapter 5, 6
12		Gram-Schmidt Process: QR – Decomposition, Least square Vector, Least Square Error	Chapter 6
13		Least Square Error, QR – Decomposition	Chapter 6
14		Orthogonal Matrices, Hessenberg's Theorem, Schur's theorem	Chapter 7
15		Elimination Matrix, $AV = b$ by LU – decomposition, The Power Method	Chapter 9
16		Singular value Decomposition (SVD), Project Presentations	Chapter 9

~~INTRODUCTION TO PSYCHOLOGY (SEIP-273)~~

~~I.Course Details~~

Credit Hours	3 (3+0)
Pre-requisites	-
Course Leader	Mr. Zohaib Ahmad
Recommended Textbook(s)	1. Feldman, R.S. (2010). Understanding psychology. New York: McGraw-Hill. 2. Myers, D. G. (2010). Introduction to Psychology (10th ed.)
Recommended Reference (Books/Websites/Articles)	1. Atkinson, R., & Smith, E. E. (2005). Introduction to Psychology (14th ed.) 2. Lahey, B. B. (2004). Psychology: An Introduction

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Identify key components in the field of Psychology	Cognitive	2	1	Q1, A1, Midterm
CLO-2	Discuss psychological concepts and techniques for practical problem solving	Cognitive	3	6	A2, Q2 Midterm, Final Term
CLO-3	Apply basic Psychological concepts in their daily life (academic and personal)	Cognitive	4	10	A3, Q3, Midterm, Final Term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Definition of Psychology, Sub-fields of Psychology, Structuralism, Functionalism, Gestalt, Neuroscience, Psychodynamic, Behavioral, Cognitive, Humanistic	
2		Research Process, Hypothesis, Theory, Archival Research, Naturalistic Observation, Survey Research, Case Study, Correlation, Research, Experimental Research	
3		Sensing the World Around Us, Absolute Thresholds: Detecting, What's Out There , Difference Thresholds: Noticing Distinctions Between Stimuli , Sensory Adaptation: Turning Down Our Responses	
4		Constructing Our View of the World , The Gestalt Laws of Organization , Top-Down and Bottom-Up Processing	
5		Depth Perception: Translating 2-D to 3-D , Perceptual Constancy , Motion Perception: As the World Turns, Perceptual Illusions: The Deceptions of Perceptions	
6	CLO-2	Definition, Factors affecting, Types of attention (sustained, selective, and divided attention)	
7		The Basics of Classical Conditioning , Applying Conditioning, Principles to Human Behavior, Extinction , Generalization and, Discrimination , Beyond Traditional Classical Conditioning:, Challenging Basic Assumptions	
8		Thorndike's Law of Effect , The Basics of Operant Conditioning:, Reinforcement and Punishment	

9	Midterm Exams		
10		Comparing classical and operant conditioning, Latent Learning , Observational Learning: Learning Through Imitation	
11	CLO-3	Sensory Memory, Short-Term Memory, Working Memory	
12		Long-Term Memory, Recalling Long-Term Memories , Retrieval Cues , Levels of Processing Theory Explicit and Implicit Memory , Flashbulb Memories , Constructive Processes in Memory: Lding the Past. Why we forget?Proactive and Retroactive Interference: The Before and After of Forgetting Memory Dysfunctions: Afflictions of Forgetting	
13		Thinking and Reasoning ,Problem Solving, What Is Intelligence?, Theories of Intelligence , The Biological Basis of Intelligence, Practical and Emotional, Intelligence Assessing Intelligence, Adaptive Testing	
14		Explaining motivation , Understanding Emotional Experiences, The Functions of Emotions , Determining the Range of Emotions: Labelling Our Feelings, The Roots of Emotions: James-Lange theory, Cannon-Bard theory, Schahchter-Singer theory	
15		Freud's Psychoanalytic Theory: Mapping the Unconscious Mind, Assessing Personality: Determining What Makes Us Distinctive	
16		Stress: Reacting to Threat and Challenge , The High Cost of Stress , Coping with Stress , Promoting Health and Wellness: Following Medical Advice , Well-Being and Happiness	

FORMAL METHODS IN SOFTWARE ENGINEERING (SEFM-262)

I.Course Details

Credit Hours	3 (3+0)
Pre-requisites	Discrete Structures
Course Leader	Dr. Nargis Fatima
Recommended Textbook(s)	Concise guide to formal methods, theory, fundamentals and industry applications, 7 th edition, Springer, 2017.
Recommended Reference (Books/Websites/Articles)	1. The way of Z practical programming with formal methods, Jonathan Jacky, 7 th Edition, Cambridge University Press, 2008. 2. Modern formal methods and applications, Hossam A. Gabbar, 8 th Edition, Springer-Verlag, 2006.

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Describe the cost, benefits, limitations, and fundamental concepts of formal methods.	Cognitive	2	2	A1, Q1, Midterm
CLO-2	Demonstrate formal models of sequential software systems using Z and VDM.	Cognitive	3	2	A2, Q2, Midterm, Final Term
CLO-3	Apply formal methods of specification and verification in the development of software systems.	Cognitive	3	3	A3, Q3, Presentation /Project, Final Term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Introduction to Formal Methods, Issues with informal specifications, Need, Purpose, and Limitations of Formal Methods, Formal Specification, Verification and automated theorem proving, Importance of Proof, Formal Specification Languages (Z, B, B-event, RAISE, VDM-SL etc.), Propositional Logic, Predicate Logic, Introducing concepts of Equality for formal specification, Formal specification and Verification, Definite description, Brief introduction to set theory, Commandments of Formal Methods, Objects definitions, Defining types, Formal Specification Structure Schema and Schema composition, Relations and Functions, Schema references, Schema texts, Predicates, Schema expressions, Generics, Sequences, modelling sequences, Defining state space, state space verification, Concept of Schema Invariant and Operations, Precondition and Post condition, Formal specifications examples, Introducing the formal specification tools for writing formal specification in Z/VDM-SL, Case Studies, Converting formal specification into code.

VI. Weekly Breakdown

Week No.	CLOs	Topics	Reference
1	CLO-1	Issues with informal specifications, need and purpose of formal methods, formal method effectiveness, limitations, and commandments.	Chapter 3
2		Levels of Formal Methods, finite state machines, Importance of Proof, Introduction to Formal Specification Languages (Z, B, B-event, RAISE, VDM etc.)	Chapter 3, Chapter 1 [Ref. Book 1]
3	CLO-2	Propositional Logic, Contradiction and Tautologies	Chapter 6
4		Predicate Logic, Quantifiers and Declarations	Chapter 6
5		Equality and Definite Description	Chapter 4
6		Formalizing and verifying formal statement (use of at most, use of at least, exactly)	Chapter 4

7		Set theory basics, set membership, extension, operations, and partitioning	Chapter 5 [Ref. Book 2]
8		Defining types, object's definitions using formal language notations, formal language specification structure and schema types	Chapter 8
9	Midterm Exams		
10	CLO-2	Relations and functions, binary relations, domain and range restrictions, relational inverse, relational image, and composition	Chapter 4,8
11		Sequences and Bags	Chapter 8
12	CLO-3	Defining state space, state space verification, concept of schema invariant and operations, pre-condition, and post condition in z-specification	Chapter 9 Chapter 12 [Ref. Book 1]
13		Introduction to VDM – SL	Chapter 9
14		VDM syntax and semantics	Chapter 9
15		Model checking, industrial tools for formal methods, tools for Z/VDM	Chapter 11
16		Formal specifications examples/Case studies, Industrial applications of Z/VDM-SL, Converting formal specification into code	Chapter 8

~~Semester IV~~

~~DATABASE SYSTEMS (SEDB-227)~~

~~I. Course Details~~

Credit Hours	4 (3+1)
Pre-requisites	-
Course Leader	Dr. Nargis Fatima
Recommended Textbook(s)	1. Modern database management, Jeffrey A. Hoffer, 12th Edition, Pearson, 2016. 2. Database systems: A practical approach to design, implementation, and management, Thomas Connolly and Carolyn Begg, 6th Edition, Pearson, 2015.
Recommended Reference (Books/Websites/Articles)	1. Database system concepts, Avi Silberschatz, Henry F. Korth and S. Sudarshan, 6th Edition, McGraw-Hill, 2010. 2. Database systems: Design, implementation and management, Carlos M. Coronel, 13th Edition, Cengage Learning, 2018.

~~I.Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Explain fundamental database concepts, core principles and significance to state its main idea	Cognitive	2	2	Q1, A1, Midterm, Presentation, Final Term
CLO-2	Construct logical conceptual and physical data schema using data models to organize and structure data , ensure data integrity and facilitate effective data management	Cognitive	3	3	Q2, A2. Midterm, Final Term, Project
CLO-3	Make use of relational and logical database design concepts to identify anomalies in database by performing normalization techniques	Cognitive	3	3	Q3, A3, Final Term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50

Total	100
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IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Fundamental database concepts, Database approach vs file based system, database architecture, three level schema architecture, data independence, types of data model (relational data model, entity relationship model), Entity Relationship diagram, entity sets, attributes, relationship, attributes, schemas, tuples, domains Enhanced entity relationship model (EER diagram), relational and logical database design, relation instances, keys of relations, integrity constraints, types of joins, functional dependencies, normal forms, Structured Query Language (SQL), data definition languages, sub-queries in SQL, Transaction Management, data mining, data warehousing, NoSQL.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Course introduction, Fundamental database concepts: Data, data versus information, data, manual file processing, traditional file processing, disadvantages of manual and traditional file processing systems.	Chapter 1 [Textbook 1]
2		Database approach vs file-based system, Advantages and disadvantages of database management system, components of DBMS environment.	Chapter 1 [Textbook 1]
3		Data Models (Relational Data Model, ER Data Model) Three level schema architecture (ANSI SPARC), external level, conceptual level, internal level, data independence, data dependence database languages overview	Chapter 2, 4 [Textbook 2]
4	CLO-2	Modeling rules process in organization (overview of business rules, scope of business rules) types of business rules structure of business rules, constraints, types of keys (primary key, composite key, surrogate key and foreign key)	Chapter 2 [Textbook 1]
5		ERD vs business rules, modelling entities and attributes (entity and entity type, Strong vs weak entity, associative entity attributes and	Chapter 2 [Textbook 1]

		types of attributes) relationship type. Degree of relationship (unary/recursive, binary and ternary relationship) structural constraints (one to one, one to many, many to many), minimum and maximum cardinality.	
6		Enhanced Entity Relationship Modeling (EERD), data modeling concepts of the Enhanced Entity Relationship model (super type, sub type, specialization and generalization)	Chapter 3 [Textbook 1]
7		Specifying constraints in super type and sub type in Enhanced Entity Relationship Modeling (EERD)	Chapter 3 [Textbook 1]
8	CLO-3	Logical database design and relational model (relations, relation keys, integrity constraints (domain constraint, entity integrity and referential integrity), transforming ERD and EERD into relations.	Chapter 4 [Textbook 1]
9	Midterm Exams		
10	CLO-3	Functional dependencies (Full functional dependency, partial functional dependency, transitive dependency)	Chapter 13 [Textbook 2]
11		Normalization process- 1NF, 2NF, 3NF, Denormalization, BCNF(optional), 4NF (optional)	Chapter 13, 14 [Textbook 2]
12	CLO-1	Relational Algebra selection, Project Cartesian product, Union, Set difference, Join operation	Chapter 4 [Textbook 2]
13		Database recovery and security OR Introduction to data mining	Internet Resource
14		Introduction to data mining (data ware housing, OLAP, OLTP)	Chapter 9 [Textbook 1]
15		NoSQL OR Database life cycle	Internet Resource
16		Transaction management (optional), Concurrency control (optional)	Chapter 20 [Textbook 2]

~~DATABASE SYSTEMS (SEDB-227) – LAB~~

~~I. Course Details~~

Credit Hours	3 (3+1)
Pre-requisites	-
Course Leader	Dr. Nargis Fatima
Recommended Textbook(s)	1. Modern database management, Jeffrey A. Hoffer, 12th Edition, Pearson, 2016. 2. Database systems: A practical approach to design, implementation, and management, Thomas Connolly and Carolyn Begg, 6th Edition, Pearson, 2015.
Recommended Reference (Books/Websites/Articles)	1. Database system concepts, Avi Silberschatz, Henry F. Korth and S. Sudarshan, 6th Edition, McGraw-Hill, 2010. 2. Database systems: Design, implementation and management, Carlos M. Coronel, 13th Edition, Cengage Learning, 2018.

~~II.Lab Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Demonstrate the task by interpreting, summarizing, and explaining key concepts/principles/relationships within the domain, enabling them to apply acquired knowledge effectively in diverse contexts.	Cognitive	2	2	Lab Task, Lab Report, Final Exam, Semester Project
CLO-2	Practice theoretical concepts for conducting the experiments/projects of varying complexities to achieve desired outcomes.	Psychomotor	3	4	Lab Task, Semester Project
CLO-3	Contribute actively and responsibly individually or in teams during laboratory activities, demonstrating a positive attitude and willingness to participate.	Affective	2	6	Lab Task, Semester Project

~~III. Lab Assessment~~

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25
Total	100

~~IV. Weekly Breakdown (All CLOs shall be assessed in every lab)~~

Week No.	Topics
1	Introduction OF Relational DBMS (Mysql or SQL Server) installation guidelines
2	Retrieving data using the SQL SELECT statement
3	Restricting using where and sorting data using order by
4	Using DDL statements to create , and manage tables
5	Creating others schema objects like table level and column level constraints
6	Using single-row character functions to customize output
7	Using single-row date functions to customize output & type conversion
8	Constructing ERD using VISIO or Draw.io
9	Midterm Exam
10	Constructing Enhanced ERD using VISIO or Draw.io
11	Aggregating data using SQL aggregate function and group functions
12	Use of Joins for displaying data from multiple tables
13	Use of subqueries
14	Open Ended Lab
15	Normalization
16	Final term assessment

~~SOFTWARE DESIGN AND ARCHITECTURE (SESD-242)~~

~~I. Course Details~~

Credit Hours	3 (2+1)
Pre-requisite(s)	Software Design & Architecture
Course Leader	Dr. Huma Hayat Khan
Recommended Book(s)	Software Engineering design theory and practices by Carles e Otero
Reference Books	1. Applying UML And Patterns: An Introduction to Object Oriented Analysis And Design And Iterative Development, Craig Larman, 3rd Ed, Pearson Education, 2005. 2. Documenting Software Architectures: Views and Beyond By Paul Clements, Felix Bachmann, Len Bass, 2nd Edition, 2002 3. Design Patterns: Elements of Reusable Object-Oriented Software, Erich Gamma, John Vlissides, Richard Helm, Ralph Johnson, 1994

~~II. Course Learning Outcomes (CLOs)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifacts
CLO-1	Explain fundamental design and architecture concepts in software systems to organize, compare, translate, and interpret the software architecture and design activities.	Cognitive	2	2	A1, Q1, Midterm exam
CLO-2	Apply UML knowledge to Comprehend the design challenges as well as to document the design of software systems	Cognitive	3	3	A2, Q2, Midterm exam, Final Term exam
CLO-3	Examine the software design patterns and architectural styles by understanding the problems to identify the best use of the architecture and design patterns.	Cognitive	4	2	A3, Q3, final term exam

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5

Midterm	25
Final Term	50
Total	100

III. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

IV. Course Contents

Software Design and Architecture Concepts, Design principles, Object-Oriented Design with UML, Architecture views, User interface design, Persistent layer design, Web applications design, State machine diagrams and modeling, Design Patterns, Architectural design issues, , Software Architecture, Architectural Structures & Styles-, Architectural Patterns.

V. Weekly Breakdown

Week No.	CLOs	Topics	Reference
1	CLO-1	Introduction to Software Engineering Design, Conceptual Overview, engineering Design, Engineering Problem Solving, Initial State, Operational State, thinking about the Problem, Problem Solution, goal State	Chapter 1
2		Introduction to Software design with UML, Conceptual Overview, what Is UML? Why Study UML? The UML's Fundamentals, Classification of UML diagrams. Designing the Architecture, the 4 + 1 View Model, user View, process View., physical View, development View, Logical View,	Chapter 2
3	CLO-2	Introduction to use case Identification, Use case Writing (fully dressed use case)	Chapter 6 [Ref. Book 1]
4		Introduction to use case modelling. Use case model generation	Chapter 6 [Ref. Book 1]
5		Introduction to Domain Model. Domain model generation	Chapter 6 [Ref. Book 1]

6		Introduction to class diagrams (logical view), Linking classes of Class Diagram through associations , class diagram generation	Chapter 16 [Ref. Book 1]
7		Introduction to Activity Diagram (process view), Fork and join Swim Lanes, Activity Diagram Generation	Chapter 28 [Ref. Book 1]
8		Introduction to Sequence diagram (process view), Message passing and its types, Loop and condition in sequence diagram, frames in sequence diagram Practice Problems	Chapter 10 [Ref. Book 1]
9	Midterm Exams		
10	CLO-2	Introduction to state diagram (process view), Naming conventions Practice problems	Chapter 13 Chapter 29 [Ref. Book 1]
11		Introduction to Development View(Component Diagram) using UML	Chapter 37 [Ref. Book 2]
12		Introduction to Physical View(Deployment Diagram) using UML	Chapter 37 [Ref. Book 2]
13	CLO-3	Patterns and Styles in Software Architecture Conceptual Overview, Architectural Styles and Patterns, History of Architectural Styles and Patterns, Architectural Pattern Classification, Data-Centered Systems, blackboard Pattern, data Flow Systems, Pipe and Filter Pattern, distributed Systems , Client-Server Pattern, broker Pattern, Interactive Systems, Model View Controller Pattern, Hierarchical Systems, Main Program and Subroutine. Layered Pattern	Chapter 4
14		Complex Computing Problem based project Report submission and Evaluation	
15-16		Presentations Week	

~~SOFTWARE DESIGN AND ARCHITECTURE (SESD-242)-LAB~~

~~I.Course Details~~

Credit Hours	3 (2+1)
Pre-requisites	Software Design & Architecture
Course Leader	Dr. Huma Hayat Khan
Recommended Textbook(s)	Software Engineering design theory and practices by Carles e Otero
Recommended Reference (Books/Websites/Articles)	1. Applying UML And Patterns: An Introduction to Object Oriented Analysis And Design And Iterative Development, Craig Larman, 3rd Ed, Pearson Education, 2005. 2. Documenting Software Architectures: Views and Beyond By Paul Clements, Felix Bachmann, Len Bass, 2nd Edition, 2002 3. Design Patterns: Elements of Reusable Object-Oriented Software, Erich Gamma, John Vlissides, Richard Helm, Ralph Johnson, 1994

~~V.Lab Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Demonstrate the task by interpreting, summarizing, and explaining key concepts/principles/relationships within the domain, enabling them to apply acquired knowledge effectively in diverse contexts.	Cognitive	2	2	Lab Task, Lab Report, Final Exam, Semester Project
CLO-2	Practice theoretical concepts for conducting the experiments/projects of varying complexities to achieve desired outcomes.	Psychomotor	3	4	Lab Task, Semester Project
CLO-3	Contribute actively and responsibly individually or in teams during laboratory activities, demonstrating a positive attitude and willingness to participate.	Affective	2	6	Lab Task, Semester Project

~~VI. Lab Assessment~~

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25
Total	100

~~VII. Weekly Breakdown (All CLOs shall be assessed in every lab)~~

Week No.	Topics
1	Introduction to UML (Unified Modeling Language), Static and Dynamic UML Diagrams,
2	Introduction to Use Case, Vocabulary, Notations, Functional Requirements, Use case Description with Examples
3	Usecase Writing (fully dressed use case)
4	use case modelling. Use case model generation
5	Domain model generation
6	class diagram generation
7	Activity Diagram Generation
8	Sequence diagram (process view) generation
9	Midterm Exams
10	state diagram (process view) generation
11	Component Diagram generation
12	Deployment Diagram generation
13	Software Architecture patterns, real work example scenarios
14	Open Ended Lab
15	Layered Architecture and MVC example
16	Final Term Exam

~~OPERATING SYSTEMS (SEOS-226)~~

~~I. Course Details~~

Credit Hours	4 (3+1)
Pre-requisites	Data Structures and Algorithms
Course Leader	Dr. Jaweria Kanwal
Recommended Textbook(s)	Operating system by Albert shilberschatz, Latest Edition, 2021
Recommended Reference (Books/Websites/Articles)	1. Modern Operating Systems, Andrew Tanenbaum, Thirteen Edition, Prentice Hall, 2021 2. Operating System by William Stalling, Latest Edition, 2021

~~II. Course Learning Outcomes (CLOs)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifacts
CLO-1	Explain the characteristics and services of different structures of operating systems to state its main ideas.	Cognitive	2	2	A1, Q1, Mid-term
CLO-2	Demonstrate basic structures, functions, and algorithms of OS to perform comparative analysis of algorithms.	Cognitive	2	2	A2, Q2, Midterm, Final Term
CLO-3	Examine algorithms of system resource management to select appropriate algorithms for resource management.	Cognitive	4	3	A3/Project, Q3, Final Term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

~~Introduction to Operating Systems, What is an OS, Single User Systems, Batch Systems, Multi-programmed Systems, Time Sharing Systems, Multiprocessor Systems, Real Time Systems Computer System Structures (Computer System Operation, I/O Structure, Storage Structure, Storage Hierarchy, Hardware Protection) Operating System Structures (Operating Systems Concepts, System Calls) Processes & Threads (Process Concept, Process Scheduling, Operation on Processes, Cooperating Processes, Threads) CPU Scheduling (Introduction to Scheduling, Scheduling Criteria, Scheduling Algorithms) Process Synchronization (The Critical Section Problem, Synchronization Hardware, Semaphores, Classical Problems of Synchronization) Deadlocks (Deadlock Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Avoidance and Detection) Memory Management (Logical vs. Physical Address Space, Swapping, Contiguous Allocation, Paging, Segmentation, Segmentation with Paging) Virtual Memory (Demand Paging, Page Replacement, Page Replacement Algorithms, Allocation of frames, Thrashing) File System Interface and Implementation (File Concept, Access Methods, Directory Structure, Protection, File System Structure, Allocation Methods, Free Space Management, Directory Implementation)~~

VI. Weekly Breakdown

Week No.	CLOs	Topics	References
1	CLO-1	Types of software, Application Software and System software, Application software categories, System software categories.	Chapter 1
2		Introduction to OS, Single user, Multi user and Network OS, H/W dependent and independent OS, Important OS components, Kernel, Shell, Shell types	Chapter 1
3		Type of services provided by OS, Uni- programming and Multi-programming, Uni-tasking and Multi-tasking, Uni- processing and Multi-processing.	Chapter 2
4		Distributed systems and its types, Reason for building distributed systems,	Chapter 2
5		SPOOLing, Buffering, Real-Time systems, Booting mechanism.	Chapter 3
6	CLO-2	Process and process states. Process Control Block, Independent and Cooperating processes, Process and Threads, Hierarchy of processes.	Chapter 3

7		Inter Process Communication, Process Synchronization, Race condition, Mutual exclusion, Critical section, Achieving process synchronization.	Chapter 3
8		Scheduler and scheduling queues, Types of schedulers, CPU and I/O bound processes, Context switching, Criteria for comparing CPU scheduling algorithms.	Chapter 4
9	Midterm Exams		
10	CLO-2	Multi-level queue scheduling, Multi-level feedback queue (MLFBO) scheduler, Multi-processor scheduling, and CPU Scheduling Algorithm evaluation.	Chapter 5
11		Introduction to Deadlocks, Deadlock characterization, Deadlock prevention, Deadlock avoidance.	Chapter 5
12		Introduction to Banker's Algorithm and importance of its usage, Deadlock detection in single resource of each resource type.	Chapter 6
13		Deadlock detection in multiple resources of each resource type, Deadlock Recovery, Techniques used to recover processes from deadlock.	Chapter 6
14	CLO-3	Introduction of memory management, Swapping, Single partition allocation and Multiple partition allocation memory management schemes, Multiprogramming with Fixed partition.	Chapter 7
15		Multiprogramming with Variable partition, First-Fit, Best-Fit, Next-Fit and Worst-Fit algorithms, External fragmentation, Compaction, Paging, Segmentation.	Chapter 7
16		Virtual Memory, Demand Paging, Page Replacement, Page Replacement algorithms, Thrashing, Introduction to File System, Directory structure, Single and Two level directory, Tree structured directory, Acyclic graph directories, General graph directories, File protection.	Chapter 8

~~OPERATING SYSTEMS (SEOS-226) – LAB~~

~~I. Course Details~~

Credit Hours	4 (3+1)
Pre-requisites	Data Structures and Algorithms
Course Leader	Dr. Jaweria Kanwal
Recommended Textbook(s)	Operating system by Albert shilberschatz, Latest Edition, 2021
Recommended Reference (Books/Websites/Articles)	1. Modern Operating Systems, Andrew Tanenbaum, Thirteen Edition, Prentice Hall, 2021 2. Operating System by William Stalling, Latest Edition, 2021

~~II. Lab Learning Outcomes (CLOs)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifacts
CLO-1	Demonstrate the task by interpreting, summarizing, and explaining key concepts/principles/ relationships within the domain, enabling them to apply acquired knowledge effectively in diverse contexts.	Cognitive	2	2	Lab Task, Lab Report, Final Exam, Semester Project
CLO-2	Practice theoretical concepts for conducting the experiments/projects of varying complexities to achieve desired outcomes.	Psychomotor	3	4	Lab Task, Semester Project
CLO-3	Contribute actively and responsibly individually or in teams during laboratory activities, demonstrating a positive attitude and willingness to participate.	Affective	2	6	Lab Task, Semester Project

~~III. Lab Assessment~~

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open-ended Lab	15
Final Term Exam	25
Total	100

~~IV. Weekly Breakdown (All CLOs shall be assessed in every lab)~~

Week No.	Topics
1	Introduction to Linux. Advantages of Linux and different versions of UNIX. Tutorial on installation of Linux. Getting started with your Linux and general overview of the OS
2	A brief introduction to a few of the basic commands of Linux.
3	Exploring the Linux file system, including the basic concepts of files and directories and their organization in a hierarchical tree structure.
4	Overview of The Bourne Again Shell
5	Performing Basic File Management (copy command (cp), move command (mv), rm, touch). Directory utilities (mkdir, rmdir) and wildcard patterns.
6	An introduction to Processing of Text Streams using Text Processing Filters.
7	Parameters passing in Linux.
8	Use of Unix Streams, Pipes and Redirects.
9	Midterm Exams
10	Implementation of Environment variables in Linux, Searching Text Files Using Regular Expressions (grep)
11	Programming Fundamentals, if-else, for, While, do while loop shell scripts.
12	Implementing Switch case structure, functions and Various Programming related exercises in Linux. fork (), getpid(), getppid(), wait(), opendir(), readdir(), closedir() system calls implementation using Linux.
13	File handling in Linux using System calls, Implementation of System calls using GCC Compiler in Linux.
14	Open Ended Lab
15	Implementation of various CPU scheduling algorithms
16	Final Lab Exam

PROBABILITY & STATISTICS (SEPS-204)

I. Course Details

Credit Hours:	3 (3+0)
Pre-requisites:	Calculus
Course Leader	Dr. Asad Husain
Recommended Textbook(s)	Probability and Statistics for Engineers and Scientists by Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying E. Ye Publisher Pearson.
Recommended Reference (Books/Websites/Articles)	1. Introduction to Statistical Theory by Sher Muhammad Chaudhry, Dr. Shahid Kamal, Ninth Edition 2013, 2. Probability and Statistics for Engineers and Scientists by Anthony J. Hayter, 3. Schaum's Outline of Probability and Statistics, by John Schiller, R. Alu Srinivasan and Murray Spiegel, McGraw-Hill; 3rd Edition (2008). ISBN-10:0071544259

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment
CLO-1	Explain fundamental concepts of probability and statistics along with graphical representation to state their main ideas.	Cognitive	2	1	A1, Q1, Mid Term
CLO-2	Apply probability formulas and distributions to solve real-world problems.	Cognitive	3	2	A2, Q2, Mid Term, Final Term
CLO-3	Apply basic statistical techniques to engineering data for solving various problems.	Cognitive	3	2	A3, Q3, Final Term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Introduction to Statistics and Data Analysis, Statistical Inference, Samples, Populations, and the Role of Probability. Sampling Procedures. Discrete and Continuous Data. Probability: Sample Space, Events, Counting Sample Points, Probability of an Event, Additive Rules, Conditional Probability, Independence, and the Product Rule, Bayes' Rule. Random Variables and Probability Distributions. Mathematical Expectation: Mean of a Random Variable, Variance and Covariance of Random Variables, Means and Variances of Linear Combinations of Random Variables, Chebyshev's Theorem. Discrete Probability Distributions. Continuous Probability Distributions. Fundamental Sampling Distributions and Data Descriptions: Random Sampling, Sampling Distributions, Sampling Distribution of Means and the Central Limit Theorem. Sampling Distribution of S^2 , t-Distribution. Tests of Hypotheses. The Use of P Values for Decision Making in Testing Hypotheses (Single Sample & One- and Two Sample Tests), Linear Regression and Correlation. Least Squares and the Fitted Model, Multiple Linear Regression and Certain, Nonlinear Regression Models, Linear Regression Model Using Matrices, Properties of the Least Squares Estimators.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Introduction to Statistics and Data Analysis: Descriptive and inferential statistics, population and sample, Observations and variables, Types of variables, Data collection	Chapter 1
2		Measure of central tendency and Measure of dispersion, quartiles for ungroup data, Graphical representation of data: dot plot, Stem leaf Display, Box and whisker plot	Chapter 1
3	CLO-2	Introduction to Probability, concept of sets, Venn diagram, operation and algebra on sets, Cartesian product	Chapter 2
4		Counting Sample Points, Sample Space, Events	Chapter 2
5		Definition of probability axioms of probability, conditional probability, independent events,	Chapter 2

6		Additive Rules and the Product Rule, Bayes' Rule	Chapter 2
7		Random variables, Mathematical Expectation: Mean of a Random Variable, Variance of Random Variables,	Chapter 3,4
8		Means and Variances of Linear Combinations of Random Variables, Chebyshev's Theorem	Chapter 3,4
9	Midterm Exams		
10	CLO-2	Discrete Probability Distributions: Binomial distribution, Poisson distribution	Chapter 5
11		Continuous Probability Distributions: Normal distribution and standard normal distribution and its properties	Chapter 6
12		Random Sampling, Sampling Distribution of Means and the Central Limit Theorem	Chapter 8
13	CLO-3	Linear Regression Model using Matrices, Least Squares and the Fitted Model	Chapter [Ref. book1]
14		Multiple Linear Regression, coefficient of correlation	Chapter [Ref. book1]
15		Tests of Hypotheses: The Use of P Values for Decision Making in Testing Hypotheses, alpha level, significance	Chapter [Ref. book1]
16		Sampling Distribution of S^2 , Chi-squared distribution, t-Distribution	Chapter [Ref. book1]

BUSINESS PROCESS ENGINEERING (SEBP-263)

I.Course Details

Credit Hours:	3 (3+0)
Pre-requisites:	
Course Leader:	Dr. Marriam Daud
Recommended Book(s):	Business process change: A business process management guide for managers and process professionals, Harmon, P, 3rd edition. Morgan/Kaufmann Publishing: Boston, 2019.
Reference Books:	1. Operations Management: An integrated Approach, Reid & Sanders, 5th Edition, John Wiley & Sons Inc. publishers, 2013. ISBN: 978-1-118-32363-2 2. Fundamentals of Total Quality Management, Jens J. Dahlgaard, Kai Kristensen, Ghopal K. Khanji, Roulledge Publishers, 2005. ISBN: 0748772936, 9780748772933, 9780203930021. 3. Total Quality Management, Gunmala Suri & Puja Chhabra Sharma, 1st Edition, Biztantra Publishers, 2013, ISBN: 978-93-5004-317-2. 4. <i>Operations Management</i> by Russell and Taylor, Prentice Hall, Fifth Edition, 2007, ISBN 0-471-67877-5 5. <i>Operations Management</i> by S. Anil Kumar & N.Suresh, New Age International Publishers, Fifth Edition, 2009, ISBN (13) : 978-81-224-2883-4 6. International Business; Competing in The Global Market Place, Charles W. L. Hill, McGraw-Hill/ Irwin Publishers, 2009, ISBN 978-0-07-338134-3 Memory Jogger - A pocket Guide of Tools of CQPI

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment Artifacts
CLO-1	Explain productivity, designing, engineering, and managing aspects of business processes to state its main ideas.	Cognitive	2	1	Q1, A1, Midterm
CLO-2	Demonstrate understanding of quality approaches for managing business operations.	Cognitive	2	2	Q2, A2, Midterm, Final term
CLO-3	Apply business process improvement techniques to solve problems in different contexts.	Cognitive	3	2	Q3, A3, Presentation / Project, Final term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentations/project	5
Mid Term	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Business process management, Manufacturing and services processes, Modelling and charting tools, Lean processes Improvement workshop techniques, Business process outsourcing, Re-engineering and improvement cases. Strategic and organizational issues of process management and the use of Enterprise Resource Planning systems (ERPs). Strategic approaches used to understand, analyze and implement efficient business processes, workflow modeling techniques, process modeling techniques and procedure models.

VI. Weekly Breakdown

Week	CLO	Topics	Reference
1	CLO1	Introduction to the course.	--
2		Introduction to important business process engineering concepts and business process improvement.	Chapter 1, RB # 4
		Efficiency, effectiveness, and productivity and their importance/ relevance in business.	
3-4		Introduction to important decisions engineering/ business design and management processes.	Chapter 1, RB # 4 Chapter 3-8, RB # 5 Reading 1: https://online.kettering.edu/news/critical-decisions-operations-management

5	CLO2	Concept of quality; feature and performance-based quality, customer and producer view of quality, various costs of delivering quality (prevention, appraisal, scrap & warranty costs).	Chapter 5 RB # 1 Chapter 2-4 RB #2, Chapter 3&4 RB # 3 Reading 2: https://www.tandfonline.com/doi/full/10.1080/14783363.2020.1844564
6		Total Quality Management (TQM) and Operations Management, customer craft quality paradigm, mass production & inspection quality paradigm, Edward Deming's 14 Points of Quality, Deming Chain Reaction.	Chapter 5 RB # 1 Chapter 2-4 RB #2, Chapter 3&4 RB # 3
7		Engineering Wastes, overproduction, inventory, repair / rejects, motion, processing, waiting, transport, line layout, long lead times, concept of mistake proofing (Poka Yoke) in BPE, importance of asking why (5 whys - getting to the roots of problem).	Reading 3: https://www.bdc.ca/en/articles-tools/operations/operational-efficiency/8-types-of-waste-to-identify-in-business
8	CLO2	Continuous Quality & Process improvement (CQPI)/ Kaizen , 5 S's of lean model.	Reading 4: https://www.wallstreetmojo.com/kaizen/
9	Midterm Exams		
10	CLO3	World Trade Organization (WTO), its role in business world	Chapter 6, RB # 6 Reading 5: https://www.wto.org/english/thewto_e/whatis_e/inbrief_e/inbr_e.htm
11		International Standard Organization (ISO), its role in business improvement, various ISO series, process of getting ISO Certification.	Reading 6: https://uk.indeed.com/career-advice/career-development/what-are-iso-certifications
12		Just In Time (JIT) systems, lean production strategy, big Vs little JIT, concept of ideal lot size of one.	Reading 7: https://www.projectmanager.com/blog/just-in-time-manufacturing
13		Quick Response Manufacturing (QRM), Time Based Competition (TBC), benefits of QRM, explaining response time spiral, key principles of restructuring a company for QRM	Reading 8: https://interpro.wisc.edu/ten-things-you-should-know-about-quick-response-manufacturing/
14-15	CLO2	Group Presentations: Basic Tools of Process Analysis: flow chart, check sheet, check list, cause and effect diagram, Pareto charts, histograms, control chart & run chart, Scatter diagram, brainstorming.	RB # 7
16	CLO1-3	Revision	-

~~Semester V~~

SOFTWARE CONSTRUCTION & DEVELOPMENT (SESC-344)

I. Course Details

Credit Hours	3 (2+1)
Pre-requisites	Software Design and Architecture
Course Leader	Dr. Huma Hayat Khan
Recommended Textbook(s)	1. Clean Code: A Handbook of Agile Software Craftsmanship, Robert C. Martin, Prentice Hall, 2008. 2. Code Complete 2nd edition: A practical handbook of software construction, published by Microsoft Press, 2004. 3. Object-Oriented Software Construction, by Bertrand Meyer, Second Edition, Published by, Prentice Hall in 1997. 4. Software Engineering by Ian Sommerville, 8th edition, Addison & Wesley. 2006.
Recommended Reference (Books/Websites/Articles)	1. Introduction To Computer Theory Daniel I. A. Cohen, Wiley, 17-Jan-1986. 2. Teach Yourself Uml In 24 Hours, Joseph Schmuller, BPB Publications, 1999. 3. Refactoring: Improving the Design of Existing Code, Martin Fowler, Addison-Wesley Professional. 1999. 4. Working Effectively with Legacy Code, Michael C. Feathers. Pearson Education, Prentice-Hall, 2004. 5. UML Distilled, Second Edition: A Brief Guide to the Standard Object Modeling Language, Fowler M. Scott K., 2000. 6. Time And Synchronization In Executable Uml https://www.modelint.com/time-and-synchronization-in-executable-uml. 7. Discrete, Continuous and Hybrid Petri Nets (David, R. and Alla, H.; 2004), in IEEE Control Systems Magazine, vol. 28, no. 3, pp. 81-84, June 2008, doi: 10.1109/MCS.2008.920445.

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment Artifact
CLO-1	Understand and Develop Object-oriented design models and refine them to reflect implementation details	Cognitive	3	2	A1, Q1, Midterm,
CLO-2	Demonstrate good and bad coding practices to ensure code quality	Cognitive	2	2	A2, Q2, MidTerm, Final Term
CLO-3	Analyze the major activities within the OO software development process to encourage modularity and reusability.	Cognitive	4	3	A3, Q3, Final term

III.Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV.Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Software construction fundamentals, Software deployment processes, Software engineering process infrastructure, Software engineering process improvement, Systems engineering life cycle models, Evolution processes and activities, Basic concepts of evolution and maintenance, Working with legacy systems, Refactoring, Error handling, exception handling, Apply a wide variety of software construction techniques and modeling, including clean code concepts, state-based and table-driven approaches to low-level design of software, configuration management, Martin Fowler's refactoring concepts and their application to small projects, legacy code concepts, Exception handling, Test driven development.

VI. Weekly Breakdown

Week No.	CLOs	Topics	Reference
1	CLO1	Course Outline, Assessment Criteria, Introduction to Construction, Key decisions of Construction and software development aspects	Chapter 1,4 [Book 2] Chapter 1 [Book 4]
2		Static Behavioral model VS. Dynamic behavioral model, Context Model, Interaction Model and Model driven engineering Systems Models in Context of Software Construction, Synchronization and timing in UML modeling	Chapter 5 [Book 4]

3		UML Models (Package, Object)	Chapter 14 [Ref Book 2] Chapter 6 [Ref Book 5]
4		UML Models (Communication and Timing)	Chapter 12,17 [Ref Book 5]
5		Introduction of Petri Nets, understanding through examples. Petri Nets vs State machine diagram	Online Source [7]
6		Formation of different Finite Automata, Transition tables Languages, regular expressions and syntax tree	Chapter 2,4,5 [Ref Book 1]
7	CLO2	Clean Code vs Bad code, Writing clean & good code practice, Meaningful Names and Standard Notations	Chapter 1,2 [Book 1]
8		Good Comments: Legal comment, clarification, explanation of intent, Amplification, informative comments	Chapter 4 [Book 1]
9	Midterm Exam		
10		Introduction to formatting in Clean Code .Introduction of Vertical Formatting. Introduction of horizontal formatting and horizontal openness	Chapter 5 [Book 1]
11		Refactoring (First make it work, then make it Right, Conclusion)	Chapter 16, [Book 1]
12		Smells and Heuristics (Comments, Functions, General)	Chapter 17 [Book 1]
13		Test driven development for measuring the software quality	Chapter 9 [Book 1]
14	CLO 3	Software Evolution and its impacts, Software Configuration management, Legacy systems and Michal Feather “Legacy” code concepts	Chapter 9 [Book 4] Chapter 1,2 [Ref Book 4]
15		Criteria of object orientation and Modularity, Approaches to reusability.	Chapter 2, 3 [Book 3] Chapter 16 [Book 4]
16		Error handling core keywords and discussion.	Chapter 7 [Book 1]

SOFTWARE CONSTRUCTION & DEVELOPMENT (SESC-344) – LAB

I.Course Details

Credit Hours	3 (2+1)
Pre-requisites	Software Design and Architecture
Course Leader	Dr. Huma Hayat Khan
Recommended Textbook(s)	1. Code Complete 2nd edition: A practical handbook of software construction, published by Microsoft Press, 2004. 2. Object-Oriented Software Construction, by Bertrand Meyer, Second Edition, Published by, Prentice Hall in 1997. 3. Software Engineering by Ian Sommerville, 8th edition, Addison & Wesley, 2006.
Recommended Reference (Books/Websites/Articles)	1. Introduction To Computer Theory Daniel I. A. Cohen, Wiley, 17-Jan-1986. 2. Teach Yourself Uml In 24 Hours, Joseph Schmuller, BPB Publications, 1999. 3. Refactoring: Improving the Design of Existing Code, Martin Fowler, Addison-Wesley Professional, 1999. 4. Working Effectively with Legacy Code, Michael C. Feathers. Pearson Education, Prentice-Hall, 2004. 5. UML Distilled, Second Edition: A Brief Guide to the Standard Object Modeling Language, Fowler M.Scott.K., 2000. 6. Time And Synchronization In Executable Uml https://www.modelint.com/time-and-synchronization-in-executable-uml. 7. Discrete, Continuous and Hybrid Petri Nets (David, R. and Alla, H.; 2004), in IEEE Control Systems Magazine, vol. 28, no. 3, pp. 81-84, June 2008, doi: 10.1109/MCS.2008.920445.

II. Lab Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Demonstrate the task by interpreting, summarizing, and explaining key concepts/principles/relationships within the domain, enabling them to apply acquired knowledge effectively in diverse contexts.	Cognitive	2	2	Lab Task, Lab Report, Final Exam, Semester Project
CLO-2	Practice theoretical concepts for conducting the experiments/projects of varying complexities to achieve desired outcomes.	Psychomotor	3	4	Lab Task, Semester Project

CLO-3	Contribute actively and responsibly individually or in teams during laboratory activities, demonstrating a positive attitude and willingness to participate.	Affective	2	6	Lab Task, Semester Project
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III. Lab Assessment

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25
Total	100

IV. Weekly Breakdown (All CLOs shall be assessed in every lab)

Week #	Topics
1	Installation of tool
2	Structural & Behavioural Modeling with one example
3	Package Diagram
4	Object Diagram
5	Communication Diagram
6	Timing Diagram
7	Petri Nets
8	Lab Assessment
9	Midterm Exam
10	Clean Code & Bad Code (Naming Guidelines)
11	Clean Code & Bad Code (Commenting & Formatting Guidelines)
12	Extract Method
13	Refactoring of code smells
14	Test Driven Development
15	Git & Git Hub
16	Exception Handling

*open ended lab after week 8

WEB ENGINEERING (SEWE-346)

I.Course Details

Credit Hours	3 (2+1)
Pre-requisites	Programming fundamentals
Course Leader	Ms. Saadia Shabbir
Recommended Textbook(s)	1. HTML & CSS: Design and Build Web Sites by Jon Duckett (1st edition 2011) 2. Javascript and jquery interactive front end web development jon duckett() 3. https://www.geeksforgeeks.org/ 4. www.w3schools.com 5. www.javatpoint.com 6. Learning PHP, MySQL & JavaScript: With JQuery, CSS & HTML5 by Robin Nixon (5th edition ,2018)
Recommended Reference (Books/Websites/Articles)	1. PHP, MySQL & JavaScript All in One, Sams Teach Yourself (6th edition 2017) 2. Responsive Web Design with HTML5 and CSS: Develop future-proof responsive websites using the latest HTML5 and CSS techniques (3rd edition 2020) 3. JavaScript: The Definitive Guide: Master the World's Most-Used Programming Language (7th edition 2020)

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Understand fundamental concepts of Web engineering to comprehend the essential principles foundational to web development and design.	Cognitive	2	2	A1, Q1, Mid Term
CLO-2	Develop front-end web applications using Html, CSS and JavaScript to structure content, incorporate styling, and integrate interactivity into web pages.	Cognitive	3	4	A2, Q2, Mid Term, Project, Final Term
CLO-3	Apply server side programming languages and integrate database functionality for generating dynamic web applications.	Cognitive	3	4	A3, Q3, Final Term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Introduction of web engineering, web terminologies, Web programming languages basics (e.g., HTML, CSS, JavaScript, PHP), Semantic elements, Links and Images, tables and forms, block level and inline elements, website design and layout, styling of html elements, CSS selectors, event driven programming, crud operations, sessions and cookies, input validation.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Course Outline, Assessment Criteria, What is Web Engineering, evolution of world-wide web (web 1.0, web 2.0, web 3.0, web 4.0 and web 5.0), what is Web application, what is web page, role of worldwide consortium, role of web server and web browser, what is world wide web, what is URL, What is domain name, parts of domain name	Online link [3]
2		Web terminologies: what is ip address, how does a web work, Role of Domain name system(DNS), what is a website, Types of websites(static, dynamic, front-end and back-end), what is web hosting, types of web hosting, features of web hosting,	Online link [5]

3	CLO-2	What is HTML, Html page structure, Anatomy of an Html element (Container element non container elements, standalone element attributes, values) Paragraphs and headings in HTML, Text Formatting elements in HTML(bold, Italic, strong, superscript, subscript, mark and delete), Lists in HTML	Chapter 1,2,3 Textbook [1] Online link[4]
4		Use of Links and images in HTML, Semantic elements (header, article, aside, figcaption, figure, header, footer, section, navigation), tables in html(tr, td, th, rowspan, colspan, table caption), using CSS properties to modify tables	Chapter 4,5,6,14,17 Textbook[1] Online link [4]
5		Inline and block level elements in html, Divs, creating forms in html (form elements, input attributes within form), using CSS properties to modify forms	Chapter 8,14 Textbook [1] Online link [4]
6		What is CSS, Types of CSS, Pros and cons of each type, CSS selectors(element, ID, Class, universal selector, grouping selector, descendant selector (space), child selector (>), adjacent sibling selector (+), general sibling selector (~), Pseudo-class selectors), CSS Specificity	Chapter 10,11,12 Textbook [1] Online link [4]
7		CSS colors(RGB, HEX, HSL), Text and font properties in CSS(font size, font weight, font style, font effects, Google fonts, text color, text alignment, text decoration, text transformation, text spacing, text shadow)	Chapter 10,11,12 Textbook [1] Online link [4]
8		Backgrounds in CSS(Background-color, Background-image, background-repeat, background positioning, borders, margins, padding, CSS Positioning	Chapter 13,15,16 Textbook [1] Online link [4]
9	Midterm Exams		
10	CLO-3	What is JavaScript, different ways of using JavaScript, input and output in javascript, variables, data types and Boolean operators in javascript, conditional statements and switch statement, Alert box, confirmation message and prompt in javascript, functions in javascript	Chapter 1,2 Textbook [2] Online link [4]
11		What is document object model(DOM), methods to find html elements in DOM (Get elements by id, class name and tag name), changing html elements and properties in DOM, changing CSS using DOM, form validation in javascript, event handling in javascript	Chapter 5,6 Textbook [2] Online link [4]
12		What is php, setting environment to work with php, variables and data types in php, operators in php(Assignment, arithmetic, comparison and logical operators), arrays, conditional statements,	Online link [4]
13		Global variables in php, Form processing in php using get method, post method and request method form validation in php(client side validation, server side validation, validation using regular expressions	Online link [4]

14		Connecting php with mysql database, usage of php my admin to establish connection of php with database, Crud operations in php(create MYSQL database, creation of MYSQL tables, Insertion of data, read data from MySQL database ,updatation of data and data and deletion of data)	Online link [4]
15		Example of Crud operations in php (Signup form or login form),session handling and cookies management in php	Online link [4]
16		Project Evaluation	

~~WEB ENGINEERING (SEWE-346) – LAB~~

~~I. Course Details~~

Credit Hours	3 (2+1)
Pre-requisites	Programming fundamentals
Course Leader	Ms. Saadia Shabbir
Recommended Textbook(s)	1. HTML & CSS: Design and Build Web Sites by Jon Duckett (1st edition 2011) 2. Javascript and jquery interactive front end web development jon duckett() 3. https://www.geeksforgeeks.org/ 4. www.w3schools.com 5. www.javatpoint.com 6. Learning PHP, MySQL & JavaScript: With JQuery, CSS & HTML5 by Robin Nixon (5th edition ,2018)
Recommended Reference (Books/Websites/Articles)	1. PHP, MySQL & JavaScript All in One, Sams Teach Yourself (6th edition 2017) 2. Responsive Web Design with HTML5 and CSS: Develop future-proof responsive websites using the latest HTML5 and CSS techniques (3rd edition 2020) 3. JavaScript: The Definitive Guide: Master the World's Most-Used Programming Language (7th edition 2020)

~~II. Lab Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Demonstrate the task by interpreting, summarizing, and explaining key concepts/principles/relationships within the domain, enabling them to apply acquired knowledge effectively in diverse contexts.	Cognitive	2	2	Lab Task, Lab Report, Final Exam, Semester Project
CLO-2	Practice theoretical concepts for conducting the experiments/projects of varying complexities to achieve desired outcomes.	Psychomotor	3	4	Lab Task, Semester Project
CLO-3	Contribute actively and responsibly individually or in teams during laboratory activities, demonstrating a	Affective	2	6	Lab Task, Semester Project

	positive attitude and willingness to participate.				
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III. Lab Assessment

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25
Total	100

IV. Weekly Breakdown (All CLOs shall be assessed in every lab)

Week No.	Topics
1	Exploring Website terminologies and architecture
2	Introduction to HTML basics
3	Exploring HTML file paths and text formatting elements
4	Exploring Lists and tables for website development
5	Exploring HTML semantic elements and form elements
6	Exploring CSS selectors, padding, margins and borders for website development
7	Exploring CSS properties for enhancing website development
8	Positioning elements using CSS
9	Midterm Exam
10	Introduction to javascript for website development
11	Implementing javascript and html dom for front end development
12	Introduction to php for website development
13	Functions arrays and global variables in PHP
14	MYSQL operations in PHP, Practicing CRUD operations on project
15	Open Ended Lab
16	Final Assessment

HUMAN COMPUTER INTERACTION (SEHC-343)

I. Course Details

Credit Hours	3(3+0)
Pre-requisites	Software Engineering
Course Leader	Dr. Marriam Daud
Recommended Textbook	D. Benyon, Designing User Experience: A Guide to HCI, UX and Interaction Design, 4th edition, Pearson, 2019.
Recommended Reference (Books/Websites/Articles)	1. Dix et al., Human Computer Interaction, 3rd edition, Pearson Education, 2003. 2. Shneiderman et al., Designing the User Interface: Strategies for Effective Human-Computer Interaction, 6th edition, Pearson Education, 2017. 3. Norman, The design of everyday things, 2nd edition, 2013. 4. J. Johnson, GUI Bloopers 2.0, 2nd edition, Morgan Kaufmann, 2007. 5. http://hcibib.org/

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment Artifacts
CLO1	Explain the essentials of designing user experience by stating its main ideas.	Cognitive	2	2	Q1, A1, Midterm
CLO2	Apply the key processes of human-centered interaction design for usable interfaces.	Cognitive	3	4	Q2, CCP, Final term
CLO3	Demonstrate interaction and experience design in the different contexts like web and mobile systems.	Cognitive	2	2	Q3, A2, Final term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Contexts for HCI, Psychology of usable things, Processes for User-Centered Design, Metrics and Measures for Evaluation, Usability heuristics and principles of Usability testing, Physical capabilities, Cognitive and social models for interaction design, Principles of good interaction design, Accessibility, Principles of GUI, Visual design elements, Data gathering, Task analysis, Prototyping, Help and user documentation, Internationalization, Usability inspection methods, Usability testing methods, New Interaction Technologies, Usability in practice, Visual Design and Typography, Icon Design, Ubiquitous, Augmented and Virtual Reality.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO1	Introduction to the course; what is plagiarism and how to avoid it	HEC Little Book of Plagiarism
2		The variety of UX; the concerns of UX; being digital	Chapter 1
3		The skills of the UX designer; why being human-centered is important	Chapter 1
4		PACT (People, Activities, Contexts, Technologies): a framework for designing interactive systems; scoping a problem with PACT	Chapter 2
5		The process of UX design; developing personas	Chapter 3
6		Developing scenarios; using scenarios throughout design; scenario based UX design	Chapter 3
7		Concepts of accessibility, usability, and acceptability; design principles	Chapter 5
8	CLO2	Finding suitable presentations; sketching for ideation; visualizing look and feel; mapping the interaction	Chapter 8
9	Midterm Exam		
10	CLO2	Wireframes; prototypes; envisionment in practice	Chapter 8
11		UX design; metaphors and blends in design	Chapter 9
12		Conceptual design; physical design; designing interactions	Chapter 9
13		Introduction to evaluation; data analytics; expert evaluation; participant-based evaluation	Chapter 10
14	CLO3	Visual interface design; interface design guidelines	Chapter 12
15		Psychological principles; information design; visualization	Chapter 12
16		Website and app development; information architecture of apps and websites; navigation design for apps and websites	Chapter 14

~~ARTIFICIAL INTELLIGENCE (SEAI-365)~~

~~I. Course Details~~

Credit Hours	3 (2+1)
Pre-requisites	Data Structures
Course Leader	Mr. Muhammad Waqar
Recommended Textbook(s)	1. Artificial Intelligence A Modern Approach, Stuart J. Russell and Peter Norvig, 3rd edition, Pearson Education, Inc., year 2016. 2. Pattern Classification by Richard O. Duda, David G. Stork, Peter E. Hart, 2nd edition, John Wiley & sons, Inc., year 2012
Recommended Reference (Books/Websites/Articles)	1. Artificial Neural Networks and Information Theory by colin fyfe, 2nd Edition, year 2000. 2. Deep learning, Ian Goodfellow, Yoshua Bengio, Aaron Courville, the MIT press, year 2018

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment
CLO-1	Explain fundamental concepts of Artificial Intelligence to interpret its main ideas.	Cognitive	2	2	A1, Q1, Mid Term
CLO-2	Apply artificial intelligence techniques to solve real-world problems.	Cognitive	3	2	A2, Q2, Mid Term, Final Term
CLO-3	Analyze artificial intelligence techniques to apply them efficiently in various situations.	Cognitive	4	3	A3, Q3, Project/Presentation, Final Term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

~~IV. Grading Policy~~

~~For students admitted in Fall 2021 and onwards~~

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

What is artificial intelligence, the foundation, and the history of Artificial Intelligence? Concept of rationality, rational agents, agents and environment, Problems in AI. Problem-solving by searching, defining problems, categories of problems, Categories of the search algorithm, tree vs graph search, informed vs uninformed search, measuring the performance of problem-solving by search, BFS, DfS, uninformed cost search, informed search best first search heuristics, A* search. Beyond classical search, optimization problems, local search algorithm, hill climbing, online and offline search agents, unknown environment Adversarial search, games, optimal decision in games, minimax algorithm, alpha-beta pruning Constraint satisfaction problems, backtracking forward filtering, Arc consistency, ordering, logical agents, first-order logic, inference Introduction to machine learning and pattern recognition decision process, feature selection. Problems in pattern classification: overfitting model selection missing features, etc. problems in machine learning, the big four problems, (density estimation regression, clustering, and classification), Bayesian decision theory with and without prior knowledge, minimum error rate, Supervised, unsupervised learning, reinforcement learning, The measure of accuracy, Receiver operating characteristic (ROC), confusion matrix, Artificial neural networks, Convolution neural networks, Deep learning.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	What is artificial intelligence, the foundation, and the history of AI. Different factors/behaviors counted as Intelligent. How can we implement it in a computer system?	Chapter 1
2		Concept of rationality, rational agents, agents and environment, Problems in AI.	Chapter 2
3		Problem-solving by searching, defining problems, categories of problems. Classical Problem-solving Approach. Generate and Test Approach. Problem. Representation through graph or tree, Component of Problem Solving The two-one puzzle Problem (example), Graph theory and Tree Terminology	Chapter 3
4		Searching and searching Algorithm. Categories of the search algorithm, tree vs graph search, informed vs uninformed search.	Chapter 3

		Data-Driven vs. Goal-driven Search. Breadth-First Search & its Algorithm, Depth First Search & Algorithm. Comparison b/w Depth-First and Breadth-First Search. Non-deterministic search	
5		Progressive deepening (An approach to improve the efficiency of depth-first search). Informed vs Uninformed Searching. Heuristics and Hill climbing to highlight the importance of heuristics. Best-first Search and Algorithm. Greedy search	Chapter 3
6	CLO-2	Heuristics and Heuristic Evaluation Function. Heuristics for tic tac toe and 8 puzzle problem. What is Optimal searching? Why is it significant? Branch and Bound technique as an optimal approach over the best-first search.	Chapter 4
7		A* procedure to improve the quality of heuristics. Constraint satisfaction problems, backtracking forward filtering,	Chapter 5 Chapter 6
8		What is Adversarial Search? What is meant by a look-ahead strategy? How look ahead strategy is implemented using Min Max? Min-Max algorithm and dry run. How Min-Max algorithm can be optimized? Alpha Beta Pruning as an efficient approach over the min-max procedure.	Chapter 5
9	Midterm Exams		
10	CLO-2	What is an expert system? Comparison of a human expert and an expert system. Roles of an expert system. Components of an Expert system. Applications of an Expert system.	Chapter 1 [Textbook 2]
11	CLO-3	What is Machine Learning and its importance? Difference between machine learning and expert systems. What are the different categories of Machine learning? Terminologies of machine learning (Datasets, features, and model) Process of Machine learning (Data collection, feature selection, training, evaluation, fine-tuning, and application) Types of Machine learning	Chapter 1 [Textbook 2]
12		Supervised learning with example, What is classification and some examples of the classifier? Different techniques to implement Machine learning? Introduction to Machine Learning Problems in pattern classification: overfitting model selection missing features, etc. problems in machine learning, the big four problems, (density estimation regression, clustering, and classification)	Chapter 1,2 [Textbook 2]
13		Bayesian decision theory with and without prior knowledge, minimum error rate, How Bayesian classifier classify input data with example (Continuous Data), KNN classifier with example	Chapter 2 [Textbook 2] Chapter 5 [Ref. Book 2]
14		Unsupervised learning, reinforcement learning Un-supervised learning with example, K- Means Clustering algorithm and it's working mathematically	Chapter 5 [Ref. Book 2]
15		The measure of accuracy, Receiver operating characteristic (ROC), confusion matrix, Model evaluation parameters, Errors and accuracy (Type I and Type II errors)	Chapter 2 [Textbook 2]

16		Biologically Inspired Algorithm: Neural Networks and its working, How they are inspired from neural system, Applications of Neural networks, Examples Artificial neural networks, Convolution neural networks. Deep learning and Its Applications Implementation in Keras (TensorFlow and Pytorch)	Chapter 1 [Ref. Book 1] Chapter 6 [Ref. Book 2]
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~~ARTIFICIAL INTELLIGENCE (SEAI-365) – LAB~~

~~I.Course Details~~

Credit Hours	3 (2+1)
Pre-requisites	Data Structures
Course Leader	Mr. Muhammad Waqar
Recommended Textbook(s)	1. Artificial Intelligence A Modern Approach, Stuart J. Russell and Peter Norvig, 3rd edition, Pearson Education, Inc., year 2016. 2. Pattern Classification by Richard O. Duda, David G. Stork, Peter E.Hart, 2nd edition, John Wiley & sons, Inc., year 2012
Recommended Reference (Books/Websites/Articles)	1. Artificial Neural Networks and Information Theory by colin fyfe, 2nd Edition, year 2000. 2. Deep learning, Ian Goodfellow, Yoshua Bengio, Aaron Courville, the MIT press, year 2018

~~II.Lab Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Demonstrate the task by interpreting, summarizing, and explaining key concepts/principles/ relationships within the domain, enabling them to apply acquired knowledge effectively in diverse contexts.	Cognitive	2	2	Lab Task, Lab Report, Final Exam, Semester Project
CLO-2	Practice theoretical concepts for conducting the experiments/projects of varying complexities to achieve desired outcomes.	Psychomotor	3	4	Lab Task, Semester Project
CLO-3	Contribute actively and responsibly individually or in teams during laboratory activities, demonstrating a positive attitude and willingness to participate.	Affective	2	6	Lab Task, Semester Project

~~III.Lab Assessment~~

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25

Total	100
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IV. Weekly Breakdown (All CLOs shall be assessed in every lab)

Week No.	Topics
1	Introduction to Artificial intelligence and Python and Installation of Python IDE
2	Python programming (Syntax, printing, data types and variables, conditional loops)
3	Python programming (loops, functions, classes)
4	Python programming (lists, tuples, strings, dictionaries)
5	Intelligent Agents
6	Graph Search: Uninformed search and Informed search
7	Introduction to NumPy, Pandas, Scikit-learn and Matplotlib Python Packages
8	Introduction to Machine Learning, Deep learning and deep learning Frameworks (TensorFlow, Keras) in Python
9	Midterm Exams
10	Supervised machine Learning: Classification with K-Nearest Neighbors
11	Supervised machine Learning: Regression with K-Nearest Neighbors
12	Unsupervised machine learning: K-mean clustering
13	Implementation of Neural Networks (NN) in Python
14	Evaluation Metrics to evaluate machine learning algorithms
15	Fuzzy Logic Systems
16	Final term assessment

COMPUTER NETWORKS (SECN-328)

I. Course Details

Credit Hours	4 (3+1)
Pre-requisites	-
Course Leader	Mr. Hasan Humayun
Recommended Textbook(s)	1. Data Communications and Networking, Behrouz A. Forouzan, 5th Edition, McGraw-Hill, 2012. 2. Computer Networking: A Top Down Approach, James F. Kurose, Keith Ross, Pearson, 2017
Recommended Reference (Books/Websites/Articles)	Data and Computer Communications, William Stallings, 10 th Edition, Pearson, 2014.

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Explain the basic network components and data representation in networks to state its main ideas.	Cognitive	2	2	A1, Q1, Mid Term
CLO-2	Make use of Data Link layer services, devices, techniques, and protocols within a network to solve related problems.	Cognitive	3	3	A2, Q2, Midterm, Final Term
CLO-3	Identify features, services, and protocols of Network and Transmission layers by applying acquired knowledge.	Cognitive	3	3	A3, Q3, Project/ Presentation, Final Term
CLO-4	Implement the technologies to create an enterprise-level network with active services.	Cognitive	3	4	CCP

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Introduction and protocols architecture, basic concepts of networking, network topologies, layered architecture, physical layer functionality, data link layer functionality, multiple access techniques, circuit switching, and packet switching, LAN technologies, MAC address, networking devices, network layer protocols, IPv4 and IPv6, DHCP, NAT, IP addressing, IP datagram, ICMP, routing protocols, UDP and (TCP), ports and sockets, connection establishment, flow and congestion control, application layer protocols, latest trends in computer networks.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Intro to Networks components, Basic concepts of networking, Network topologies, network layers, internet standards, and architecture.	Chapter 1
2		The concept of layered architecture modeling. TCP/IP protocol suite, comparison, protocols and OSI.	Chapter 2
3		The distinction between Data and Signals of Analog and Digital Transmission Impairments. Network performance measures including throughput, and delays. Data vs. signaling rates. Channel bandwidth and capacity, delay product.	Chapter 3
4		Switching Technologies: Circuit Switching, Packet Switching, datagram, and Virtual Circuit Switching: Phases, efficiency, and delays.	Chapter 8
5	CLO-2	Data link layer (DLL), Nodes and links, services, framing, flow, error, congestion control, data link addressing, ARP, and communication using DLL.	Chapter 9

6		Error Detection and Correction, Types of Error, Redundancy, Error Detection, Parity, checksum, VRC, LRC, cyclic codes, Detection versus Correction, Forward Error Correction (Hamming codes, Hamming distance) versus Retransmission	Chapter 10
7		Data link control services, Flow, and error control (simple protocol & stop N wait protocol), PAP and CHAP Protocols, Go back N (GBN) and Selective Reject.	Chapter 11 & Chapter 23
8		Multiple Control Access: Random access protocols (Aloha, CSMA, CSMA/CD, CA), Wireless hidden and exposed node problems, and Controlled access protocols (Reservation, Polling, and Token Passing)	Chapter 12
9		Midterm Exam	
10		Ethernet, Evolution, Standard Ethernet addressing, access method Fast Ethernet Access, Mac sub-layer.	Chapter 13
11		WLANs: architecture, characteristics, 802.11 sublayers, Exposed and hidden node problem.	Chapter 15
12	CLO3	Connecting devices: Hub, Switch and Router, Virtual Lans, Spanning tree Algorithm. IP addressing, classful, classless routing, local and global addresses including APIPA.	Chapter 17 & Chapter 18
13		IP addressing format, fragmentation, ICMP v4: error reporting and query messages.	Chapter 19
14		Routing: Least cost, Routing Algorithms: Distance vector and link state, RIP and RIP messages, algorithm, and timers.	Chapter 20
15		UDP Error control and Flow control, congestion. TCP Features, segments, connections, pushing, handshaking, windows in TCP.	Chapter 24
16		Flow control: opening and closing, silly window syndrome. Error Control: acknowledgments, retransmissions. TCP Congestion control, detection, and policies.	Chapter 24

COMPUTER NETWORKS (SECN-328) – LAB

I. Course Details

Credit Hours	4 (3+1)
Pre-requisites	-
Course Leader	Mr. Hasan Humayun
Recommended Textbook(s)	Cisco CCNA Exam #640-607 Certification Guide
Recommended Reference (Books/Websites/Articles)	1. Data Communications and Networking by Behrouz A. Forouzan 2. Data Communications by William Stallings.

II. Lab Learning Outcomes (CLOs)

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment Artifacts
CLO-1	Demonstrate the task by interpreting, summarizing, and explaining key concepts/principles/relationships within the domain, enabling them to apply acquired knowledge effectively in diverse contexts.	Cognitive	2	2	Lab Task, Lab Report, Final Exam, Semester Project
CLO-2	Practice theoretical concepts for conducting the experiments/projects of varying complexities to achieve desired outcomes.	Psychomotor	3	4	Lab Task, Semester Project
CLO-3	Contribute actively and responsibly individually or in teams during laboratory activities, demonstrating a positive attitude and willingness to participate.	Affective	2	6	Lab Task, Semester Project

III. Lab Assessment

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25
Total	100

~~IV. Weekly Breakdown (All CLOs shall be assessed in every lab)~~

Week No.	Topics
1	Preparing Ethernet Cables
2	Understanding Network Commands
3	IP Addressing and Subnetting
4	Configuring basic Switch and Router: Switch and Router configuration, Speed, and Duplex
5	Securing the Switch and Router: Setting up telnet, MOTD banner, etc, Test telnet connection, Line VTY, Cisco Password Encryption, Cisco port security
6	Virtual LAN (VLANs): Create VLANs, Router On A Stick, Native VLAN, and Mismatch
7	Cisco Discovery Protocol (CDP) and Servers: DHCP server, Web Server, CDP Protocol
8	Open Ended Lab
9	Midterm Exams
10	Switch Redundant connections and Loops (STP)
11	Configuring and understanding Static Routes
12	Setting up Wide Area Network (WAN) connections and Default Routes
13	Implementing and analyzing Router Information Protocol (RIP) ver. 1 & 2
14	Simulation of IoT based Home Automation
15	Programming and controlling the IoT devices using MCU and SBC
16	Setting up an Enterprise level Network and testing connectivity
17	End Term Exam

~~Semester VI~~

INFORMATION SECURITY (SEIS-329)

I. Course Details

Credit Hours	3(3 + 0)
Pre-requisites	
Course Leader	Mr. Hasan Humayun
Recommended Textbook(s)	Cryptography and Network Security by William Stallings, 5e 2. Principles of Information Security by Michael E. Whitman
Recommended Reference (Books/Websites/Articles)	1. Computer Security: Art and Science by Mathew Bishop 2. CISSP-all-in-one-exam-guide-seventh-edition

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Explain security components, protocols and standards to achieve security services.	Cognitive	2	2	A1, Q1, Midterm
CLO-2	Identify appropriate techniques to tackle and solve problems in the discipline of information security.	Cognitive	3	3	A2, Q2, Midterm, Final Term
CLO-3	Apply various security and risk management tools for achieving information security and privacy.	Cognitive	3	3	A3, Q3, Final Term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
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%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Information security foundations, security design principles; security mechanisms, symmetric and asymmetric cryptography, encryption, hash functions, digital signatures, key management, authentication and access control; software security, vulnerabilities and protections, malware, database security; network security, firewalls, intrusion detection; security policies, policy formation and enforcement, risk assessment, cybercrime, law and ethics in information security, privacy and anonymity of data.

VI. Weekly Breakdown

Week No.	CLOs	Topics	Reference
1	CLO-1	Main topics and Course Objectives, Definitions of Information Security, Computer Security, Network and Internet Security, Components of an Information System, CNSS Security Model, Operational Model of Security, Key Concepts Security	Chapter 1
2		Architecture: Security attacks, Security Services, Security Mechanisms, Exhaustive Key search, Principles of Counting, Modular arithmetic,	Chapter 1
3		Ceaser Cipher, Mono alphabetic Ciphers, Frequency analysis attacks, ROT13, affine cipher, Vigenère Cipher	Chapter 2
4		Substitution Techniques: Polyalphabetic ciphers, one-time Pad, Transposition Ciphers, Columnar Cipher, Product ciphers: Rotor Machines	Chapter 2
5		History and implementation of DES	Chapter 3
6		History and implementation of AES	Chapter 5
7	CLO-2	RSA algorithm	Chapter 9
8		End to end, Link to Link, Key distribution using Secret key, cryptography, For link encryption, For E2E encryption, Key, Management, Diffie-Hellman Key Exchange	Chapter 10
9	Midterm Exams		
10	CLO-2	Digital signatures	Chapter 13
11		Security Assessment and internal Audit for Passwords: Good and bad passwords, Password Policy Guidelines, Password Cracking Tools, Password Files, Password Collision, Dictionary attacks, Hashing and	Chapter 15

		Salting techniques, Biometrics, Security Tokens, Two Factor Authentication, Kerberos, X.509 Certificates	Ref book (CISSP) 2 Chapter 5 and 6
12	CLO-3	Web Security Considerations, Web Security (SSL)	Chapter 16
13		Transport layer security (TLS), TLS protocol stack, HTTP, and SSH, Cross site scripting (XSS), Cross-site request forgery (CSRF)	Chapter 16 and external resources for XSS and CSFR
14		Software Development Security Common software development issues, Software development life cycles, Change control and configuration management, Security of code repositories, Programming language types, Database concepts and security issues	Reference book 2 (CISSP), Chapter 8
15		Access Control, Types of Firewalls, Attacks and countermeasures, Application-Level Gateways, Bastion Host, Configuration DMZ Network, Intrusion Detection & Response	External resources
16		Types of Access Controls, Discretionary, Mandatory, Access Control, Bell La Padula, Access Biba	External resources

~~TECHNICAL & BUSINESS WRITING (SETB-317)~~

~~I. Course Details~~

Credit Hours	3 (3+0)
Pre-requisites	FYP
Course Leader	Mr. Safee Ullah Safee
Recommended Textbook(s)	Technical report Writing Today by Daniel G. Riordon-8th edition
Recommended Reference (Books/Websites/Articles)	Technical and Business Writing for working Professionals by Ray E. Hardesty

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Explain the purposes and qualities of reports to state main ideas about reports	Cognitive	2	7	A1, Q1, Midterm
CLO-2	Illustrate different aspects of report writing to comprehend and write effectively	Cognitive	2	7	A2, Q2, Midterm, Final Term
CLO-3	Apply different techniques and principles to prepare and demonstrate a report, paraphrase and summarize	Cognitive	3	7	A3, Q3, Final Term, Presentation

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

~~IV. Grading Policy~~

~~For students admitted in Fall 2021 and onwards~~

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Technical writing-an essential job skills, Basic Concepts about report writing. Types and purposes, qualities, advantages and disadvantages of reports, the flow chart of technical writing process, primary and secondary research, tools of data collection, writing process at work-outlining, paraphrasing, sequencing. Short reports-IMRD reports, Recommendation reports, Proposals-types, purposes and Characteristics Proposals for research reports. Summarizing skills-writing summary, abstract-types and principles process of revising-editing, plagiarism, paraphrasing skills, long reports, aspects and format of FYP(objectives, existing knowledge/review of literature, statement of problem, referencing skills etc.), Plagiarism-definition, types, consequences and how to avoid it.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Technical writing-an essential job skills, types and purposes of reports	[Textbook 3]
2		The research process, primary and secondary research	[Textbook 3]
3	CLO-3	Advantages and disadvantages of report writing The flow chart of Technical writing process	[Textbook 3]
4	CLO-1	Proposals-types and Characteristics	[Textbook 3]
5	CLO-2	Proposals for research reports	[Textbook 3]
6	CLO-1	Writing proposals for solutions/bids for technical projects	[Textbook 3]
7	CLO-2	Short reports-types, IMRD reports and progress reports and format	[Textbook 3]
8	CLO-2	Process of revising-editing, sentence errors	[Textbook 3]
9	Midterm Exams		
10	CLO-3	Effective mechanics: spellings, abbreviations, caps	
11		Writing process at work outlining, paraphrasing, sequencing	
12		Plagiarism-definition, types and techniques to avoid it.	
13		Long reports contents, parts and format	

14		Aspects of FYP	
15		Summarizing skills-writing summary or an abstract Paraphrasing Skills	
16		Final presentations	

SOFTWARE QUALITY ENGINEERING (SESQ-345)

I.Course Details

Credit Hours	3 (3+0)
Pre-requisites	Software Engineering
Course Leader	Dr. Sumaira Nazir
Recommended Textbook(s)	1. Software Quality Assurance, Integrating Testing, Security, and Audit , Abu Sayed Mahfuz, CRC Press 2021. 2. D. Galin, Software Quality: Concept and Practice, D. Galin, John Wiley & Sons, 2018.
Recommended Reference (Books/Websites/Articles)	1. Software Testing, A Craftsman's approach, P. Jorgensen, 5th edition, CRC Press, 2021. 2. Introduction to Software Testing, Paul Ammann and Jeff Offutt, Cambridge University Press, 2017. 3. Introduction to Software Quality, Gerard O'Regan, Springer International, 2014. 4. https://fisip.unpatti.ac.id/wp-content/uploads/2019/08/ISO-9001-2015.pdf

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Explain different software quality concepts and quality management system to state its main ideas	Cognitive	2	2	A1, Q1, Midterm
CLO-2	Compare software quality assurance product and process models to ensure the quality of software products.	Cognitive	2	3	A2, Q2 Midterm, Final Term
CLO-3	Apply quality assurance and testing techniques which are relevant for a particular case to ensure quality of software products.	Cognitive	3	3	A3, Q3, Presentation /Project, Final Term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Software Quality, Quality Engineering, Software Quality Assurance, Software error, fault and failure, causes of software errors, Principles of SQA Benefits of Quality, Software Quality Challenges, Deming's Chain reaction, Software Testing Vs Software Quality Assurance, Software Quality Assurance Vs Quality Control, Software Quality Engineering Function, SQA Process Implementation Activities, Software quality factors and attributes, Software Product Quality Models (MacCall's, Boehm's, ISO/IEC 25010:2011, ISO 9126 etc.), Software Product Quality Models, SQA plan and project plan, Inspections, Review, and Audits IEEE 1028-2008 Software Verification and Validation Method, Software quality metrics, product metrics and process metrics, Goal Question Metric Model, CMM, Cost of Software Quality, Classical Model of Cost of Software Quality, Extended Model of Cost of Software Quality, Quality Management System, Methodologies for Quality Management, ISO:9001:2015. Software testing concepts, Software Testing Life Cycle, Test Maturity Model. Testing strategies/Approaches/Types/Testing Levels: Unit, Integration, system, acceptance testing, Top-down, bottom-up testing, White box, Black box, and grey box testing etc.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Introduction to Software Quality, Quality Engineering, Software Quality Assurance, Software error, fault and failure, causes of software errors, Principles of SQA Benefits of Quality	Chapter 1 [Textbook 1 & 2]
2		Software Quality Challenges, Deming's Chain reaction, Software Testing Vs Software Quality Assurance, Software Quality Assurance Vs Quality Control, Software Quality Engineering Function	Chapter 3 [Textbook 2]
3	CLO-2	SQA Process Implementation Activities: Establishing SQA processes, Coordinating SQA processes with related software processes.	Chapter 6 [Textbook 2]
4			

5		Software quality factors and attributes, Software Product Quality Models (MacCall's, Boehm's, ISO/IEC 25010:2011, ISO 9126 etc.)	Chapter 1 Part II [Textbook 1], Chapter 2 [Textbook 2]
6		SQA plan and project plan: Process of preparing an SQA plan, SQA plan Updating activities. Process of preparing a project plan, Design SQA plan IEEE 730: 2014	Chapter 7 [Textbook 2]
7		Software quality metrics, product metrics and process metrics, Goal Question Metric Model, CMM	Chapter 16 [Textbook 2]
8		Cost of Software Quality, Classical Model of Cost of Software Quality, Extended Model of Cost of Software Quality	Chapter 9 [Textbook 2]
9	Midterm Break		
10		Inspections, Walkthrough, Management review, technical reviews, and Audits IEEE 1028-2008 Software Verification and Validation Method.	Chapter 1 Part IV [Textbook 1],
11		Quality Management System. Methodologies for Quality Management, ISO:9001:2015.	Ref. 4
12		Software testing concepts, Software Testing Life Cycle, Test Maturity Model.	Chapter 4 [Textbook 1]
13	CLO-3	Testing strategies/Approaches/Types/Testing Levels: Unit, Integration, system, acceptance testing, Top-down, bottom-up testing	Chapter 3 [Textbook 1]
14		White box, Black box, and grey box testing	Chapter 14 [Textbook 2]
15		Software Test Planning Process, Testing Documentation, Test Case Design	Chapter 3 [Textbook 1]
16		Presentations	-

~~VISUAL PROGRAMMING (SEVP-364)~~

~~I. Course Details~~

Credit Hours	3 (2+1)
Pre-requisites	Object Oriented Programming
Course Leader	Ms. Saadia Sabbir
Recommended Textbook(s)	1. Deitel, Harvey, and Paul Deitel. Visual C# How to Program. Prentice Hall Press, 6th Edition, (2016) 2. Pro C# 7 With .NET and .NET Core — Eighth Edition — Andrew Troelsen Philip Japikse, 2017. 3. Joseph Albahari, C# 10 in a Nutshell The Definitive Reference. " O'Reilly Media, Inc.", 2022.
Recommended Reference (Books/Websites/Articles)	1. Griffiths, Ian. Programming C# 8.0: Building Cloud, Web, and Desktop Applications. " O'Reilly Media, Inc.", 2020. 2. https://www.nuget.org/packages/Newtonsoft.Json/ 3. https://www.newtonsoft.com/json/help/html/SerializingJSON.html

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Understand the key concepts of visual programming language for generating simple and interactive visual applications.	Cognitive	2	2	A1, Q1, Mid Term
CLO-2	Apply data access and handling concepts in .Net applications using ADO.Net and Entity Framework to enhance problem solving abilities.	Cognitive	3	4	Q2, A2, Mid Term Exam, End Term Exam
CLO-3	Choose an Asp.NET Core to enable applications to exchange data easily and securely using Model View Controller Model.	Cognitive	3	5	Q3, A3, End Term Exam, Project

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	5
Presentation/Project	10
Midterm	25
Final Term	50

Total	100
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IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Introduction to problem solving, a brief review of Von-Neumann architecture, Introduction to programming, role of compiler and linker, introduction to algorithms, basic data types and variables, input/output constructs, arithmetic, comparison and logical operators, conditional statements and execution flow for conditional statements, repetitive statements and execution flow for repetitive statements, lists and their memory organization, multi-dimensional lists, introduction to modular programming, function definition and calling, stack rolling and unrolling, string and string operations, pointers/references, static and dynamic memory allocation, File I/O operations.

VI. Weekly Breakdown

Week No.	CLOs	Topics	Reference
1	CLO-1	Visual Programming Basics, Introduction to Events; Fundamentals of Event-driven Programming Message handling, user interfaces, graphics device interface, Common Controls, Windows Management Introduction to Windows Form Application using VisualStudio. Simple Calculator and Scientific Calculator using Windows Form Application.	Chapter: 14, Chapter 15 [Textbook 1]
2		Database Connectivity of Windows Form application with SQL server. Use of ADO.Net for Insert, Update and Delete Records. Entity Data Model, Querying Database using LINQ to SQL, Data binding, Use of Data Grid View. Retrieving data from multiple tables.	Chapter 21 Chapter 22 [Textbook 1]
3		Designing Centralized Database operations using DbConnclass, Search records using Connection-less approach Using Data Set,	Chapter 22 [Textbook 2]

		Data Table and Data Adapter using Connection-less Approach	
4	CLO-2	Web Applications, Master Page, Web Pages, Introduction to ASP.net Life Cycle, Web Form Application Development Overview of Front-end pages and applying a template using master pages	Online link for Master Pages
5		Dynamic link libraries, Building Class Libraries, Using References, Three-tier architecture including Properties, Presentation Layer, Data Access Layer	Online link for 3-tier architecture
6		Creating Stored Procedures for CRUD operations and calling in visual application. Populating List boxes and Combo boxes with data tables. Using Data Readers with Connection Oriented Approach	Online link for stored procedures
7		Assemblies, Private Assembly, Shared Assembly, Configuration Overview, Reporting mechanism using RDLC reports, Login and Signup using Web Form development, Manage the User Access Level and design appropriate Master page and webpages. Session Management on multiple forms	Chapter 14 [Text Book 2]
8		Local and global variables, Introduction to Arrays and defining Arrays. Arrays initialization, Operations, Passing to functions, Multidimensional Arrays.	Online link for shopping cart
9	Midterm Exams		
10	CLO-3	Entity Framework, Code first approach, Add/Update Model classes, EF Migrations, Retrieve, edit, insert and delete records.	Chapter 22, [Text book 2]
11		Threading and Synchronization, Delegates, Lambda Expressions, State Management, Debugging application, Tracing Event Logs, Using the Boolean Switch and Trace Switch Classes, Using Listeners, and Implementing Custom Listeners	Chapter 21, [Text book 1] Chapter 15 [Text book 2]
12		Model View Controller, Designing classes and properties in Model to generate Database using code first approach Index, Create, edit, delete and Details Actions, Model binding	Chapter 29 [Text book 1]
13		Collection Framework, Use of LINQ to take data from collections and populate on Grid or List View.	Chapter 7 [Text book 3] Chapter 22 [Textbook 1]
14		Load json data from any Web API and use LINQ to JSON to read and display on grid. Save the record to database Table, WCF / Web Service creation, deployment, debugging and calling at client side	Online links Chapter 23 [Text book 2]

15	Dot net Core, Entity Framework Core, Asp.NET Core web applications, applications and request life cycle with MVC and code-first approach	Chapter 31 Chapter 32 Chapter 33 [Text book 2] Online links
16	ASP.net core Web APIs and its applicability for cross platform application, Introduction to Graph Query Language and its applicability, Introduction to Micro Service and its application2] scenarios ,Introduction Containerization(Docker)	Chapter 34 [Text book Online Links

~~VISUAL PROGRAMMING (SEVP-364)- LAB~~

~~I.Course Details~~

Credit Hours	3 (2+1)
Pre-requisites	Object Oriented Programming
Course Leader	Ms. Saadia Shabbir
Recommended Textbook(s)	1. Deitel, Harvey, and Paul Deitel. Visual C# How to Program. Prentice Hall Press, 6th Edition, (2016) 2. Pro C# 7 With .NET and .NET Core — Eighth Edition — Andrew Troelsen Philip Japikse, 2017. 3. Joseph Albahari, C# 10 in a Nutshell The Definitive Reference. " O'Reilly Media, Inc.", 2022.
Recommended Reference (Books/Websites/Articles)	1. Griffiths, Ian. Programming C# 8.0: Building Cloud, Web, and Desktop Applications. " O'Reilly Media, Inc.", 2020. 2. https://www.nuget.org/packages/Newtonsoft.Json/ 3. https://www.newtonsoft.com/json/help/html/SerializingJSON.html

~~II.Lab Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Demonstrate the task by interpreting, summarizing, and explaining key concepts/principles/relationships within the domain, enabling them to apply acquired knowledge effectively in diverse contexts.	Cognitive	2	2	Lab Task, Lab Report, Final Exam, Semester Project
CLO-2	Practice theoretical concepts for conducting the experiments/projects of varying complexities to achieve desired outcomes.	Psychomotor	3	4	Lab Task, Semester Project
CLO-3	Contribute actively and responsibly individually or in teams during laboratory activities, demonstrating a positive attitude and willingness to participate.	Affective	2	6	Lab Task, Semester Project

III. Lab Assessment

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25
Total	100

IV. Weekly Breakdown (All CLOs shall be assessed in every lab)

Week No.	Topics
1	Create a child class and parent class for calculator and scientific Calc, Create windows forms with Menus, buttons, etc. Create Account class with public and private variables and props, Design Windows to get values and store in props and pass
2	Database Connectivity of Windows Form application with SQL server. Use of ADO.Net for Insert, Update and Delete Records. Entity Data Model, Querying Database using LINQ to SQL, Data binding, Use of Data Grid View. Retrieving data from multiple tables.
3	Develop an employee page to perform Insert and Search the records, Design database in SQL server, Designing Centralized Database operations using DbConn class, Search records using Connection-less approach, Using Data Set, Data Table and Data Adapter using Connection-less Approach
4	Web Applications, Master Page, Web Pages, Introduction to ASP.net Life Cycle, Develop a MasterPage in ASP.net, Apply AJAX and provide the data to web pages. Add pages in different Master Pages
5	Dynamic link libraries, Building Class Libraries, Using References, Three-tier architecture including Properties, Presentation Layer, Business Layer, Data Access Layer
6	Create a stored procedure and execute it in sql server to proof results of execution, Write code behind insert button to call stored procedure
7	Create an rdlc file, add table and select columns, design header and footer of report. Write code behind view report button to display list of all employees and print it.
8	Using Data Readers using Connection Oriented Approach, Add functionality for Shopping Cart along with Add to Cart, View cart and delete item
9	Midterm Exam
10	Entity Framework, Code first approach, Add/Update Model classes, Retrieve, edit, insert and delete records.
11	Threading and Synchronization, Using Listeners, and Implementing Custom Listeners
12	Create a view class to show the results, Create controller class to utilise model and view, finally create a web form to utilize mvc.

13	LINQ to Objects, LINQ to SQL, LINQ to DataSet, LINQ to XML, JSON to connect with XML and JSON based Data, like in Mongo DB etc. Load JSON from Web API, Save the record to database Table WCF
14	Open Ended Lab
15	Dot net Core, Entity Framework Core, Asp.NET Core web applications, applications and request life cycle with MVC and code-first approach
16	ASP.net Core Web APIs and its applicability for cross-platform applications, Introduction to Graph Query Language and its applicability

~~MOBILE APPLICATION DEVELOPMENT (SEMA-366)~~

~~I. Course Details~~

Credit Hours	3 (2+1)
Pre-requisites	Object Oriented Programming
Course Leader	Ms. Saadia Shabbir
Recommended Textbook(s)	1. Professional Android Application Development, Reto Meier, Wrox Programmer to Programmer, 3rd Edition (2014) 2. iOS Programming: The Big Nerd Ranch Guide, Conway, J., Hillegass, A., & Keur, C., 5th Edition (2015)
Recommended Reference (Books/Websites/Articles)	Android Programming: The Big Nerd Ranch Guides, Phillips, B. & Hardy, B., 3rd Edition (2017)

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Understand the key concepts of Android application development for generating creative and efficient mobile applications.	Cognitive	2	2	A1, Q1, Mid Term
CLO-2	Construct mobile applications layout to optimize user interaction.	Cognitive	3	4	A2, Q2, Mid Term, Final Term
CLO-3	Create mobile applications for comprehensive systems using latest trends and practices to address complex user needs.	Cognitive	6	5	A3, Q3, Project, Final Term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

~~Mobiles Application Development Platform, Android OS: Architecture, Framework and Application Development; iOS: Architecture, Framework; Application Development with Windows Mobile: Eclipse; Fragments; Calling Built-in Applications using Intents; Displaying Notifications; Components of a Screen; Adapting to Display Orientation; Managing Changes to Screen Orientation; Utilizing the Action Bar; Creating the User Interface; Listening for UI Notifications; Views; User Preferences; Persisting Data; Sharing Data; Sending SMS Messages; Getting Feedback; Sending E-mail; Displaying Maps; Consuming Web Services Using HTTP; Web Services: Accessing and Creating; Threading; Publishing, Android Applications; Deployment on App Stores; Mobile Programming Languages; Challenges with Mobility and Wireless Communication; Location-aware Applications; Performance/Power Tradeoffs; Mobile Platform Constraints; Emerging Technologies.~~

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Mobile Application Development Platforms, System installation (Android Studio)	Chapter 1 [Textbook 1]
2		Android Architecture and framework, What is Android The various Android devices on the market, The Android Market application store, How to develop your first Android application	Chapter 2 [Textbook 1]
3		Components of Android Project (Application Structure) Using code Completion, Debugging Application, What activities are?	Chapter 2 [Textbook 1]
4	CLO-2	Android Studio Layout editor, component tree, Constraint view, layout properties, XML files (string, colors, values, dimen) in the project, Resource folders	Chapter 3 [Textbook 1]
5		Activities, Layouts, layout attributes, Views, View Groups, layout Orientation	Chapter 4 [Textbook 1]
6		Intents in Android	Chapter 5 [Textbook 1]
7		Calling build in intents, communication between activities	Chapter 5 [Textbook 1]

8		Fragments, Display Notifications	Chapter 4, 10 [Textbook 1]
9	Midterm Exams		
10	CLO-2	Design interface with views, Displaying Pictures and menus with views	Chapter 7, 8 [Textbook 1]
11	CLO-3	Data Persistence, Content providers	Chapter 8 [Textbook 1]
12		Messages (SMS, EMAIL), Location based services (Maps)	Chapter 5, 13 [Textbook 1]
13		Networking, Establishing an HTTP Connection, Consuming Web Services using HTTP, Consuming JSON Services	Chapter 9 [Textbook 1]
14		Android Services, Create Own Services, Establishing Communication between a Service and an Activity Binding Activities to Services, Understanding Threading	Chapter 9 [Textbook 1]
15		Introduction to iOS Architecture and framework, Introduction to Xamarin/Flutter for cross platform development	Chapter 1 [Textbook 2]
16		Publishing, Android Applications, Mobile Programming Languages, Emerging Technologies.	Chapter 19 [Textbook 1]

MOBILE APPLICATION DEVELOPMENT (SEMA-366) – LAB

I.Course Details

Credit Hours	3 (2+1)
Pre-requisites	Object Oriented Programming
Course Leader	Ms. Saadia Shabbir
Recommended Textbook(s)	1. Professional Android Application Development, Reto Meier, Wrox Programmer to Programmer, 3 rd Edition (2014) 2. iOS Programming: The Big Nerd Ranch Guide, Conway, J., Hillegass, A., & Keur, C., 5th Edition (2015)
Recommended Reference (Books/Websites/Articles)	Android Programming: The Big Nerd Ranch Guides, Phillips, B. & Hardy, B., 3rd Edition (2017)

II. Lab Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Demonstrate the task by interpreting, summarizing, and explaining key concepts/principles/relationships within the domain, enabling them to apply acquired knowledge effectively in diverse contexts	Cognitive	2	2	Lab Task, Lab Report, Final Exam, Semester Project
CLO-2	Practice theoretical concepts for conducting the experiments/projects of varying complexities to achieve desired outcomes.	Psychomotor	3	4	Lab Task, Semester Project
CLO-3	Contribute actively and responsibly individually or in teams during laboratory activities, demonstrating a positive attitude and willingness to participate.	Affective	2	6	Lab Task, Semester Project

III. Lab Assessment

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25

Total	100
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II. ~~Weekly Breakdown (All CLOs shall be assessed in every lab)~~

Week No.	Topics
1	Install android studio
2	the layout editor, design interactive ui – edit text, text views, buttons
3	intents in android
4	activity lifecycle and state
5	implicit intents
6	fragments, display notifications
7	user interactions, user navigation
8	recycler view
9	Midterm Exams
10	Open ended lab
11	menus and pickers
12	preference and settings
13	storing data with sqlite database
14	broadcast receivers
15	networking, establishing an http connection, consuming web services using http, consuming json services (part-1)
16	networking, establishing an http connection, consuming web services using http, consuming json services (part-2)

~~USER EXPERIENCE DESIGN (SEUS-366)~~

~~I.Course Details~~

Credit Hours	3(2+1)
Pre-requisites	Human Computer Interaction
Course Leader	Dr. Marriam Daud
Recommended Textbook(s)	1. The Design of Everyday Things, Don Norman, Revised Version, Basic Books, 2013 2. Android UI Design by Jessica Thornsby, 2016 3. D. Benyon, Designing Interactive Systems: A Comprehensive Guide to HCI, UX and Interaction Design, 4th edition, Pearson, 2019.
Recommended Reference (Books/Websites/Articles)	Human Computer Interaction by Alan Dix, 3rd edition, 2004

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Explain the user-centered requirement gathering techniques by stating its main ideas.	Cognitive	2	2	Q1, A1, Midterm
CLO-2	Apply UX guidelines for designing web and mobile user interfaces.	Cognitive	3	4	Q2, A2, Midterm, Final term
CLO-3	Analyze usability evaluation of user interfaces by following the standard methods.	Cognitive	4	4	Q3, A3, Presentation / Project, Final term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Background to Human-computer interaction. Underpinnings from psychology and cognitive science, More background. Evaluation techniques: Heuristic evaluation, More evaluation techniques: Videotaped user testing; cognitive walkthroughs, Task analysis. User-centered design, Conceptual models and metaphors, Usability engineering processes; conducting experiments, designing interface of mobile, desktop, and special devices: Coding techniques using color, fonts, sound, animation, etc. Screen layout, response time, feedback, error messages, etc. Internationalization, help systems, etc. User interface software architectures, Expressing design rationale for user interface design.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Introduction to different types of users and data, Terms and concepts, features of good design, model of interaction	Chapter 1 Study resource
2		Requirement gathering techniques – 1. Naturalistic observation, 2. Surveys, 3. Focus groups	Chapter 7 [Textbook 3]
3		Requirement gathering techniques – 4. Interviews, Presenting the results, Participant observation, interviewing writing techniques, additional need finding techniques.	Chapter 7 [Textbook 3]
4	CLO-2	Creating design goals, review design goals, design alternatives, Comparing alternates	Chapter 12 [Textbook 3]
5		Visual designing, typography, grids and navigations, reading and navigation.	Chapter 12 [Textbook 3]
6		Type of mobile Apps, mobile screen types, enhance mobile interaction, step to make powerful mobile UI	Chapter 1 [Textbook 2]
7		Representation matters, Distributing cognition.	Study resource
8		Design principles and guidelines for mobile and web interface	Chapter 2 [Textbook 2]
9	Midterm Exams		

10	CLO-2	Android UI layout	Chapter 2 [Textbook 2]
11	CLO-3	Prototyping basics, intro to Balsamic prototyping tool	Chapter 7 [Textbook 2]
12		Story boarding, Paper/video prototyping	Chapter 8 [Textbook 3]
13		Faking using Wizard of OZ, Evaluation basics -	Chapter 6
14		Design Evaluation Heuristics — Schneiderman, Nelson, Jakob Nielsen	Chapter 4 [Textbook 2]
15		Heuristic evaluation and usability testing	Chapter 9 Alan Dix
16		Testing a user interface	Chapter 6 [Textbook 2]

~~USER EXPERIENCE DESIGN (SEUS-366) – LAB~~

~~I.Course Details~~

Credit Hours	3(2+1)
Pre-requisites	Human Computer Interaction
Course Leader	Dr. Marriam Daud
Recommended Textbook(s)	1. The Design of Everyday Things, Don Norman, Revised Version, Basic Books, 2013 2. Android UI Design by Jessica Thornsby, 2016 3. D. Benyon, Designing Interactive Systems: A Comprehensive Guide to HCI, UX and Interaction Design, 4th edition, Pearson, 2019.
Recommended Reference (Books/Websites/Articles)	Human Computer Interaction by Alan Dix, 3rd edition, 2004

~~II. Lab Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Demonstrate the task by interpreting, summarizing, and explaining key concepts/principles/relationships within the domain, enabling them to apply acquired knowledge effectively in diverse contexts.	Cognitive	2	2	Lab Task, Lab Report, Final Exam, Semester Project
CLO-2	Practice theoretical concepts for conducting the experiments/projects of varying complexities to achieve desired outcomes.	Psychomotor	3	4	Lab Task, Semester Project
CLO-3	Contribute actively and responsibly individually or in teams during laboratory activities, demonstrating a positive attitude and willingness to participate.	Affective	2	6	Lab Task, Semester Project

~~III. Lab Assessment~~

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25

Total	100
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IV. Weekly Breakdown (All CLOs shall be assessed in every lab)

Week No.	Topics
1	Learning how to Identify the problem space and choosing the briefs
2	How to prepare for survey and develop Interview and questionnaires
3	Defining the need for the problem space, understanding the examples
4	Understanding the point of view and identifying inspiration,
5	Understanding tools for developing prototype
6	What are storyboards and how to create storyboards?
7	Understanding Balsamic and Web flow for developing prototype
8	How to pick a concrete direction and Create 2 rapid interactive prototypes for web and mobile design interactive prototypes
9	Midterm Exams
10	Discussion on creating rapid interactive prototypes and showing examples from the Internet.
11	How to Conduct a Heuristic Evaluation
12	How to prepare a development plan and going Online and hosting Home screen
13	Examples of home screen and development plan
14	How to develop user evaluation plan
15	Guide to Completing the development plan
16	Guide to Get ready for testing, How to create test materials and Making User tests and analysis findings

~~SOCIAL SERVICE (SESS-367)~~

~~I. Course Details~~

Credit Hours	1 (1+0)
Pre-requisites	-
Course Leader	Mr. Zohaib Ahmad
Recommended Textbook(s)	Social Work and Social Policy. An Introduction, Jonathan Dickens, 2nd edition, Taylor and Francis, 2016.
Recommended Reference (Books/Websites/Articles)	1. Modern Social Work Theory, Malcolm Payne, 4th edition, Palgrave Macmillan, 2016. 2. Social Work Fields of Practice, Catherine N. D. & Karen M.S, John Wiley & Sons, 2012. 3. Social Work: A Beginner's Text written, Andrea Bernstein & Mel Gray Juta, 1997.

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Define knowledge of social work theories, principles and concepts related to social work.	Cognitive	1	10	Midterm
CLO-2	Discuss societal, health, poverty, safety, legal issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.	Cognitive	2	8	Midterm, Final Term
CLO-3	Demonstrate social work theories and practices to address real societal issues.	Cognitive	3	8	Midterm, Final Term, Project (Proposal, Presentation, Final Report)

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Project (Proposal, Presentation, Final Report)	25
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

This course introduces students to get the understanding about social work. It will be helpful for them to get the insight about fundamental norms and value system of the society and teach them to be empathetic about society and individual problems. The course explores social welfare through social work and human services lenses. This includes their shared values, historical foundations, and critical perspectives. Once some approaches and theories, practices, processes, and levels have been introduced, students develop a personal philosophy statement of social welfare that self-assesses their fit in these fields. The students learn to apply their foundational learning to specific populations. At the students integrate material from the entire course and begin applying it to specific scenarios.

VI. Weekly Breakdown

Week No.	CLOs	Topics	Reference
1	CLO-1	Introduction and Philosophical background of Social Work, Use of models: social work diamond, the social construction of social work theory.	Chapter 1
2		Principles of Social Work, Voluntary and Involuntary Social Work, Professional Social Work, Social Policy, social policy triangle.	Chapter 2
3	CLO-2	Modern Concepts of Social Work: Rehabilitation, Stress Management, Preventive etcetera. Role of State	Chapter 3
4		Primary and Secondary Research Methods in Social Work, Defining social needs.	Chapter 4
5		Field Work in Social Work and Field Ethics, Values & Ethics of the social work profession, Code of Ethics	Chapter 4
6		Exploration of specific arenas of practice, including key challenges, populations, and setting. Poverty.	Chapter 5

7		Rights, types of human rights, human right Act. A debate between rights and duties.	Chapter 6
8		Aspects of inequality, Slogan under Human Rights Agenda.	Chapter 7
9	Midterm Exams		
10	CLO-2	Problem Identification, Funding and spending for social work	Chapter 12
11			
12	CLO-3	Case work example: Crisis and task-centered practice	Chapter 5 [Ref Book 1]
13		Case work example: Cognitive-behavioral practice	Chapter 6 [Ref Book 1]
14		Working with Special Populations	-
15		Working with Special Populations	-
16		Project Presentations	-

~~Semester VII~~

SOFTWARE RE-ENGINEERING (SESR-448)

I.Course Details

Credit Hours:	3 (3+0)
Pre-requisites:	Software Construction and Development
Instructor:	Dr. Jaweria Kanwal
Recommended Book(s):	Software Evolution and Maintenance: A Practitioner's Approach by Priyadarshi Tripathy, Kshirasagar Naik, John Wiley & Sons, 2015
Reference Books:	1. Re-engineering legacy software, David Lorge Parnas, Chris Birchall, Safari Books, Shelter Island, NY, 2016 2. Reengineering, Priyadarshi Tripathy and Kshirasagppar Naik, John Wiley & Sons, Inc.2015 3. Software Maintenance and Evolution: a Roadmap, K.H.Bennett and V.T Rajlich, The Future of Software Engineering, ACM Press 2000.

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment Artifact
CLO-1	Explain the concepts and techniques of software reengineering to support software evolution	Cognitive	2	2	A1, Q1, Midterm
CLO-2	Demonstrate reengineering techniques to analyze evolution related issues of software systems	Cognitive	3	2	A2, Q2, Midterm, Final Term
CLO-3	Analyze reengineering techniques to upgrade and modify legacy software systems	Cognitive	4	3	A3, Q3, Final Term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Salient topics include the terminologies pertaining to legacy systems, software aging, software evolution, taxonomy of software maintenance and evolution, software evolution and maintenance models, fundamental re-engineering techniques to modernize legacy systems including, general model for software reengineering, software reengineering strategies, variations, process, approaches, source code reengineering reference model, code reverse engineering, reverse engineering tools, re-engineering economics, legacy information systems, wrapping, migration, impact analysis, impact analysis process, analysis of traceability graph, refactoring, refactoring process activities, program comprehension, reuse and domain engineering

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO1	Introducing basic concepts and preliminaries, for instances legacy systems, software aging, software evolution and maintenance, re-engineering	Chapter 1
2	CLO1	Taxonomy of Software Maintenance and Evolution: Intention-Based Classification of Software Maintenance, Activity-Based Classification of Software Maintenance, Categories of Maintenance Concepts, Maintained Product, Maintenance Types, Maintenance Organization Processes, Peopleware, Evolution of Software Systems,	Chapter 2
3	CLO2	Evolution and Maintenance Models: Reuse-Oriented Model, Staged Model for Closed Source Software, Staged Model for Free, Libre, Open-Source Software, Change Mini-Cycle Model, IEEE/EIA, Maintenance Process, ISO/IEC 14764 Maintenance Process	Chapter 3
4	CLO2	Fundamental re-engineering techniques to modernize legacy systems: reengineering concepts, a general model	Chapter 4

		for software reengineering, Horse shoe model of software reengineering, reengineering approaches.	
5	CLO2	Fundamental re-engineering techniques to modernize legacy systems: Source Code reengineering reference model, Phase reengineering model.	Chapter 4
6	CLO2	Fundamental re-engineering techniques to modernize legacy systems: Code Reverse Engineering, Techniques Used for Reverse Engineering, Lexical Analysis, Syntactic Analysis, Control Flow Analysis, Data Flow Analysis, Program Slicing, Visualization, Program Metrics	Chapter 4
7	CLO2	Fundamental re-engineering techniques to modernize legacy systems: De-compilation Versus Reverse Engineering, Data Reverse Engineering, Reverse Engineering Tools,	Chapter 4
8	CLO2	Re-engineering Economics	https://doi.org/10.1002/smr.4360030304
9	Midterm Exam		
10	CLO2	Legacy Information System: general concepts, Wrapping, types of wrapping, Levels of Encapsulation, constructing a Wrapper, Adapting a Program for Wrapper	Chapter 5
11	CLO2	Legacy Information System: Migration process and phases, Migration Planning	Chapter 5
12	CLO3	Impact Analysis: Impact Analysis Process (Identifying the SIS, Analysis of Traceability Graph, Identifying the Candidate Impact Set), Traceability links, Dependency-Based Impact Analysis	Chapter 6
13	CLO3	Introducing Refactoring, Activities in a Refactoring Process, Types of Refactoring, Determine Which Refactoring Should be Applied	Chapter 7
14	CLO3	Ensure that Refactoring Preserves the Behavior of the Software, Apply the Refactoring to the Chosen Entities, Evaluate the Impacts of the refactoring on Quality, Maintain Consistency of Software Artifacts	Chapter 7
15	CLO3	Program Comprehension: Understanding Code, Cognition Models for Program Understanding such as Letovsky Model, Shneiderman and Mayer Model	Chapter 8
16	CLO3	Program Comprehension: Protocol Analysis, Visualization for Comprehension	Chapter 8
17	CLO3	Reuse and Domain Engineering: Benefits of Reuse, Reuse Models, Factors Influencing Reuse, Success Factors of Reuse, Economic Models of Software Reuse.	Chapter 9

SOFTWARE PROJECT MANAGEMENT (SESP-447)

I. Course Details

Credit Hours	3 (3+0)
Pre-requisites	Software Engineering
Course Leader	Dr. Sumaira Nazir
Recommended Textbook(s)	Information technology project management, Kathy Schwalbe 9 th Edition, Cengage, 2019
Recommended Reference (Books/Websites/Articles)	Mastering software project management: Best practices, tools and techniques Chemuturi, M. & Cagley, T. M. (2010)

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Explain fundamental concepts of project management knowledge areas and process groups to state their main idea.	Cognitive	2	2	A1, Q1, Midterm
CLO-2	Demonstrate the ability to compare software project management process areas and knowledge areas	Cognitive	2	3	A2, Q2-Midterm, Final Term
CLO-3	Apply acquired knowledge of software project management knowledge areas and process groups to develop various software project management plans on small-scale project.	Cognitive	3	3	A3, Q3, Presentation /Project, Final Term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Introduction to Project Management. The Project Management Process Groups. Project Integration Management. Project Scope Management. Project Time Management. Project Cost Management. Project Quality Management. Project Human Resource Management. Project Communications Management. Project Risk Management. Project Procurement Management. Controlling and tracking

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Course introduction, software crises, project failure, what is a project, what is project management, Program and Project Portfolio Management, Role of project manager, project management ethics and certifications, project management profession	Chapter 1 [Textbook]
2		Software project acquisition from external client (RFP, proposal, negotiation, contract acceptance) Software project acquisition from internal client (feasibility study, preparing and finalizing proposal)	Chapter 3 [Ref. Book]
3		PMI Framework, Project management process groups, Mapping the Process Groups to the Knowledge Areas, Case study (JWD consultancy intranet site project) of project management process groups	Chapter 3 [Textbook]
4		Software project initiation, project management office level activities, software project manager level activities. Common pitfalls in project initiation	Chapter 4 [Ref. Book]
5	CLO-2	What is Project Integration Management, Strategic Planning and Project Selection, Methods for Selecting Projects, Developing a Project Charter, Developing a Project Management Plan, Monitoring and Controlling Project Work	Chapter 4 [Textbook] chapter 5 [Ref. Book]

6		Project Scope Management, Planning Scope Management, collecting requirements, Defining Scope Creating the Work Breakdown Structure, Validating Scope Controlling Scope	Chapter 5 [Textbook] Chapter 5 [Ref. Book]
7		The Importance of Project Schedule, Planning Schedule Management, Defining Activities, Sequencing Activities, Estimating Activity Resources and Activity Durations (PERT, CPM), Developing the Schedule, Controlling the Schedule,	Chapter 6 [Textbook] Chapter 5 and 9 [Ref. Book]
8		Project Cost Management, Basic Principles of Cost Management, Planning Cost Management, Estimating Costs, controlling costs	Chapter 7 [Textbook]
9	Midterm Exams		
10	CLO-3	Project Quality Management, The Importance of Project Quality Management ,Planning Quality Management Performing Quality Assurance, Controlling Quality , Tools and Techniques for Quality Control	Chapter 8 [Textbook]
11		What is Project Human Resource Management, Keys to Managing and Leading People, Developing the Human Resource Plan, Acquiring the Project Team, developing the Project Team, managing the Project Team	Chapter 9 [Textbook] Chapter 10 [Textbook]
12			
		The Importance of Project Communications Management, Keys to Good Communications, Planning Communications Management ,Managing Communications, Controlling Communications	
13		The Importance of Project Risk Management Planning Risk Management, Identifying Risks, Performing Qualitative Risk Analysis, Performing Quantitative Risk Analysis, controlling risks	Chapter 11 [Textbook]
14		Planning Procurement Management, Conducting Procurements, Controlling Procurements	Chapter 12 [Textbook]
15		Project tracking and control, earned value analysis	Chapter 7 [Ref. Book]
16		Project closure (best practices, lessons learnt, project post mortem, releasing the team, releasing the contracts)	Chapter 10 [Ref. Book]

~~OPERATIONS RESEARCH (SEOR-468)~~

~~I. Course Details~~

Credit Hours:	3 (3+0)
Pre-requisites:	Calculus & Analytical Geometry, Linear Algebra
Instructor:	Dr. Asad Husain
Recommended Textbook(s)	Operations Research-An Introduction, Hamdy A. Taha, 10th Ed. Pearson Education, 2016
Recommended Reference (Books/Websites/Articles)	1-Introduction to Operations Research, B.E Gillett, Tata McGraw Hill Publishing Company Ltd., New Delhi, 1979 2-Operations Research, F.S Hillier and G. J. Liebraman, CBS Publishers and Distributors, New Delhi, 1974

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment Artifact
CLO-1	Explain the basic concept of optimization, modeling and linear modeling (LP)	Cognitive	2	1	A1, Q1, Midterm
CLO-2	Construct LP models for various types of problems and solve a problem modeled with LP	Cognitive	5	2	A2, Q2, Midterm, Final Term
CLO-3	Distinguish the feasible solution, optimal and basic feasible solution make the duality and sensitivity analysis of the LP model	Cognitive	4	2	Q3, Project, Final Term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Introduction to Linear Programming (Operations Research) with examples, Graphical solutions to Mathematical Model with Special Cases, Simplex Algorithm and its different cases, Big M Method and Two-phase Method, Scheduling and Blending Problems, The transportation Problems, The Transshipment Problems, The Assignment Problems, integer Programming, network Models, Inventory Models, Dynamic Programming and Queuing Theory. The geometric method for solving a linear program, Simplex method, two-phase method. Degenerate solution revised simplex method, Dual linear optimization problems and sensitivity analysis, Applications in transport and network analysis.

VI. Weekly Breakdown

Week No.	CLO	Topics
1	CLO-1	Introduction of OR, Meaning and Definition of OR, Features or Characteristic of OR, Scope of OR, Limitation of OR
2		Linear Programming (LP), Modelling with LP, Two-variable LP model, Graphical Solution of LP and Special Cases
3		LP Model in Equation Form, Transition from Graphical to Algebraic Solution with Examples
4	CLO-2	Iterative Nature of the Simplex Method, Computational Details of the Simplex Algorithm, Summary of the Simplex Method
5		Different Cases of Simplex Method including Unbounded, Infeasible, Multiple and Degeneracy with Examples
6		Two-Phase Method with Applications for Two-variables and Three-variables LP Models
7		Big M Method with Applications for Two-variables and Three-variables LP Models

8	CLO-3	The Transportation Problems (TPs), Solutions via North-West Corner, Least-Cost and Vogel's Approximation Method
9	Midterm Exams	
10	CLO-3	Optimal Solutions of TPs via MODI/U-V Method, Unbalanced Problem, Degeneracy, Transshipment Problems
11		Assignment Problems, Unbalanced Assignment Problems, Hungarian Method, Integer Linear Programming
12		Application of Integer Linear Programming Using Branch and Bound Method, Gomory's Cutting Plane Methods
13		Network Models and Applications Via Minimal Spanning Tree and Shortest-Route Algorithms
14		Analysis and Applications of Maximal Flow and Critical Path Algorithms
15		Revision, Discussion
16		Project Presentations

~~FINAL YEAR PROJECT (SEFP-499)~~

~~I.Course Details~~

Credit Hours:	6
Pre-requisites:	Programming Fundamentals, OOP, VP, Web Programming, DBMS
Course Leader	Dr. Jaweria Kanwal
Recommended Textbook(s)	
Recommended Reference (Books/Websites/Articles)	As prescribed by the teacher depending on the project's domain

~~II. Course Learning Outcomes~~

CLOs	Description	Domain-Taxonomy level	PLOs	Assessment Artifacts
CLO-1	Identify the problem or deficiency in the existing software product, system or concept	C-2	1.Academic knowledge	Proposal presentation /Proposal report
CLO-2	Apply knowledge of computing to formulate a problem statement and requirements (functional and non-functional) based on research literature and existing systems.	C-3	2. Knowledge for Solving Computing Problems	Proposal presentation /Proposal report
CLO-3	Propose the different components of a software project and model requirements to generate system design based on the proposed components.	C-5	3: Problem Analysis	Progress presentation / Progress report
CLO-4	Construct the proposed software project using suitable tools based on the appropriate algorithms/APIs/DB schema	P-5	5: Modern Tool Usage	Final report/ Prototype
CLO-5	Produce test results (based on test cases) of various modules of the developed project	P-4	4: Design/ Development of Solutions	Final report

CLO-6	Justify the application of the proposed solutions for the betterment of Society	C-6	8: Computing Professionalism and Society	Presentation
CLO-7	Demonstrate development of solutions effectively through written and oral mode with the aid of multimedia tools.	A-4	7: Communication	Project code/Presentation
CLO-8	Demonstrate the originality and relevance of document's content to the project	A-3	9: Ethics	Final report
CLO-9	Show professional team coordination and communication	A-3	6: Individual and Team Work	Presentation
CLO-10	Demonstrate and defend understanding of tools and techniques used in the development of the developed project	C-3	10: Life-long Learning	Presentation

III. ~~Marks weightage of CLOs according to Rubrics~~

CLOs	Project Proposal (40)	Project Progress (40)	Final Project (90)	Supervisor evaluation (30)
CLO-1	4			
CLO-2	12	6		
CLO-3	2	6		
CLO-4	4	16	8	
CLO-5			44	
CLO-6	4			
CLO-7	8	8	8	5
CLO-8	4	2	6	10
CLO-9	2	2	8	10
CLO-10			16	5

IV. ~~Grading Policy~~

~~For students admitted in Fall 2021 and onwards~~

Grade	A+	A	B+	B	C+	C	D+	D	F
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%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

~~V. Catalogue description~~

~~This course is designed as finale project which requires students to demonstrate technical, theoretical and presentation skills at levels which are aligned with professional software engineering practices. It is needed that students will apply their knowledge of computing throughout the course such as development of requirements, design and implementation to develop a software solution to a real-world problem from conception to completion. In this part, students propose a practically useful project, and develop a document which consists of project plan and software design document.~~

TECHNICAL ENTREPRENEURSHIP (SETE-469)

I. Course Details

Credit Hours	2 (2+0)
Pre-requisites	-
Course Leader	Dr. Nargis Fatima
Recommended Textbook(s)	Entrepreneurship: Successfully launching new ventures, Bruce R. Barringer, 4th Edition, Pearson, 2012.
Recommended Reference (Books/Websites/Articles)	1. Innovation and entrepreneurship, Peter F. Drucker, Harper Collins, 2006. 2. Patterns of entrepreneurship, Jack M Kaplan, Wiley, 2006. 3. Entrepreneurship-A recipe for economic development, Naqi, Dr. S. M, 2002. 4. Entrepreneurship-A contemporary approach, Kuratko Donald F, Hodgetts, Richard M, Thomas Learning, 2001.

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Define the core concepts of entrepreneurship and entrepreneurial processes to state their main idea.	Cognitive	1	1	A1, Q1, Midterm
CLO-2	Explain the impact of a sustainable engineering solution in societal and environmental context.	Cognitive	2	8	A2, Q2, Midterm, Final Term
CLO-3	Apply concepts of entrepreneurship and entrepreneurial processes to design an appropriate business plan and select suitable business model.	Cognitive	3	3	A3, Q3, Final Term, Project/Presentation

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25

Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Entrepreneurship Introduction, Meanings/Definition, Nature, Features, Scope, Pros and Cons, Commercialize New Ideas, Differences and Similarities between a Businessman and an Entrepreneur, Competitive Advantage of organizations having Entrepreneurial Edge, identifying and validating good opportunities and then creating, communicating, and capturing value from those opportunities over time, Firms in corporate and non-profit settings, Real World Examples, Prospects for Entrepreneurs in Pakistan, Need for Entrepreneurial Education and Activities.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Introduction to Entrepreneurship, Nature and Importance of Entrepreneurship, Myths about Entrepreneurship	Chapter 1
2		Types of entrepreneurial ventures, Process of entrepreneurship	Chapter 1
3	CLO-2	Developing Successful Business Ideas, Recognizing Opportunities and Generating Ideas, Finding gaps	Chapter 2
4		Techniques for generating ideas and the process of generating creative ideas.	Chapter 3,4
5		Industry and Competitor Analysis, Industry Trends	Chapter 5
6		The Five Forces Model and Competitor Analysis	Chapter 5
7		Writing a Business Plan, The Business Plan and prepare Outline of the Business Plan	Chapter 6

8		Presenting the Business Plan to Investors	Chapter 8
9	Midterm Exams		
10	CLO-3	Building a New Venture Team and Creating New Venture Team	Chapter 9
11		Rounding Out the Team, Customer Advisory Board	Chapter 10
12		Getting Funding or Financing, The Importance of Funding or Financing and Sources of Equity Funding	Chapter 10
13		Sources of Debt Financing, Creative Sources of Financing	Chapter 11
14		The Importance of Intellectual Property, Patents and TradeMarks	Chapter 12
15		Copyrights and Trade Secrets	Chapter 12
16		What is franchising and how does it work?, Establishing a franchise system, Buying a franchise	Chapter 15

~~WEB ENGINEERING II (SEWE-470)~~

~~I.Course Details~~

Credit Hours	3 (2+1)
Pre-requisites	Web Engineering
Instructor	Ms. Saadia Shabbir
Recommended Textbook(s)	Web Development with JQuery, Richar York, Wiley, 2015
Recommended Reference (Books/Websites/Articles)	1. PHP Web Services, O'Reilly Media, Inc, (2nd edition) 2013 2. The Road to Learn React: Your Journey to Master Plain Yet Pragmatic React. Js, Robin Wieruch, (2020 edition), 2020 3. https://react.dev/blog/2023/03/16/introducing-react-dev 4. https://www.w3schools.com/nodejs/

~~II.Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Understand asynchronous programming concepts for improving user interactions within web applications.	Cognitive	2	2	A1, Q1, Mid Term
CLO-2	Make use of React JS concepts for front end development.	Cognitive	3	4	A2, Q2, Mid Term, Final Term
CLO-3	Develop dynamic web applications by using MERN components to achieve specified functionalities.	Cognitive	3	4	A3, Q3, Project, Final Term

~~III.Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

AJAX, Web services, React JS, Mongo DB, Express, Node.js

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO 1	Course Outline, Assessment Criteria, Introduction to Advanced Web Course, Intro to basic components.	
2		Intro to AJAX, Pagination with AJAX	Chapter 7 [Ref. Book 1]
3		Asynchronous Data Fetching (Using fetch API for making HTTP requests, Handling responses and errors, Working with JSON data)	Chapter 8 [Ref. Book 1]
4	CLO 2	Intro to React, Fundamentals of React	Chapter 1 [Ref. Book 2]
5		Data handling with React: Data fetching	Chapter 1 [Ref. Book 2]
6		Data handling with React: React States	Chapter 1 [Ref. Book 2]
7		React UI Styling using States	Chapter 3 [Ref. Book 2]
8		Working with React API Fetch()	Chapter 1 [Ref. Book 2]
9	Midterm Exams		
10	CLO 2	Working with React Routers	Chapter 1 [Ref. Book 2]
11	CLO 3	Introduction to MongoDB	https://docs.mongodb.com/manual/introduction/

12		Working with Data in MongoDB	https://docs.mongodb.com/manual/tutorial/getting-started/
13		Introduction to Express	https://expressjs.com/en/starter/installing.html
14		Express Middleware	https://expressjs.com/en/guide/writingmiddleware.html
15		Intro to Node.js, concepts, React Testing	https://www.w3schools.com/nodejs/
16		Creating Sample CRUD Application with MERN stack	

WEB ENGINEERING II (SEWE-470) – LAB

I. Course Details

Credit Hours	3 (2+1)
Pre-requisites	Web Engineering
Instructor	Ms. Saadia Shabbir
Recommended Textbook(s)	Web Development with JQuery, Richar York, Wiley, 2015
Recommended Reference (Books/Websites/Articles)	1. PHP Web Services, O'Reilly Media, Inc, (2nd edition) 2013 2. The Road to Learn React: Your Journey to Master Plain Yet Pragmatic React. Js, Robin Wieruch, (2020 edition), 2020 3. https://react.dev/blog/2023/03/16/introducing-react-dev 4. https://www.w3schools.com/nodejs/

II. Lab Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Demonstrate the task by interpreting, summarizing, and explaining key concepts/principles/relationships within the domain, enabling them to apply acquired knowledge effectively in diverse contexts	Cognitive	2	2	Lab Task, Lab Report, Final Exam, Semester Project
CLO-2	Practice theoretical concepts for conducting the experiments/projects of varying complexities to achieve desired outcomes.	Psychomotor	3	4	Lab Task, Semester Project
CLO-3	Contribute actively and responsibly individually or in teams during laboratory activities, demonstrating a positive attitude and willingness to participate.	Affective	2	6	Lab Task, Semester Project

~~III. Lab Assessment~~

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25
Total	100

~~IV. Weekly Breakdown (All CLOs shall be assessed in every lab)~~

Week No.	Topics
1	Introduction to Advanced Web Concepts
2	Pagination with AJAX
3	Tools Installation (Visual studio, Node JS)
4	Intro to React, Fundamentals of React
5	Data handling with React: Data fetching
6	Data handling with React: React States
7	React UI Styling using States
8	Introduction to Advanced Web Concepts
9	Midterm Exams
10	Working with React API Fetch()
11	React routers
12	Introduction to MongoDB
13	Data handling practice in MongoDB, Introduction to Express
14	Open ended lab
15	Express Middleware
16	Final assessment

SOFTWARE TESTING (SEST- 470)

I. Course Details

Credit Hours	3 (2+1)
Pre-requisites	Software Engineering
Course Leader	Dr. Sumaira Nazir
Recommended Textbook(s)	1. The Art of Software Testing, Third Edition Glenford J. Myers, Corey Sandler, Tom Badgett, 2012. 2. Software Testing: principles and practices, Naresh Chauhan Oxford University Press, 2010.
Recommended Reference (Books/Websites/Articles)	1. Introduction to software testing, Paul Ammann and Jeff Offutt, Cambridge University Press, 2008. 2. www.Onestoptesting.com .

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Explain the software testing concepts and principles to state their main ideas	Cognitive	2	2	A1, Q1, Midterm
CLO-2	Construct test case and test suites for testing different aspects of a software system to ensure quality of software.	Cognitive	3	3	A2, Q2 Midterm, Final Term
CLO-3	Choose the most suitable state-of-the-art software testing techniques for a specific system to ensure comprehensive and effective testing.	Cognitive	3	3	A3, Q3, Presentation/ Project, Final Term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

To discuss the distinctions between validation testing and defect testing, To describe the principles of software testing and component testing, To describe strategies for generating system test cases, To understand the essential characteristics of tool used for test automation, Wrong requirement: not what the customer wants Missing requirement, Requirement impossible to implement, Faulty design, Faulty code, Improperly implemented design Technical Reviews, Informal Reviews, Testing Strategies, Testing Scope, Testing Approaches for Different Domains of software

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Course outline, assessment criteria, introduction to software testing and advantages of software testing	Chapter 1
2		Types of software testing, objectives and types: black box vs. white box, correctness and defects, a classification scheme for quality assurance activities	Chapter 2
3		Static testing (traditional human manual testing methods and features)	Chapter 6 [Textbook 2]
4		Dynamic testing: white-box testing techniques	Chapter 6 [Textbook 2]
5	CLO-2	Unit testing; how it's done in practice. introduction of test driven development(tdd)	Chapter 5
6		Test management(test planning and analysis procedure, test suite formation	Chapter 9 [Textbook 2]
7		Validation activities (integration testing, function testing, system testing)	Chapter 7 [Textbook 2]
8		Regression testing (progressive testing vs regression testing, regression testing objectives)	Chapter86 [Textbook 2]
9	Midterm Exams		

10	CLO-2	Testing process maturity models, measurement and improvement testing processes(tim)	Chapter 14 [Textbook 2]
11		Mobile application testing (model-based testing), mobile environment test design considerations, test scripting for real devices	Chapter 11
12		Testing object-oriented software, object-oriented modeling and UML, conventional testing and object-oriented testing (OOT), verification and validation	Chapter 16 [Textbook 2]
13	CLO-3	Testing web-based systems, web technology evolution (2 tier web system to n tier architecture), challenges in web testing	Chapter 17 [Textbook 2]
14		Test case design and higher order testing, development process with intermediate verification steps.	Chapter 4,6
15		Automation and testing tools, test completion criteria	Chapter 16 [Textbook 2]
16		The psychology of testing, foundations of testing and general rules of testing, Real time systems testing and different cases studies	[Ref. Website 2]

~~SOFTWARE TESTING (SEST- 470) – LAB~~

~~I. Course Details~~

Credit Hours	3 (2+1)
Pre-requisites	Software Engineering
Course Leader	Dr. Sumaira Nazir
Recommended Textbook(s)	1. The Art of Software Testing, Third Edition Glenford J. Myers, Corey Sandler, Tom Badgett, 2012. 2. Software Testing: principles and practices, Naresh Chauhan Oxford University Press, 2010.
Recommended Reference (Books/Websites/Articles)	1. Introduction to software testing, Paul Ammann and Jeff Offutt, Cambridge University Press, 2008. 2. www.Onestoptesting.com.

~~II. Lab Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Demonstrate the task by interpreting, summarizing, and explaining key concepts/principles/relationships within the domain, enabling them to apply acquired knowledge effectively in diverse contexts	Cognitive	2	2	Lab Task, Lab Report, Final Exam, Semester Project
CLO-2	Practice theoretical concepts for conducting the experiments/projects of varying complexities to achieve desired outcomes.	Psychomotor	3	4	Lab Task, Semester Project
CLO-3	Contribute actively and responsibly individually or in teams during laboratory activities, demonstrating a positive attitude and willingness to participate.	Affective	2	6	Lab Task, Semester Project

~~III. Lab Assessment~~

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25

Total	100
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~~IV. Weekly Breakdown (All CLOs shall be assessed in every lab)~~

Week No.	Topics
1	Introduction to Quality Engineering, how to perform QE with demonstration on a case study.
2	How to perform QA at different phases of SDLC. Perform QA at requirement phase
3	How to test design. Perform testing on a design
4	What is test plan document, develop a test plan document.
5	What is test data, how to generate test data. Generate test data for the given system.
6	What is test case, how to generate case. Generate case for the given system.
7	Manually test the system through black box testing.
8	Testing tools introduction, Open ended lab
9	Midterm Exams
10	How to install testing tools, How to use testing tool
11	Test the system using tools (Black Box testing)
12	White box testing, case study of white box testing.
13	Calculate cyclomatic complexity of the code
14	Generate test cases for white box testing
15	How to make a bug report (Trello, email, excel, word, Git)
16	Lab Exam

~~Semester VIII~~

~~FINAL YEAR PROJECT (SEFP-499)~~

~~I. Course Details~~

Credit Hours	6
Pre-requisites	Programming Fundamentals, OOP, VP, Web Programming, DBMS
Course Leader	Dr. Jaweria Kanwal
Recommended Textbook(s)	
Recommended Reference (Books/Websites/Articles)	As prescribed by the teacher depending on the project's domain

~~II. Course Learning Outcomes~~

CLOs	Description	Domain-Taxonomy level	PLOs	Assessment Artifacts
CLO-1	Identify the problem or deficiency in the existing software product, system or concept	C-2	1.Academic knowledge	Proposal presentation /Proposal report
CLO-2	Apply knowledge of computing to formulate a problem statement and requirements (functional and non-functional) based on research literature and existing systems.	C-3	2. Knowledge for Solving Computing Problems	Proposal presentation /Proposal report
CLO-3	Propose the different components of a software project and model requirements to generate system design based on the proposed components.	C-5	3: Problem Analysis	Progress presentation /Progress report
CLO-4	Construct the proposed software project using suitable tools based on the appropriate algorithms/APIs/DB schema	P-5	5: Modern Tool Usage	Final report/ Prototype
CLO-5	Produce test results (based on test cases) of various modules of the developed project	P-4	4: Design/ Development of Solutions	Final report
CLO-6	Justify the application of the proposed solutions for the betterment of Society	C-6	8: Computing Professionalism and Society	Presentation

CLO-7	Demonstrate development of solutions effectively through written and oral mode with the aid of multimedia tools.	A-4	7: Communication	Project code/Presentation
CLO-8	Demonstrate the originality and relevance of document's content to the project	A-3	9: Ethics	Final report
CLO-9	Show professional team coordination and communication	A-3	6: Individual and Team Work	Presentation
CLO-10	Demonstrate and defend understanding of tools and techniques used in the development of the developed project	C-3	10: Life-long Learning	Presentation

III. Marks weightage of CLOs according to Rubrics

CLOs	Project Proposal (40)	Project Progress (40)	Final Project (90)	Supervisor evaluation (30)
CLO-1	4			
CLO-2	12	6		
CLO-3	2	6		
CLO-4	4	16	8	
CLO-5			44	
CLO-6	4			
CLO-7	8	8	8	5
CLO-8	4	2	6	10
CLO-9	2	2	8	10
CLO-10			16	5

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Catalogue description

This course is designed as finale project which requires students to demonstrate technical, theoretical and presentation skills at levels which are aligned with professional software engineering practices. It is needed that students will apply their knowledge of computing throughout the course such as development of requirements, design and implementation to develop a software solution to a real-world problem from conception to completion. In this part, students propose a practically useful project, and develop a document which consists of project plan and software design document.

FOREIGN LANGUAGE (SEFL-472)

I. Course Details

Credit Hours	2 (2+0)
Pre-requisites	-
Course Leader	Mr. Zohaib Ahmed
Recommended Textbook(s)	1. Menschen A1.1 Deutsch Als Fremdsprache
Recommended Reference (Books/Websites/Articles)	1. Studio d, A1

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Understand familiar, everyday expressions and very simple sentences, which relate to the satisfying of concrete needs	Cognitive	2	2	Q1, Midterm
CLO-2	Utilize words of everyday use to introduce him/herself and others as well as ask others about themselves.	Cognitive	3	3	A1, Q2 Midterm, Final Term
CLO-3	Apply different sentences to communicate about where they live, who they know and what they own, and what they need etc.	Cognitive	3	3	A2, A3, Q3, Q4 Midterm, Final Term

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50

GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00
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For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO 1	Oral Drilling & Speaking (Greetings,, Counting 1-10, Intro questions, Questions)	1 st
2		Oral Drilling & Speaking (Times of day, Counting 21-100, Days of Week, Months, Seasons of a Year,)	1 st
3		Ordinals numbers 0-20, Years i.e 1990, Questions, Alphabets, Combination of alphabets and their sounds	1 st
4		Combination of alphabets and their sounds	1 st
5		Grammar: Writing & Revision, Personal Pronouns & exercises, Verbs and conjugation	1 st
6		Grammar: Verbs and conjugation, Articles (Definite & indefinite, Possessive) & Exercises, Family	1 st
7	CLO 2	Possessive article and exercise	1 st
8		Speaking & Writing (How to make simple questions, Time and preposition of time) & exercises	1 st
9	Midterm Exams		
10	CLO 2	Revision, Speaking & Writing (Modal Verbs), Exercises, Negation	1 st
11		Grammar (Intro to Akkusativ, Exercises)	1 st
12	CLO 3	Formal Email & Form Filling	1 st and 2 nd
13		Revision	1 st and 2 nd
14		Presentations	1 st and 2 nd
15		Presentations	1 st and 2 nd
16		Exams	1 st and 2 nd

~~PROFESSIONAL PRACTICES (SEPP-416)~~

~~I. Course Details~~

Credit Hours	3 (3+0)
Pre-requisites	-
Course Leader	Dr. Nargis Fatima
Recommended Textbook(s)	Computer Ethics by Deborah G. Johnson, Pearson (Latest Edition). ISBN 10: 0131112414 2. Professional Issues in Software Engineering by Frank Bott, Allison Coleman, Jack Eaton and Diane Rowland, CRC Press
Recommended Reference (Books/Websites/Articles)	1. A Gift of Fire: Social, Legal, and Ethical Issues for Computing and the Internet by Sara Baase, Prentice Hall. ISBN-10: 0136008488 2. Applied Professional Ethics by Gregory R. Beabout, University Press of America (1993). ISBN-10: 0819193747 3. Ethics in information technology, Reynolds, G., 6th Edition (2018), Cengage, 2018. 4. Ethical and social issues in information age, Kizza J. M., 6th Edition, Springer-Verlag, 2017. 5. A gift of fire, social, legal, and ethical issues for computing technology, Baase, S., 5th Edition, Pearson, 2017.

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Explain various terms in Computer and General Ethics.	Cognitive	2	9	A1, Q1, Midterm
CLO-2	Apply the IEEE / ACM Code of Ethics on various situations.	Cognitive	3	8	A2, Q2, Final exam
CLO-3	Analyze given situations for finding ethical, legal and religious issues and their solutions.	Cognitive	4	10	Q3, Midterm, Final Exam

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50

Total	100
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IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

A Computing graduate as professional has some responsibilities with respect to the society. It identifies key sources for information and opinion about professionalism and ethics. Historical, social context of Computing (SE, CS & IT). Professional activities; professional societies; professional ethics; professional competency and life-long learning. Uses, misuses, risks of software; information security & privacy. Business practices; intellectual property & software law (cyber law). Social responsibilities, software related contracts, Software Houses Management. Professional ethics methodologies, computer ethics, ethical issues and social impact of these ethical issues in our lives. Analyze, evaluate, and assess ethical and professional computing case studies.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Introduction to ethics, ethics considerations in decision making, and information technology, Technology and Ethics w.r.t Islam Computer Ethics vs morality.	
2		Computer Crime, criminals, fraud, Deep Fake, Dark Web	
3		Hackers and Hacking, Cracking, Computer Sabotage, Security, Legislation, and Education.	
4		ACM Code of Ethics and Professional Conduct, Overview of all four parts. General Ethical Principles for IT Users, IT Professionals' Responsibilities, IT Leaders' Responsibilities	
5		Copyright, Patent, The Basis of property law, Legal Aspects	
6		Intellectual Property Rights, Protecting Intellectual Property, Free Software, Adware & Spyware, Pirated software, Software Piracy The Moral Implications of Software Piracy, Islamic Perspective of software piracy, Piracy vs Plagiarism.	

7	CLO-2	Privacy and Anonymity, The Invasion of Privacy, The Right of Privacy, Methods of Privacy Violation, Surveillance Technologies, Neighborhood Surveillance, Protecting Privacy	
8		Computer Reliability, Can we trust computers?, What is Computer reliability?, Professional Responsibility, Software Liability, Solutions to the Problems	
9	Midterm Exams		
10	CLO-2	Computer Crime and Security, Objectionable Material: What is it? How to protect children from accessing it. · Vulgarity: Pornography, Unethical Videography and Cyber Laws, Punishment. · Violence and Hatred: What is it? Cyber Laws, Punishment. Blasphemy: What is it? Cyber Laws, Punishment. Bitcoins: What is it? Cyber Laws, Punishment	
11	CLO-3	The Structure of Organizations: Anatomy of a Software House, Finance and Accounting, The Framework of Employee Relations Law and Changing Management Practices, Accountability and Auditing, Human Resource Management and IT, JD (Job Descriptions) & Contracts	
12		Computers and Health & Safety at Work, Medical Robots, Telemedicine	
13		Ethical Concerns in: Off-shore employment (anonymous), Free lancing, FIA (roles)	
14		Bioinformatics and computational genomics, Introduction & Basic issues, Judgments about individuals (Employment/ Insurance), Ethical concern on ownership of genetic data	
15		Bioinformatics and computational genomics, Ethical concerns in plants and animal's genomics, Cloning, Genetic Engineering	
16		Bioinformatics and computational genomics: Ethical concerns in plants and animal's genomics, Cloning, Genetic Engineering	

SOFTWARE PROCESS IMPROVEMENT (SEPI-471)

I. Course Details

Credit Hours	3 (3+0)
Pre-requisites	Introduction to Software Engineering
Course Leader	Dr. Nargis Fatima
Recommended Textbook(s)	1. Introduction to Software Process Improvement, Regan, G., Springer, 2011. 2. Software Process Definition and Management, Münch, J., Armbrust, O., Kowalczyk, M., Soto, M., Springer, 2012. Practical process improvement Robert fantana 2005 3. Sustaining a Culture of Process Control and Continuous Improvement The Roadmap for Efficiency and Operational Excellence, Philip Gisi, Taylor & Francis Group, 2018
Recommended Reference (Books/Websites/Articles)	1. Software Engineering, A Practitioner's Approach, Pressman R. S. & Maxi R., 8th Edition, McGraw-Hill, 2015 2. Concise Guide to Software Engineering Gerard O'Regan 2017

II. Course Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy level	PLOs	Assessment Artifact
CLO1	Explain fundamental concepts of software processes improvement to improve the processes in software development organizations	Cognitive	2	2	Q1, A1, Midterm
CLO2	Demonstrate software process improvement models for appropriate implementation in software development organizations.	Cognitive	2	2	Q2, A2, Midterm, Final Term
CLO3	Apply software process improvement approaches and models for problem analysis and process improvement.	Cognitive	3	3	Q3, A3, Final Term, Project/Presentation

III. Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

~~Software process, software process improvement, SPI approaches, SPI process, software process improvement models, Process Mapping, Process Improvement Initiatives, Software Process Improvement Models CMM, CMMI, SPICE, etc. Process Improvement in Requirements Engineering, process improvement in project management, process improvement in Quality assurance, Model-Based Improvement Approaches(ISO/IEC 15504 (SPICE)), Continuous Improvement Approaches (PDCA Cycle, Total Quality Management ,Zero Defect Program ,Six Sigma ,The Quality Improvement Paradigm,)~~

VI. Weekly Breakdown

Wee k No.	CLO	Topics	Reference
1	CLO1	Course Introduction and overview, Software Process, Software Process Improvement (SPI), SPI motivation, Historical aspect of SPI Motivation for Software Process Improvement, Benefits of Software Process Improvement	Chapter 1 [Textbook 1]
2		Triggers and Types of Process Improvement, Process Improvement Initiatives, Barriers to Successful SPI, Process Mapping.	Chapter 1 [Textbook 1] Chapter 8 [Textbook 3]
3	CLO2	Swim Lane, Lean, Wastes, Value Adding Vs Non Value Adding, Steps for SPI Process, Value Stream mapping	Chapter 6, 9 [Textbook 3] Chapter 37 [Ref Book 1]
4		SPI Framework, Process Models/Standards for SPI	Chapter 1 [Text Book 2]
5		CMMI, CMMI for Development, (CMMI-DEV) – CMMI for Services (CMMI-SVC) – CMMI for Acquisition (CMMI-ACQ), CMMI-DEV Maturity levels, Capability Levels,	Chapter 3 [Textbook 1] Chapter 15 [Ref Book 2]

		Staged Representations, Continuous Representations, CMMI Vs CMM	
6		Setting up CMMI initiative: CMMI Improvement Structure and Teams (Setting up SEPG team, setting up steering group, Setting Up Dedicated Improvement Sub-teams, Role of the CMMI Project Manager)	Chapter 4 [Text book1]
7		Requirements Engineering Process Improvement and Related Models. RE-CMM, MDREPM – Market Driven Requirement Engineering PM, RE Process Maturity	Chapter 4 [Text book 1]
8		Continuous Improvement Approaches (PDCA Cycle)	Chapter 5 [textbook 2]
9	Midterm Exams		
10	CLO2	Model-Based Improvement Approaches (ISO 9001, ISO/IEC 15504 (SPICE)	Chapter 15 [Ref. Book 1]
11	CLO3	Personal software process, Implementation of the Personal Software Process	Chapter 7
12		Team software process, Managing Software Projects with Team Software Process (TSP)	Chapter 4 [Textbook 2]
13		Six sigma, Root cause analysis (RCA), Balanced score card, DMAIC and DMADV methodologies.	Chapter 15 [Textbook 2]
14		Reducing risks with SPI, Resistance Factors in Software Processes Improvement	Chapter 6 [Textbook 1]
15		Total Quality Improvement Paradigm, Software Process Improvement for Small and Very Small Enterprises case study	Chapter 9 [Textbook 1] Chapter 12 [Textbook 3]
16		SPI return on investment, SPI trends	Chapter 37 [Textbook 2]

~~GAME APPLICATION DEVELOPEMNT (SEGD-471)~~

~~I. Course Details~~

Credit Hours	3 (2+1)
Pre-requisites	-
Course Leader	Ms. Saadia Shabbir
Recommended Textbook(s)	1. Learning C# by Developing Games with Unity 2020 by Harrison Ferrone, 2020. 2. Introduction to Game Design, Prototyping, and Development: From Concept to Playable Game with Unity and C#, 2nd Edition, 2017
Recommended Reference (Books/Websites/Articles)	Game Design: Theory and Practice (2nd Edition), 2005

~~II. Course Learning Outcomes (CLO)~~

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Understand game development concepts, to comprehend the fundamental principles underlying game design and development.	Cognitive	2	2	A1, Q1, Mid Term
CLO-2	Develop engaging games with captivating storylines and immersive gaming environments to enhance player immersion.	Cognitive	3	4	A2, Q2, Mid Term, Final Term
CLO-3	Create games with modern gaming engines using Unity for designing and implementing interactive gaming experiences.	Cognitive	6	5	A3, Q3, Project, Final Term

~~III. Course Assessment~~

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	100

V. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

VI. Course Contents

Students will work on various aspects of game design and production: storyboarding, level design, texturing & materials, programming, terrain, physics, animation, 2D & 3D rendering, and user interface. Students will learn the Unity game engine and the C# programming language.

VI. Weekly Breakdown

Week No.	CLO	Topics	Reference
1	CLO-1	Course introduction, Fundamental game development concepts: a brief history of how games have been developed	Chapter 1 [Textbook 1]
2		Basic components of a game Sprites, animations, coordinate systems. Introduction to GML	Chapter 1 [Textbook 1]
3		Types of Games, Introduction to Game frameworks. TheMDA Framework, 6-11 framework.	Chapter 2, 4 [Textbook 2]
4	CLO-2	Introduction to Game Design Document and its format. How to develop a Game Design Document.	Chapter 2 [Textbook 1]
5		Introduction to Game Design Document and its format. How to develop a Game Design Document.	Chapter 2 [Textbook 1]
6		2D games platforms: Building a sample 2D Game	Chapter 2 [Textbook 1]
7		Introduction to 3D Modelling in Blender	Chapter 3 [Textbook 1]
8		Introduction to character modelling in Blender 3D	Chapter 3 [Textbook 1]
9	Midterm Exam		
10	CLO-3	Multiple Gaming engines. Introduction to 3D Games, Handling lights, particle systems, fire, smoke, water and more.	Chapter 13 [Textbook 2]
11		Introduction to Unity 3D, Interface, object rotation, scaling etc.	Chapter 13, 14

			[Textbook 2]
12		Using Terrains, mountains and landscapes, placing objects like trees, stones and texturing them to create custom worlds.	Chapter 4 [Textbook 2]
13		Creating a sample 3D Game in Unity 3D Part 1	Internet Resource
14		Creating a sample 3D Game in Unity 3D Part 2	Chapter 9 [Textbook 1]
15		Creating a sample 3D Game in Unity 3D Part 3	[Ref. Book 1]
16		Advance Gaming concepts: AI in Gaming, Network Games etc.	Chapter 20 [Ref. Book 2]

GAME APPLICATION DEVELOPMENT (SEGD-471) – LAB

I. Course Details

Credit Hours	3 (2+1)
Pre-requisites	-
Course Leader	Ms. Saadia Shabbir
Recommended Textbook(s)	1. Learning C# by Developing Games with Unity 2020 by Harrison Ferrone, 2020. 2. Introduction to Game Design, Prototyping, and Development: From Concept to Playable Game with Unity and C#, 2nd Edition, 2017
Recommended Reference (Books/Websites/Articles)	Game Design: Theory and Practice (2nd Edition), 2005

II. Lab Learning Outcomes (CLO)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifact
CLO-1	Demonstrate the task by interpreting, summarizing, and explaining key concepts/principles/relationships within the domain, enabling them to apply acquired knowledge effectively in diverse contexts	Cognitive	2	2	Lab Task, Lab Report, Final Exam, Semester Project
CLO-2	Practice theoretical concepts for conducting the experiments/projects of varying complexities to achieve desired outcomes.	Psychomotor	3	4	Lab Task, Semester Project
CLO-3	Contribute actively and responsibly individually or in teams during laboratory activities, demonstrating a positive attitude and willingness to participate.	Affective	2	6	Lab Task, Semester Project

III. Lab Assessment

Evaluation Methods	Weight (%)
Internal Evaluation / Lab Report	60
Project / Open ended Lab	15
Final Term Exam	25
Total	100

IV. Weekly Breakdown (All CLOs shall be assessed in every lab)

Week No.	Topics
1	Intro to Game Maker Studio
2	Creating First Game in Game Maker Studio
3	Adding Elements to Game in Game Maker Studio
4	Publishing game in Game Maker Studio
5	Introduction to Blender
6	Introduction to Unity
7	Introduction to GDD
8	Intro to Game Maker Studio
9	Midterm Exam
10	First Game in Unity C# Part I
11	First Game in Unity C# Part II
12	Creating Improved Visuals
13	Unity Playground Part I
14	Unity Playground Part II
15	Unity Playground Part III, IV
16	Project evaluation and lab assessment

*open ended lab after week 8